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Jun et al.

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(54) **IMAGE ENCODING/DECODING METHOD AND DEVICE, AND RECORDING MEDIUM STORING BITSTREAM**

(58) **Field of Classification Search**
CPC H04N 19/00
See application file for complete search history.

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(21) Appl. No.: **17/733,615**

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Foreign Application Priority Data

Nov. 16, 2017 (KR) 10-2017-0153191

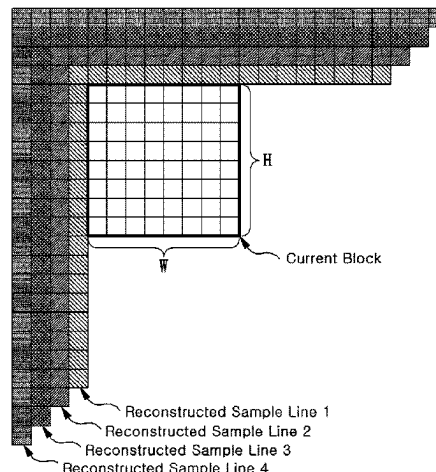
(51) **Int. Cl.**
H04N 19/00 (2014.01)
H04N 19/11 (2014.01)
(Continued)

(57) **ABSTRACT**

An image encoding/decoding method and apparatus for performing template matching-based intra prediction are provided. An image decoding method may comprise deriving a first intra-prediction mode for a current block, generating a first intra-prediction block corresponding to the first intra-prediction mode, deriving a second intra-prediction mode for the current block, generating a second intra-prediction block corresponding to the second intra-prediction mode, and generating a final intra-prediction block by using a weighted sum of the first intra-prediction block and the second intra-prediction block.

(52) **U.S. Cl.**
CPC **H04N 19/159** (2014.11); **H04N 19/11** (2014.11); **H04N 19/176** (2014.11)

12 Claims, 24 Drawing Sheets



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H04N 19/176 (2014.01)

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FIG. 1

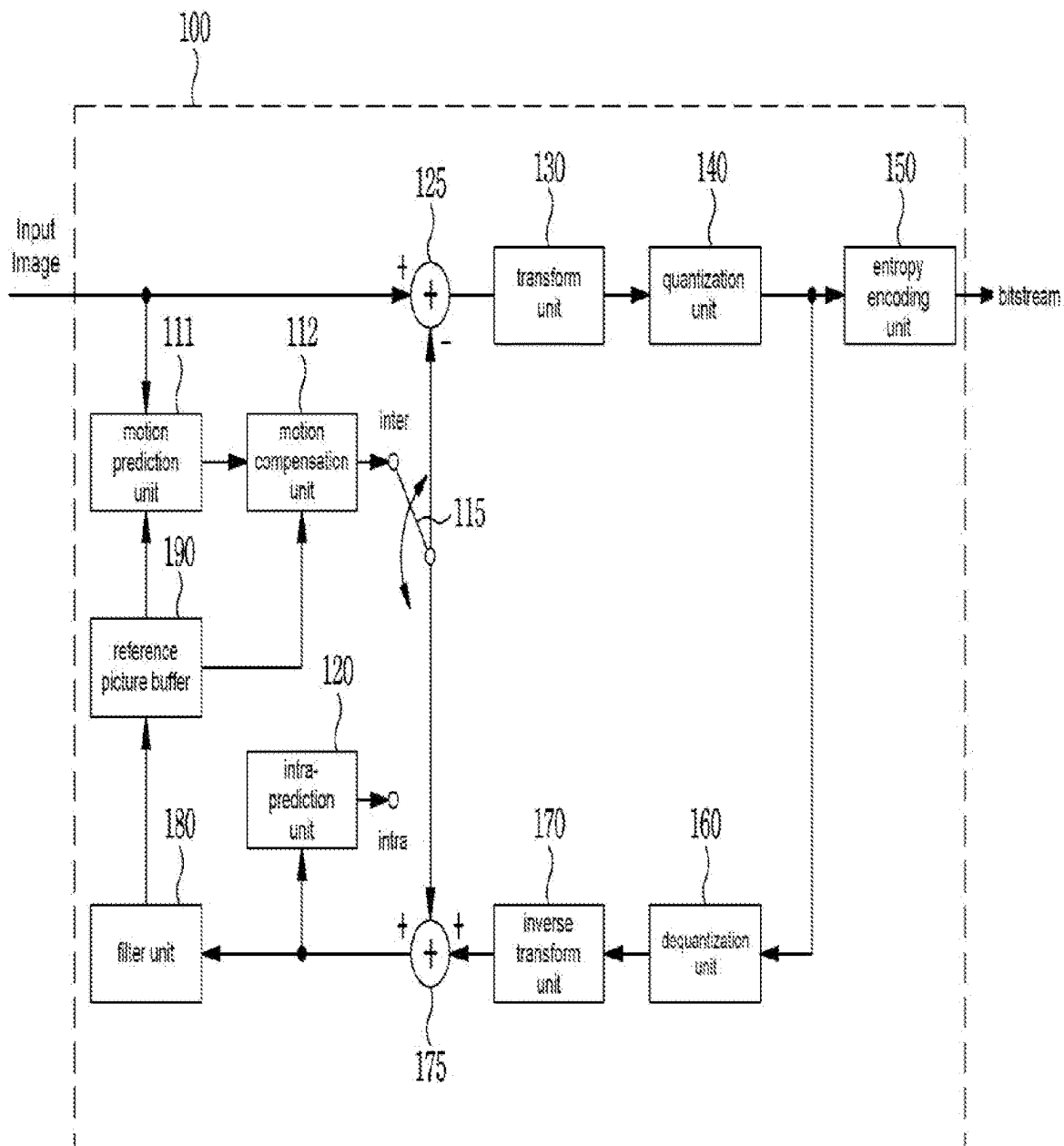


FIG. 2

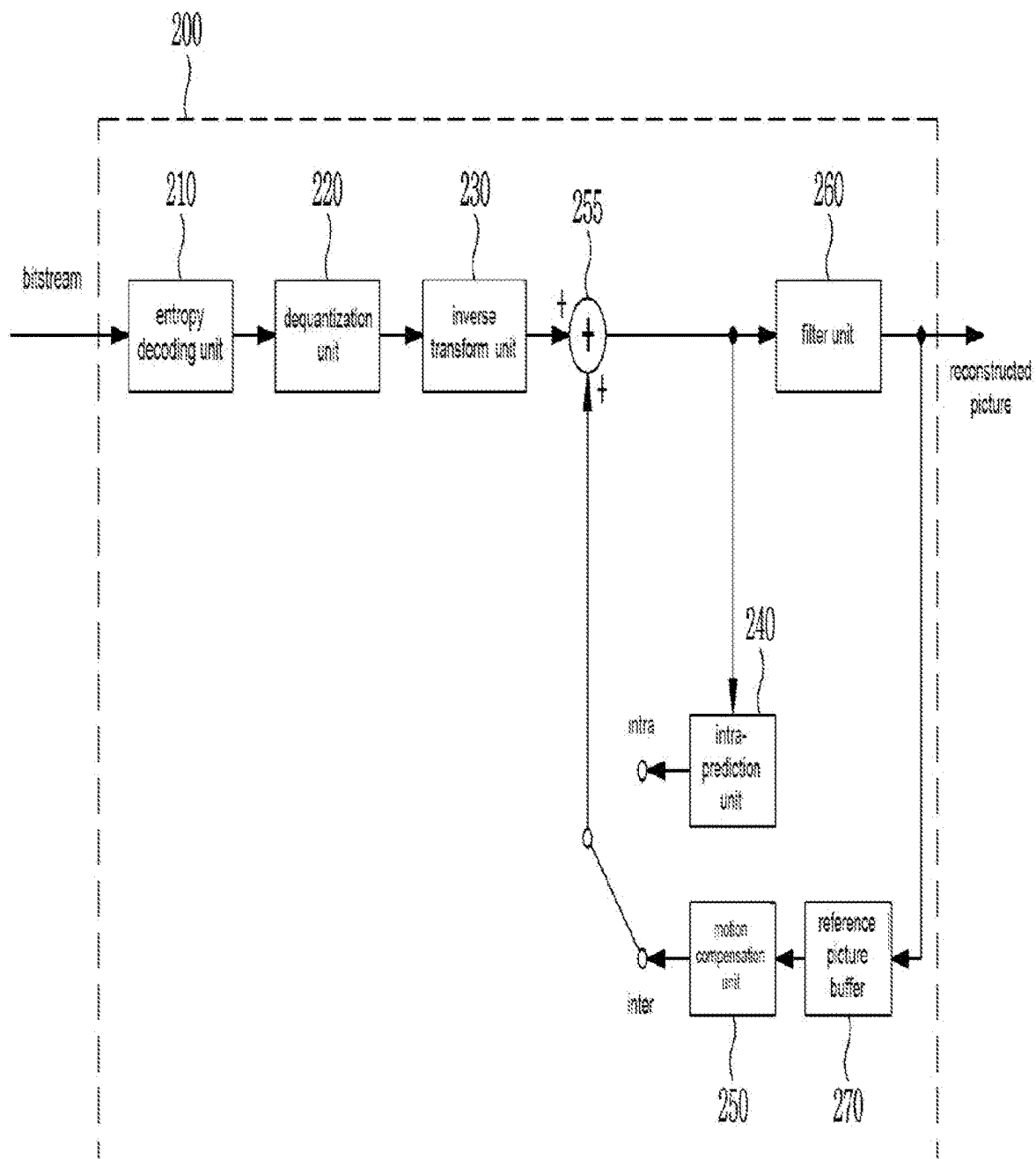


FIG. 4

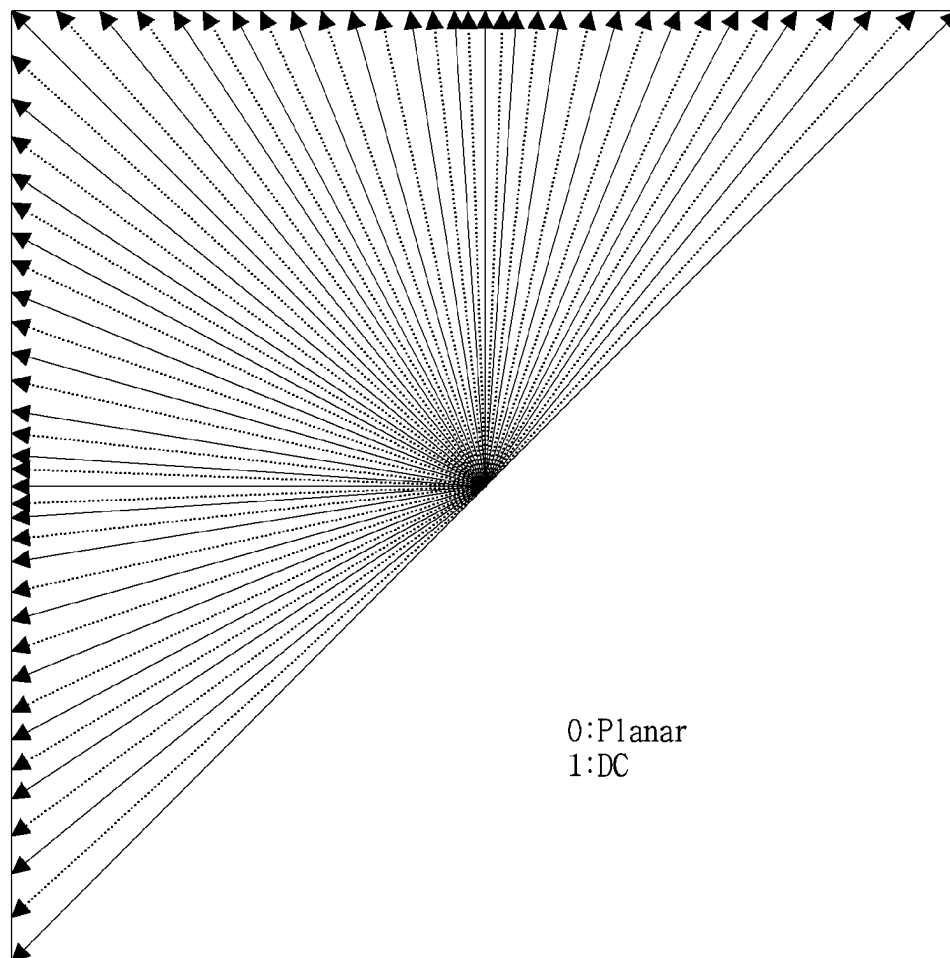


FIG. 5

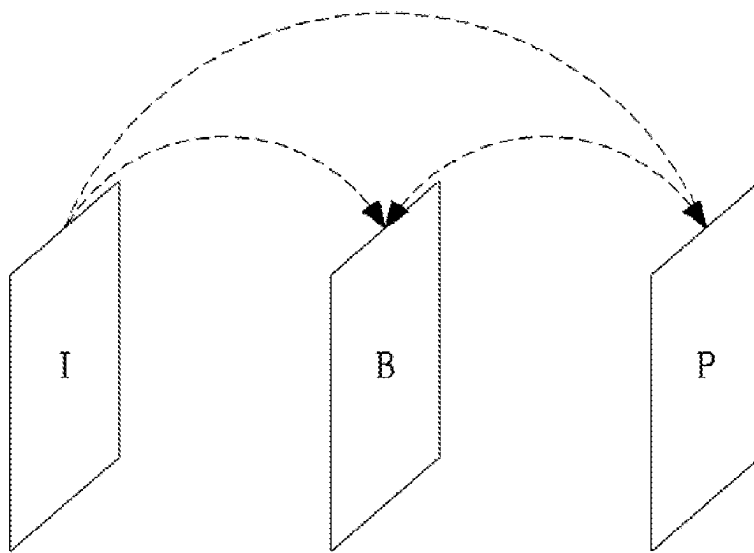


FIG. 6

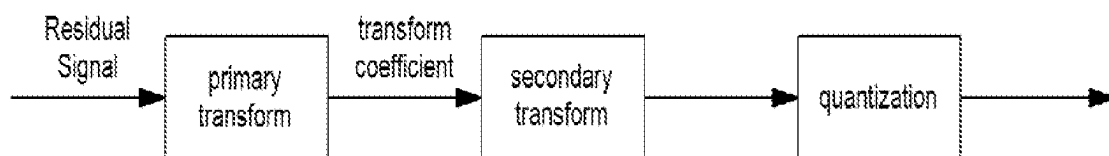


FIG. 7

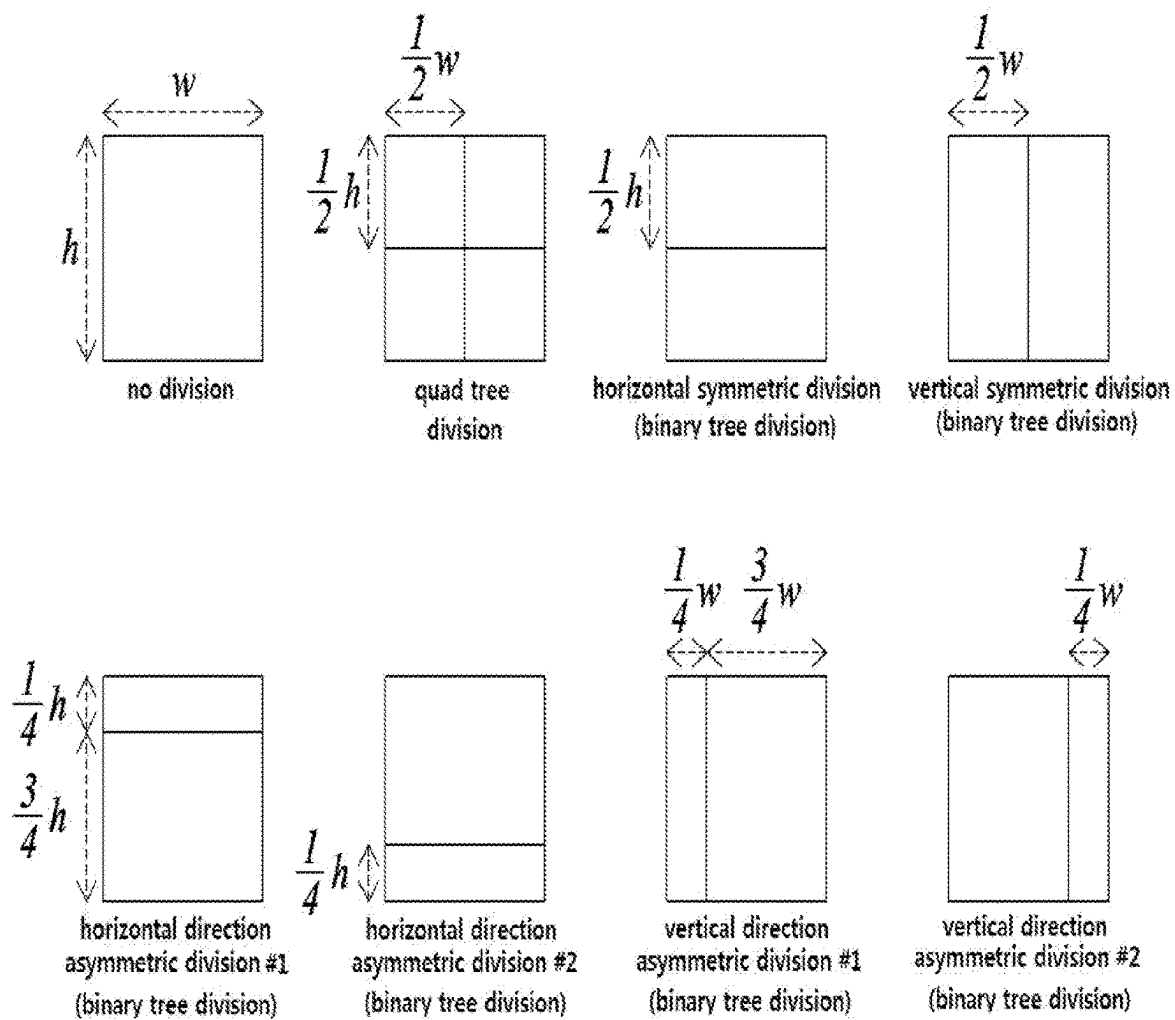


FIG. 8

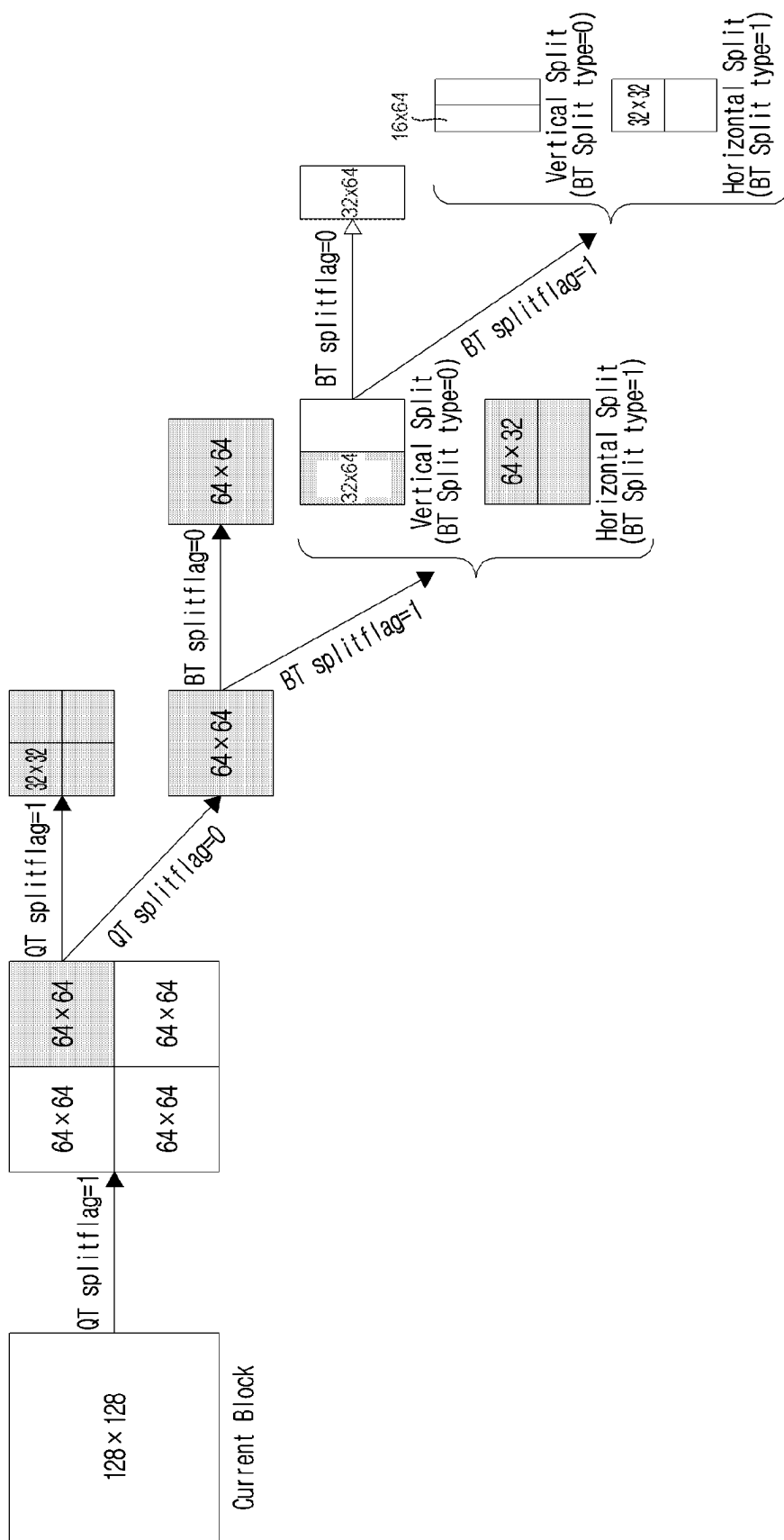


FIG. 9

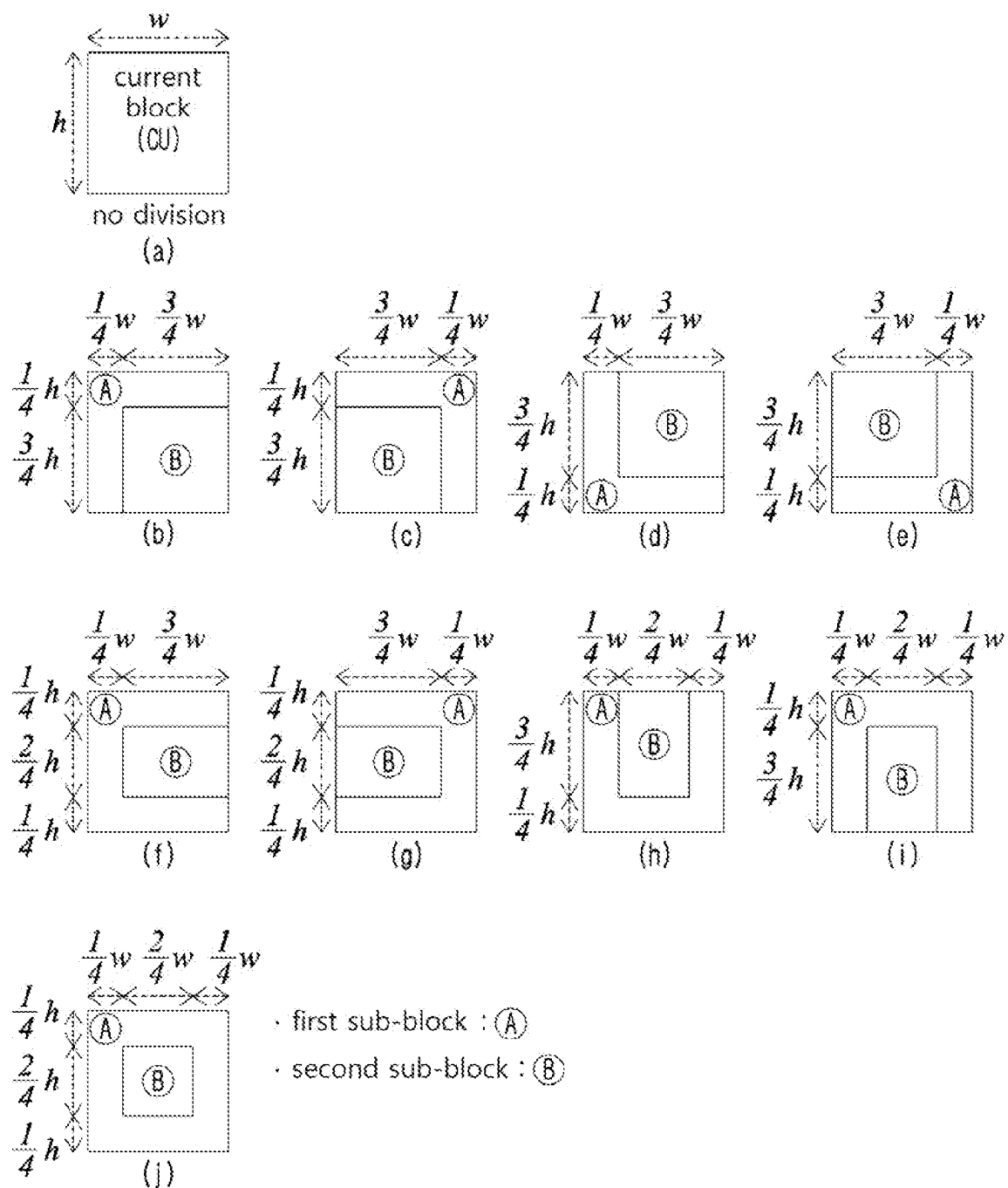


FIG. 10

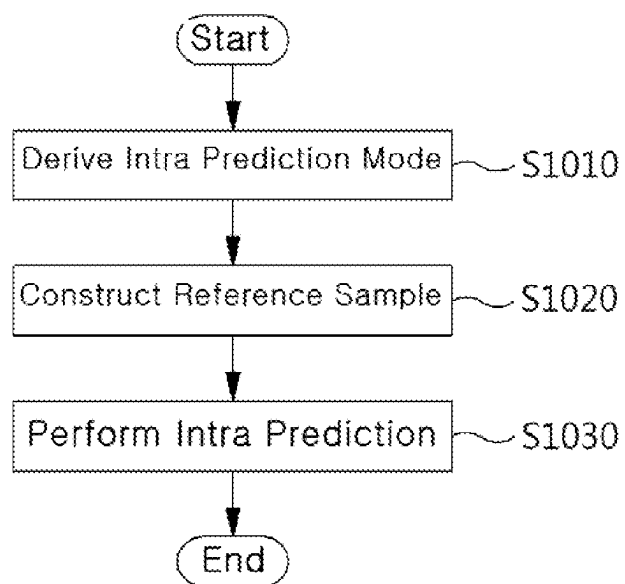


FIG. 11

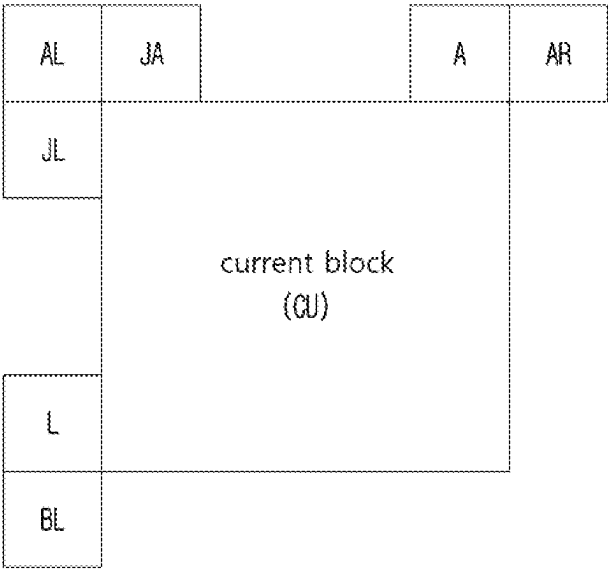


FIG. 12

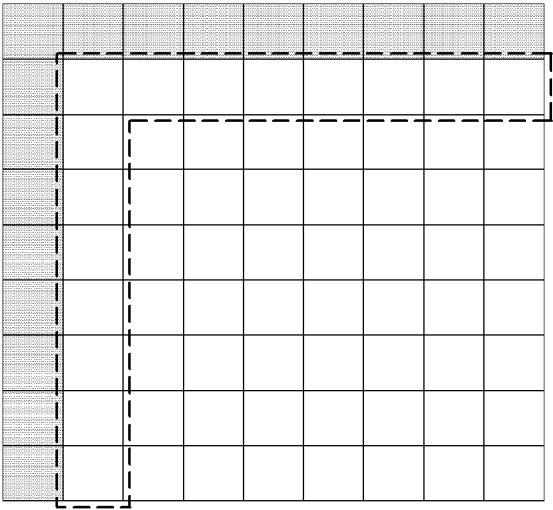


FIG. 13

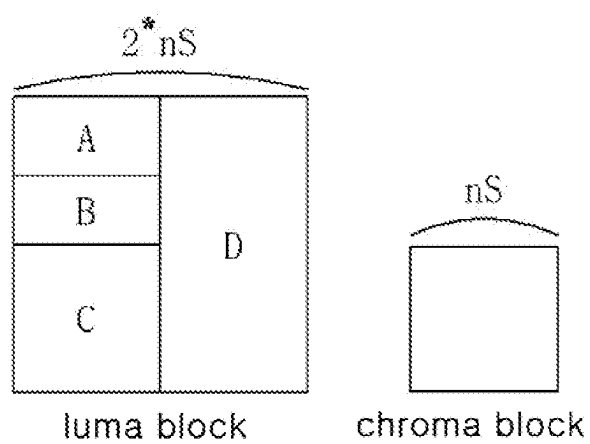


FIG. 14

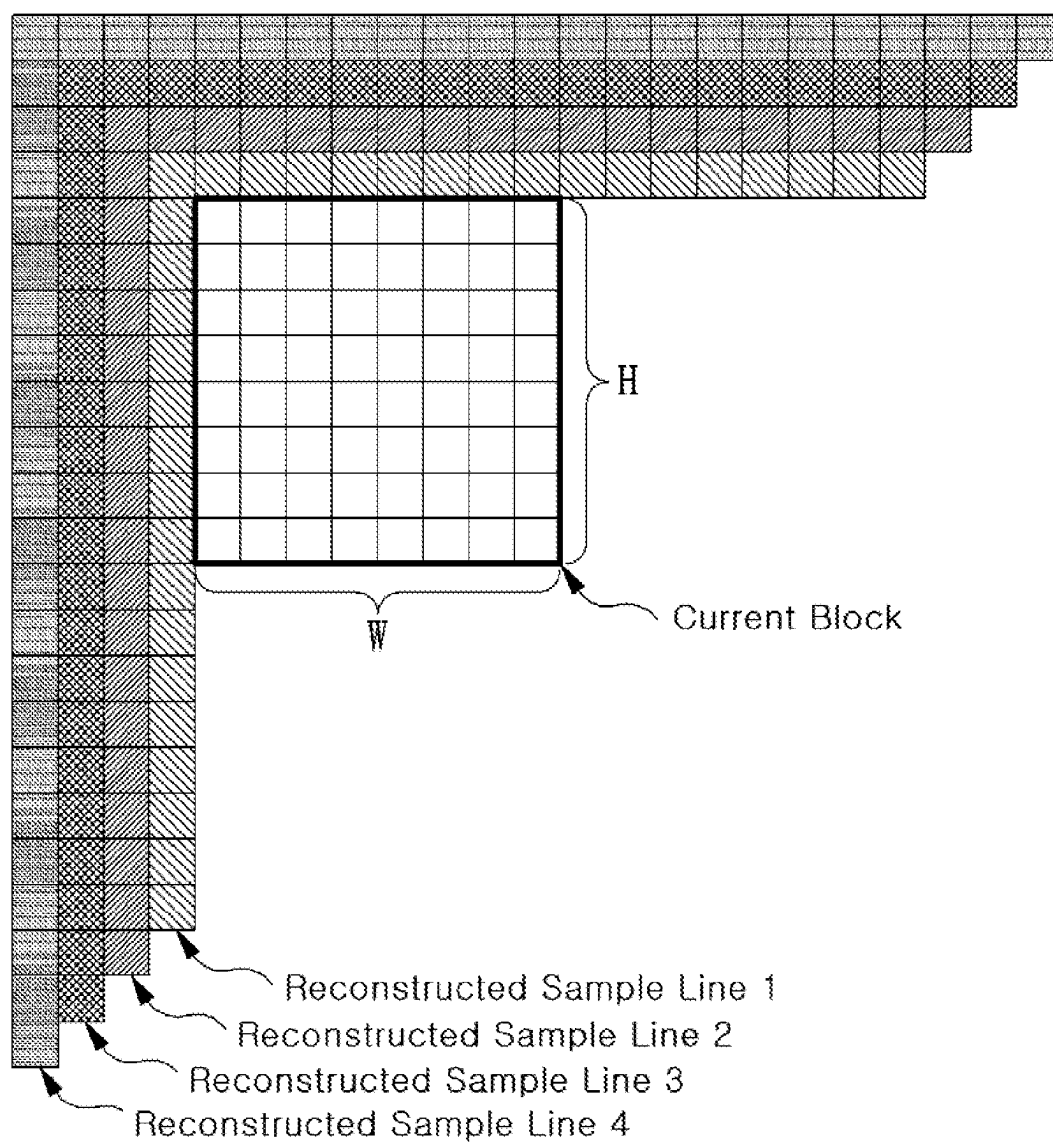


FIG. 15

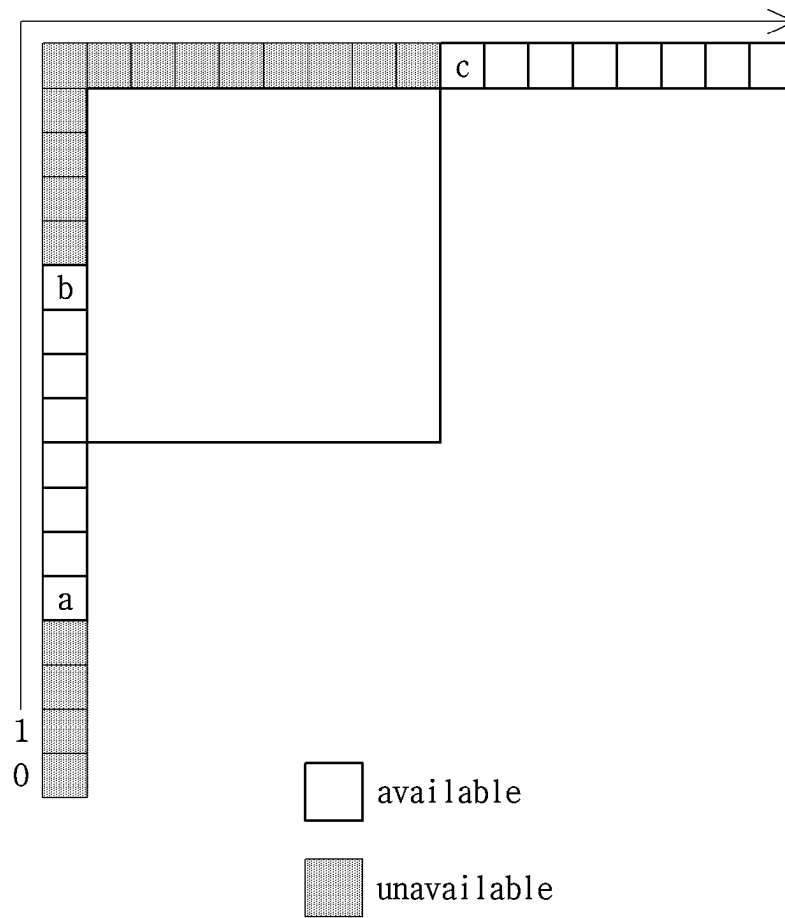


FIG. 16

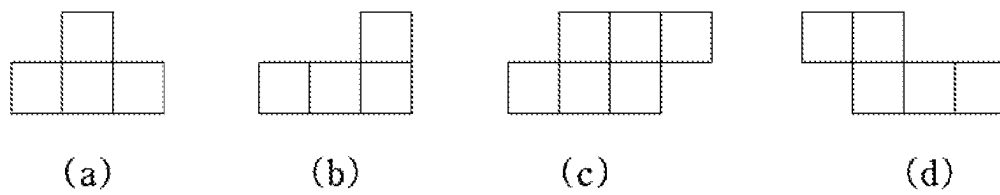


FIG. 17

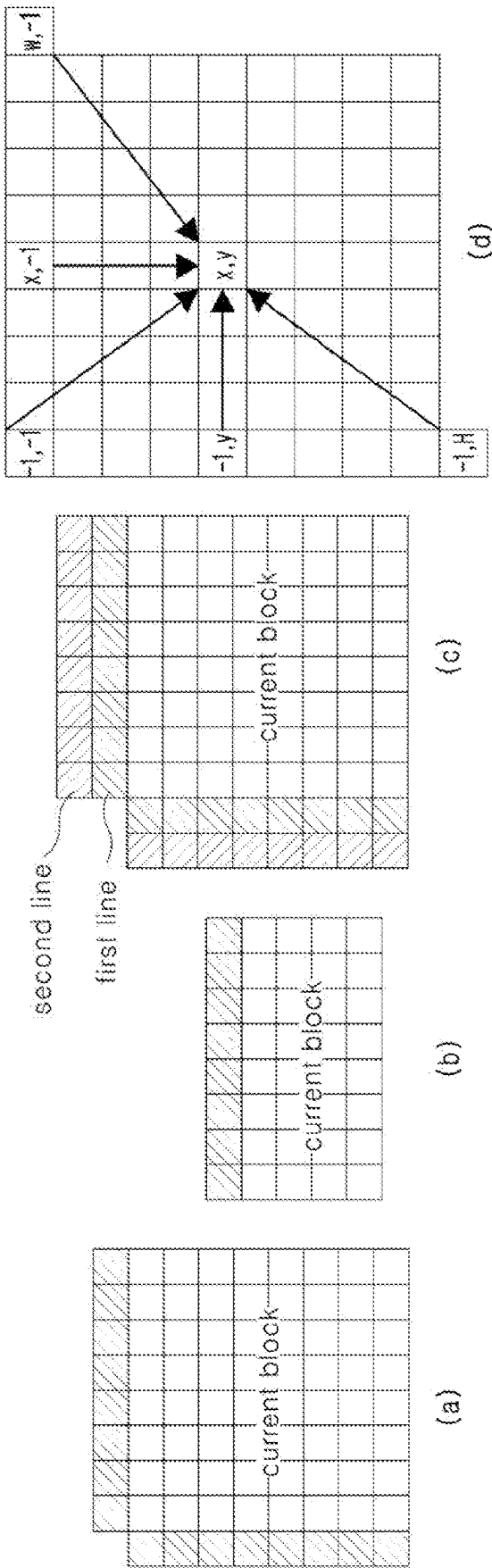


FIG. 18

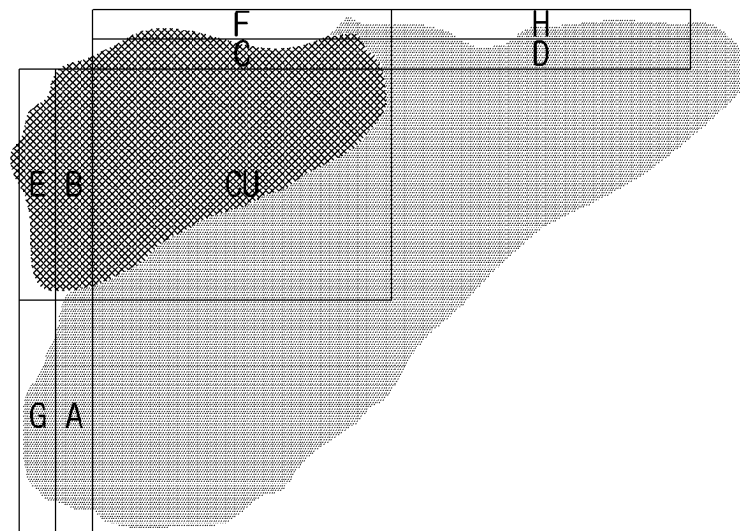
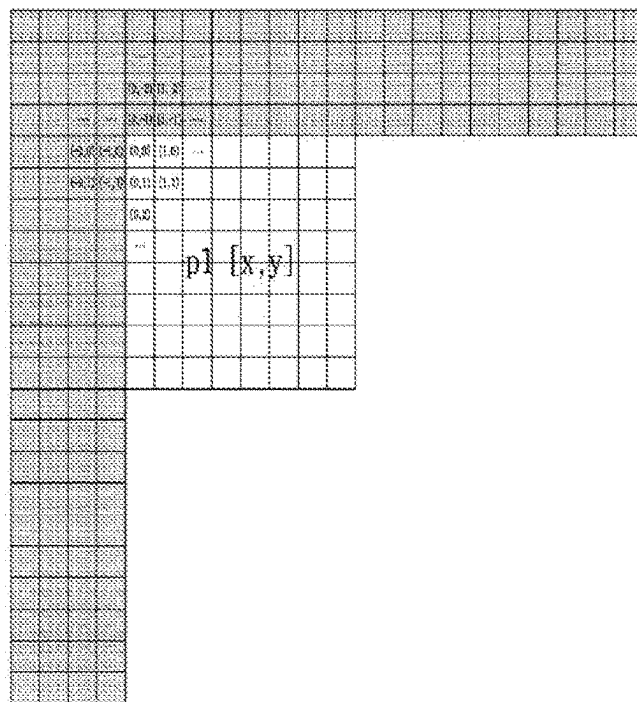
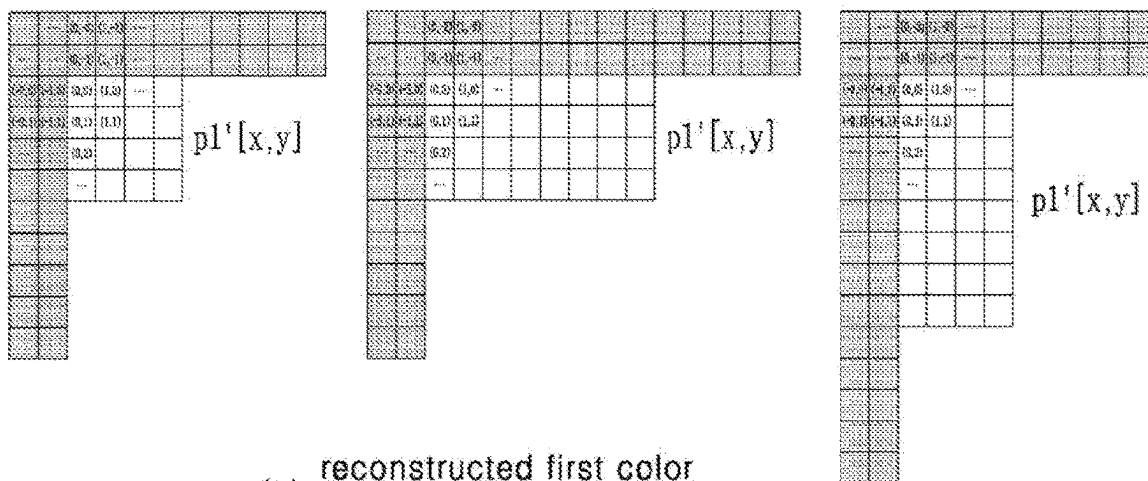


FIG. 19



(a) first color component block



(b) reconstructed first color component block

FIG. 20

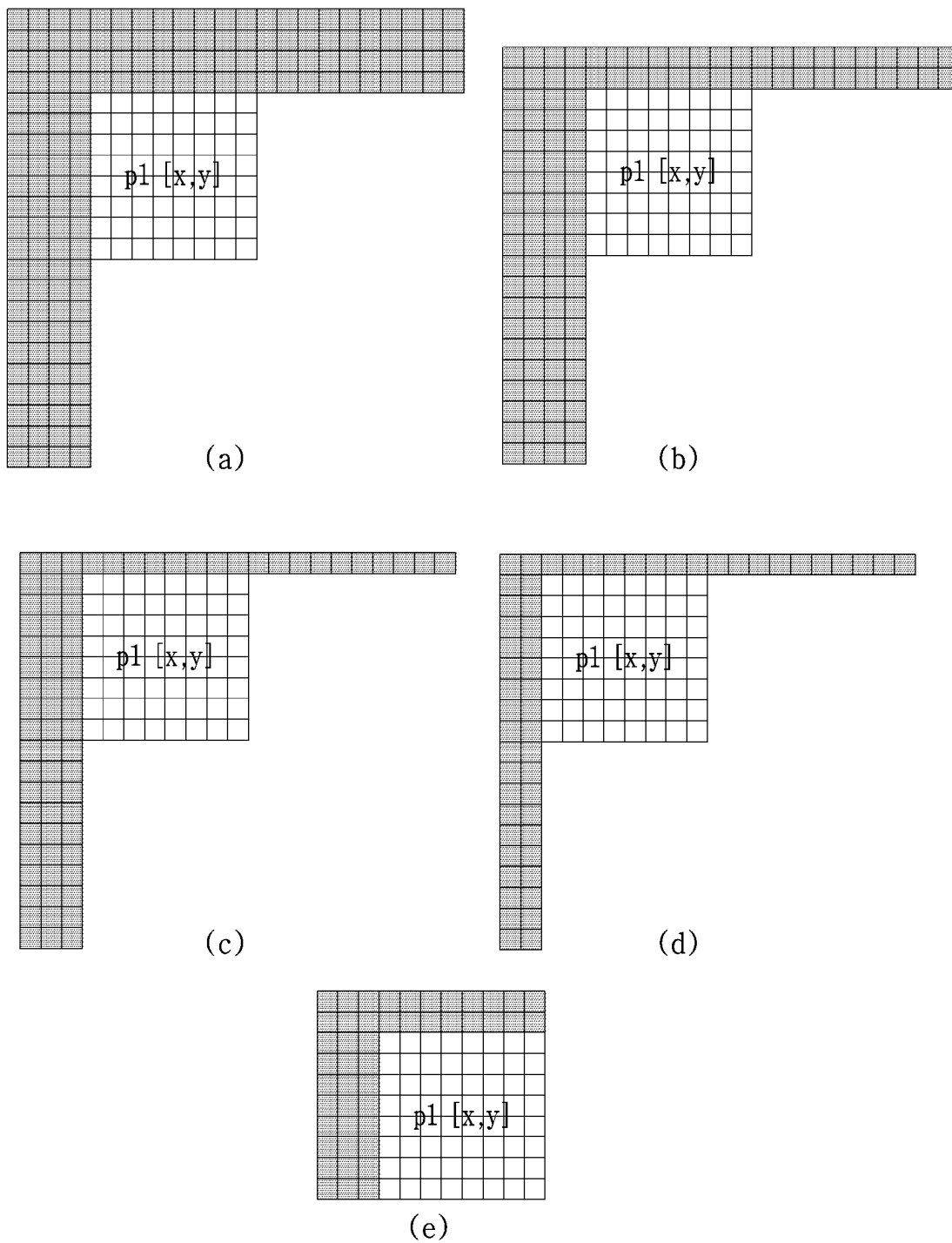
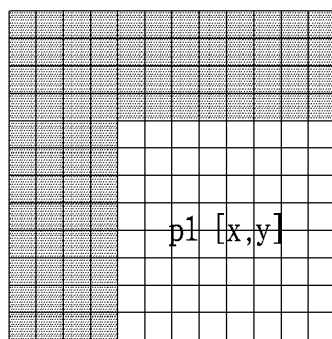
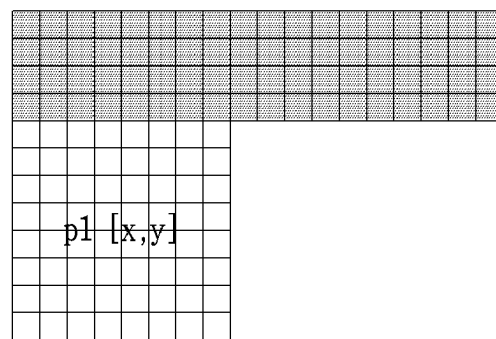


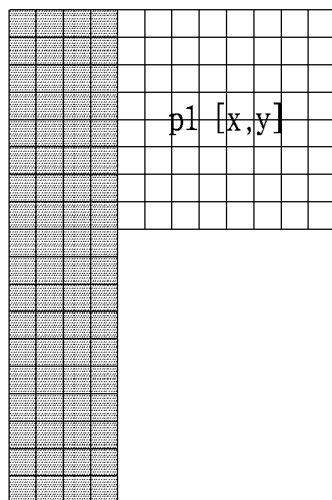
FIG. 21



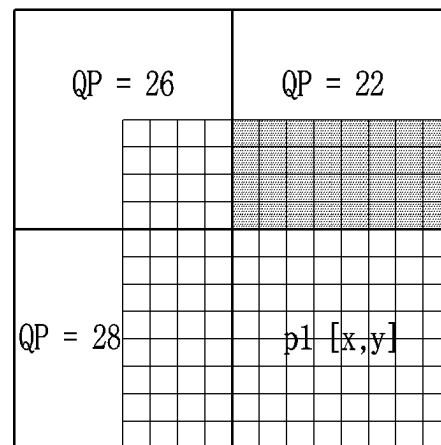
(a)



(b)



(c)



(d)

FIG. 22

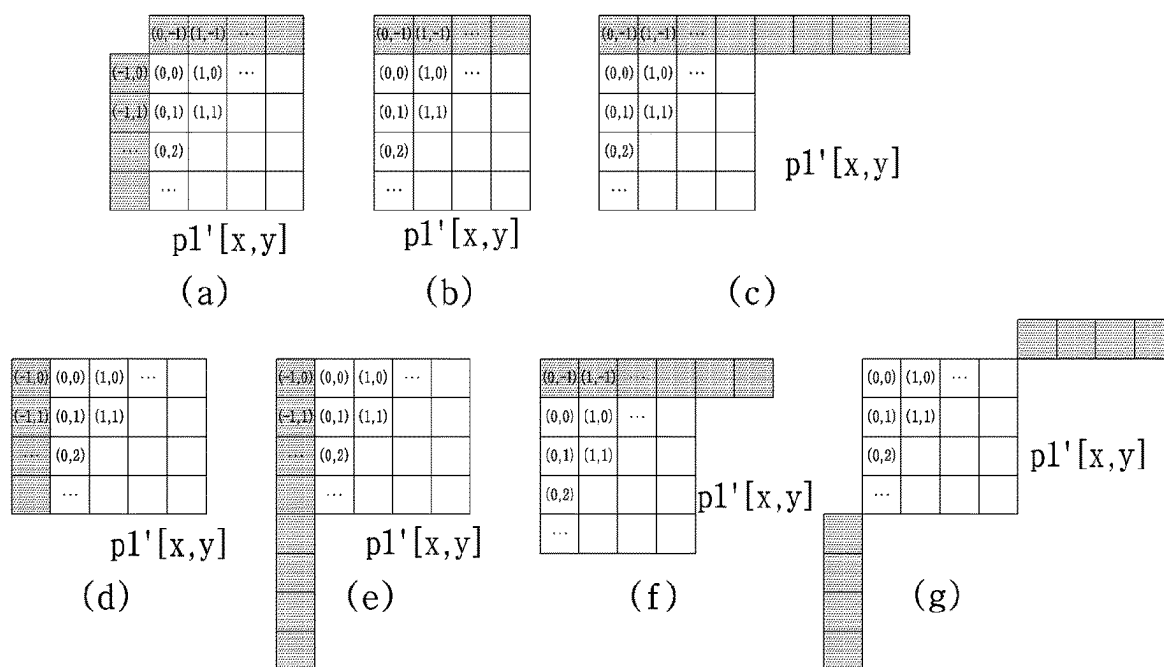


FIG. 23

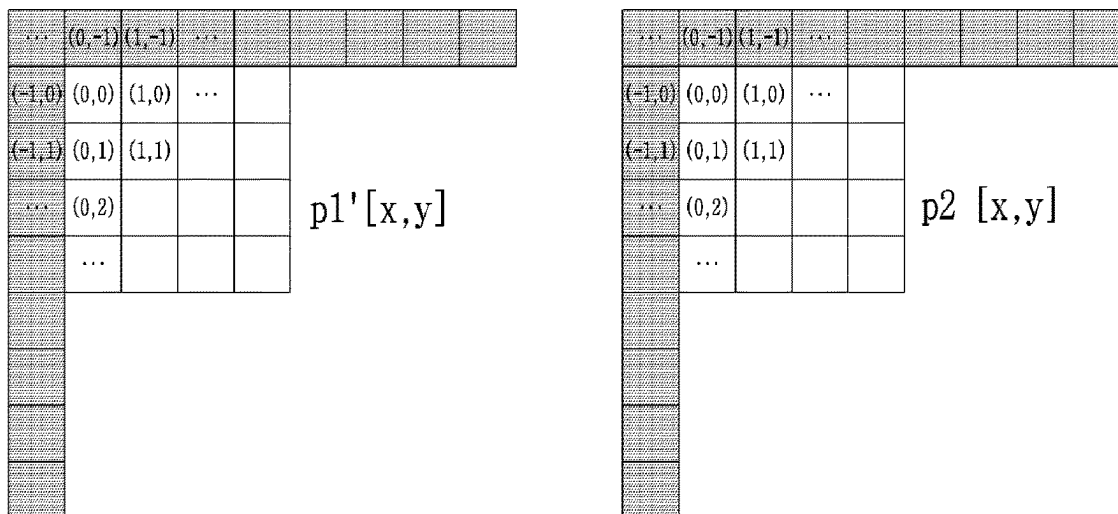


FIG. 24

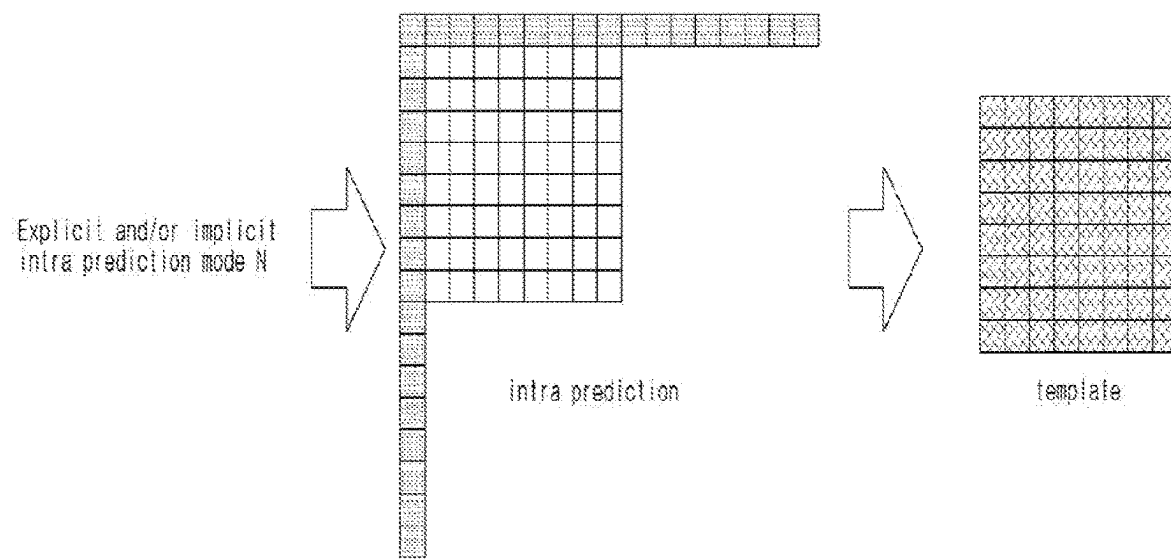


FIG. 25

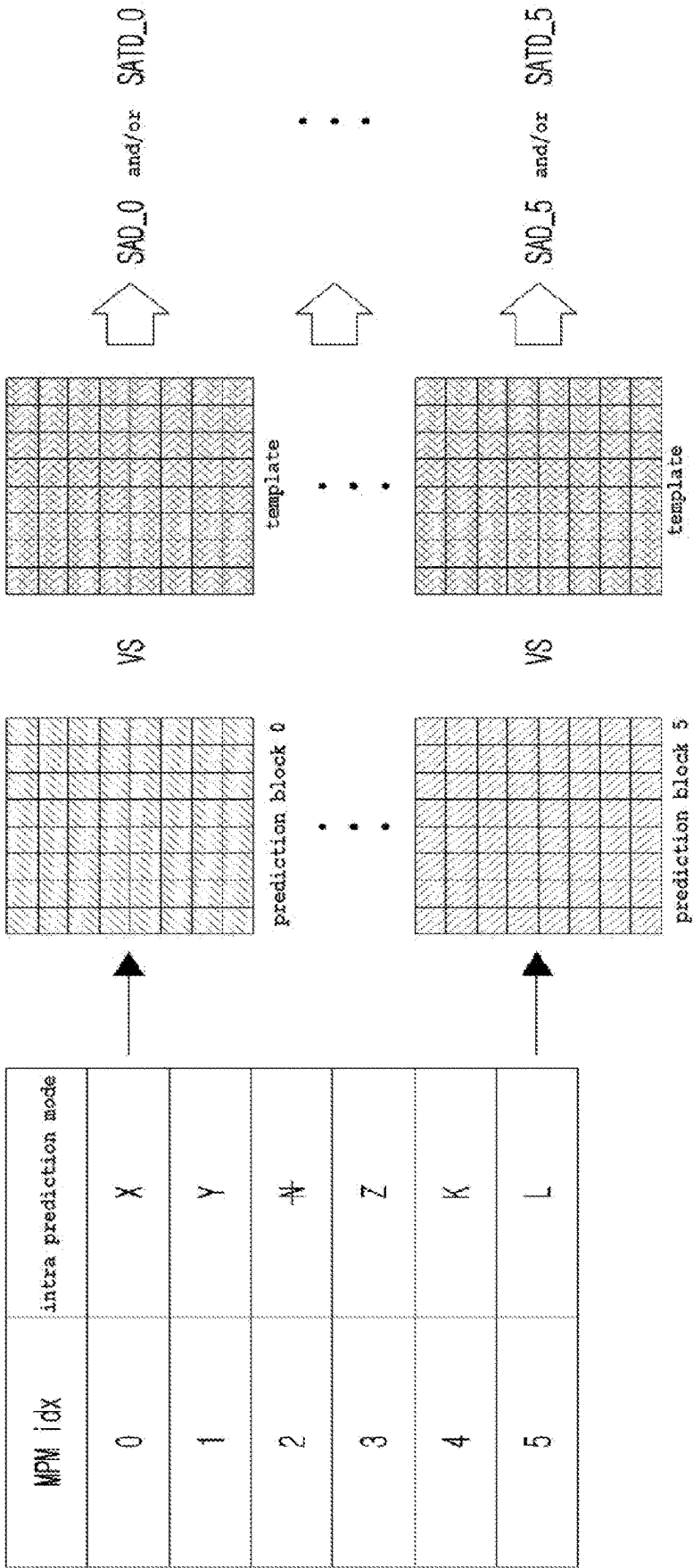


FIG. 26

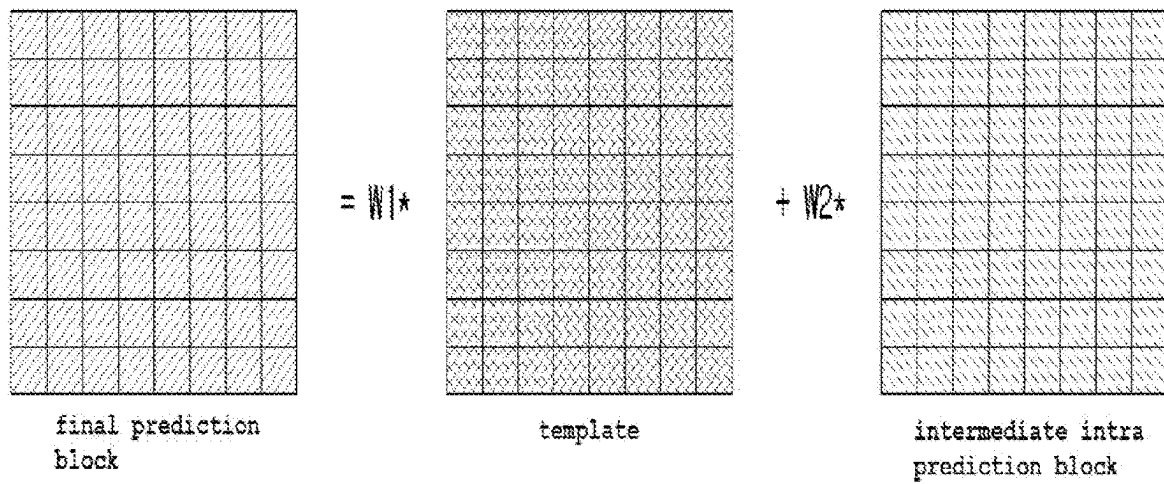


FIG. 27

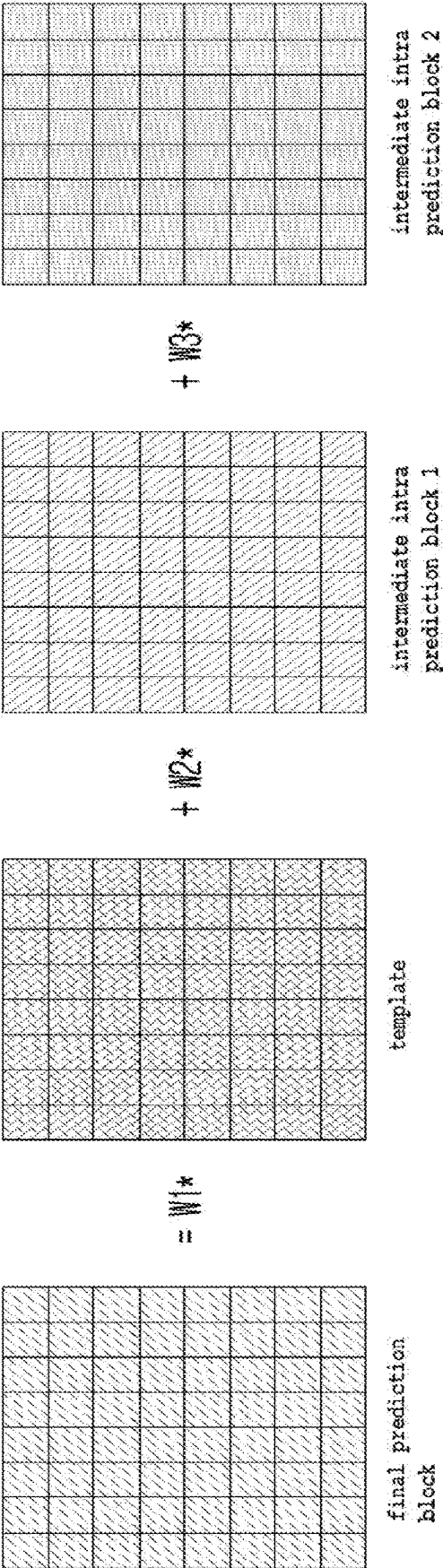
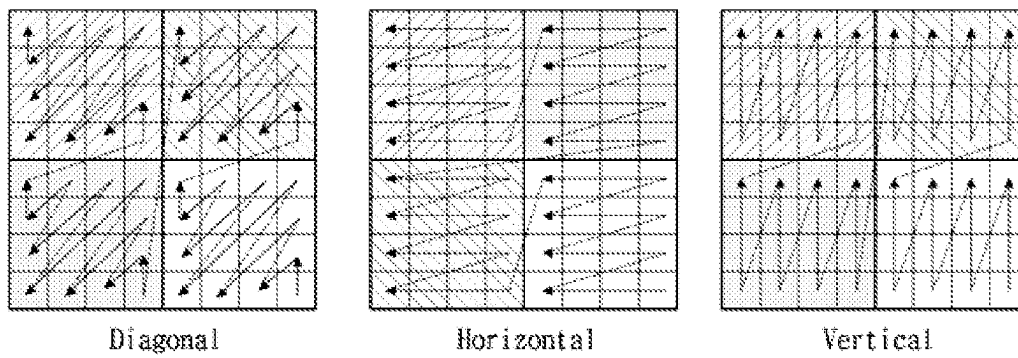


FIG. 28



1

IMAGE ENCODING/DECODING METHOD AND DEVICE, AND RECORDING MEDIUM STORING BITSTREAM

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation application of U.S. application Ser. No. 16/764,784, filed on May 15, 2020, which was the National Stage of International Application No. PCT/KR2018/014111 filed on Nov. 16, 2018, which claims priority to Korean Patent Applications: KR10-2017-0153191, filed on Nov. 16, 2017, with the Korean Intellectual Property Office, which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present invention relates to a method and apparatus for encoding/decoding an image and a recording medium storing a bitstream. Particularly, the present invention relates to a method and apparatus for encoding/decoding an image using intra prediction and a recording medium storing a bitstream generated by an image encoding method/apparatus of the present invention.

BACKGROUND ART

Recently, demands for high-resolution and high-quality images such as high definition (HD) images and ultra high definition (UHD) images, have increased in various application fields. However, higher resolution and quality image data has increasing amounts of data in comparison with conventional image data. Therefore, when transmitting image data by using a medium such as conventional wired and wireless broadband networks, or when storing image data by using a conventional storage medium, costs of transmitting and storing increase. In order to solve these problems occurring with an increase in resolution and quality of image data, high-efficiency image encoding/decoding techniques are required for higher-resolution and higher-quality images.

Image compression technology includes various techniques, including: an inter-prediction technique of predicting a pixel value included in a current picture from a previous or subsequent picture of the current picture; an intra-prediction technique of predicting a pixel value included in a current picture by using pixel information in the current picture; a transform and quantization technique for compressing energy of a residual signal; an entropy encoding technique of assigning a short code to a value with a high appearance frequency and assigning a long code to a value with a low appearance frequency; etc. Image data may be effectively compressed by using such image compression technology, and may be transmitted or stored.

Intra-prediction is a prediction technique that allows only spatial reference and refers to a method of predicting a current block by referring to samples that have already been reconstructed around a block to be currently encoded. Neighboring reference samples referred to in the intra-prediction are not a brightness value of the original video but a brightness value of a video reconstructed by prediction and restoration before post-filtering is applied. Since the neighboring reference samples have been previously encoded and restored, they can be used as reference samples in the encoder and decoder.

2

Intra-prediction is conceptually effective only in a flat area with continuity with respect to the surrounding reference signal and an area with a constant directionality, in which the area without characteristics in a video has significantly lower encoding efficiency than inter-prediction. Especially, in the video encoding, since the video must be encoded only using intra-prediction for the first picture, random access, and error robustness, there is an increased need for a method of enhancing encoding efficiency of the intra-prediction.

DISCLOSURE

Technical Problem

An object of the present invention is to provide a method and apparatus for encoding and decoding an image to enhance compression efficiency.

Another object of the present invention is to provide a method and apparatus for encoding and decoding an image using intra prediction to enhance compression efficiency.

Another object of the present invention is to provide a recording medium storing a bitstream generated by an image encoding method/apparatus of the present invention.

Technical Solution

A method of decoding a video according to an embodiment of the present invention may comprise deriving a first intra-prediction mode for a current block, generating a first intra-prediction block corresponding to the first intra-prediction mode, deriving a second intra-prediction mode for the current block, generating a second intra-prediction block corresponding to the second intra-prediction mode, and generating a final intra-prediction block by using a weighted sum of the first intra-prediction block and the second intra-prediction block.

In the method of decoding a video according to the present invention, the first intra-prediction mode may be derived on the basis of at least one candidate mode included in a MPM list of the current block.

In the method of decoding a video according to the present invention, the first intra-prediction mode may be derived on the basis of intra-prediction modes of one or more neighboring blocks to the current block.

In the method of decoding a video according to the present invention, the second intra-prediction mode may be one mode derived from candidate modes included in a MPM list of the current block.

In the method of decoding a video according to the present invention, the deriving of the second intra-prediction mode may include generating candidate intra-prediction blocks corresponding to the candidate modes, calculating a matching degree between each of the candidate intra-prediction blocks and the first intra-prediction block, and deriving a candidate mode for a candidate intra-prediction block having the highest matching degree among the candidate intra-prediction blocks, as the second intra-prediction mode.

In the method of decoding a video according to the present invention, the matching degree may be calculated using a sum of absolute difference (SAD) or a sum of absolute transformed difference (SATD), and the candidate intra-prediction block having the highest matching degree may be a block with the SAD or the SATD being the smallest.

In the method of decoding a video according to the present invention, the generating of the candidate intra-

3

prediction blocks and the calculating of the matching degree may be skipped for a same mode as the first intra-prediction mode among the candidate modes.

In the method of decoding a video according to the present invention, a weight for the first intra-prediction block or the second intra-prediction block corresponding to the first intra-prediction mode or the second intra-prediction mode that are same as a predetermined mode may be higher than a weight for an intra-prediction block corresponding to a mode other than the predetermined mode.

In the method of decoding a video according to the present invention, the predetermined mode may be a first candidate mode in the MPM list.

In the method of decoding a video according to the present invention, when the number of the second intra-prediction modes and the second intra-prediction blocks corresponding thereto is n (n is an integer of 2 or more), n candidate modes in order of decreasing matching degree may be derived as the second intra-prediction modes.

A method of encoding a video according to another embodiment of the present invention may comprise deriving a first intra-prediction mode for a current block, generating a first intra-prediction block corresponding to the first intra-prediction mode, deriving a second intra-prediction mode for the current block, generating a second intra-prediction block corresponding to the second intra-prediction mode, and generating a final intra-prediction block by using a weighted sum of the first intra-prediction block and the second intra-prediction block.

In the method of encoding a video according to the present invention, the first intra-prediction mode may be derived on the basis of at least one candidate mode included in a MPM list of the current block.

In the method of encoding a video according to the present invention, the first intra-prediction mode may be derived on the basis of intra-prediction modes of one or more neighboring blocks to the current block.

In the method of encoding a video according to the present invention, the second intra-prediction mode may be one mode derived from candidate modes included in a MPM list of the current block.

In the method of encoding a video according to the present invention, the deriving of the second intra-prediction mode may include generating candidate intra-prediction blocks corresponding to the candidate modes, calculating a matching degree between each of the candidate intra-prediction blocks and the first intra-prediction block, and deriving a candidate mode for a candidate intra-prediction block having the highest matching degree among the candidate intra-prediction blocks, as the second intra-prediction mode.

In the method of encoding a video according to the present invention, the matching degree may be calculated using a sum of absolute difference (SAD) or a sum of absolute transformed difference (SATD), and the candidate intra-prediction block having the highest matching degree may be a block with the SAD or the SATD being the smallest.

In the method of encoding a video according to the present invention, the generating of the candidate intra-prediction blocks and the calculating of the matching degree may be skipped for a same mode as the first intra-prediction mode among the candidate modes.

In the method of encoding a video according to the present invention, a weight for the first intra-prediction block or the second intra-prediction block corresponding to the first intra-prediction mode or the second intra-prediction mode that are same as a predetermined mode may be higher

4

than a weight for an intra-prediction block corresponding to a mode other than the predetermined mode.

In the method of encoding a video according to the present invention, when the number of the second intra-prediction modes and the second intra-prediction blocks corresponding thereto is n (n is an integer of 2 or more), n candidate modes in order of decreasing matching degree may be derived as the second intra-prediction modes.

A computer readable recording medium according to another embodiment of the present invention may store a bitstream generated by an image encoding method according to the present invention.

Advantageous Effects

According to the present invention, a method and apparatus for encoding and decoding an image to enhance compression efficiency may be provided.

According to the present invention, a method and apparatus for encoding and decoding an image using intra prediction to enhance compression efficiency may be provided.

According to the present invention, a recording medium storing a bitstream generated by an image encoding method/apparatus of the present invention may be provided.

DESCRIPTION OF DRAWINGS

FIG. 1 is a block diagram showing configurations of an encoding apparatus according to an embodiment of the present invention.

FIG. 2 is a block diagram showing configurations of a decoding apparatus according to an embodiment of the present invention.

FIG. 3 is a view schematically showing a partition structure of an image when encoding and decoding the image.

FIG. 4 is a view for explaining an embodiment of a process of intra prediction.

FIG. 5 is a view for explaining an embodiment of a process of inter prediction.

FIG. 6 is a view for explaining a process of transformation and quantization.

FIG. 7 is a diagram illustrating various embodiments for dividing one block into a plurality of blocks.

FIG. 8 is a diagram illustrating an embodiment in which one block is divided by performing a binary-tree after quad-tree division and information on whether or not division is performed and/or a type of division is signaled.

FIG. 9 is a diagram illustrating various embodiments of dividing a current block into two sub-blocks.

FIG. 10 is a view showing intra-prediction according to the present invention.

FIG. 11 is a diagram illustrating spatial neighboring blocks of a current block used when configuring an MPM list.

FIG. 12 is a view illustrating an embodiment for rearranging candidate modes in the MPM list.

FIG. 13 is an exemplary diagram illustrating the relationship between a luma block and a chroma block.

FIG. 14 is a diagram for describing a plurality of reconstructed sample lines.

FIG. 15 is a diagram for describing a process of replacing an unavailable sample with an available sample.

FIG. 16 illustrates various filter shapes.

FIG. 17 is a diagram for describing intra prediction according to the shapes of the current block.

5

FIG. 18 is a diagram for describing neighboring samples of a current block used to derive the parameters of the models.

FIG. 19 is an exemplary diagram illustrating a process of restructuring a color component block.

FIG. 20 is a diagram illustrating an embodiment performing restructuring by using a plurality of upper-side reference sample lines and/or a plurality of left-side reference sample lines.

FIG. 21 is an exemplary diagram illustrating reference samples used for the restructuring in accordance with an intra prediction mode or a coding parameter of a corresponding block.

FIG. 22 is a diagram illustrating an exemplary restructured first color component corresponding block when a second color component prediction target block is a 4×4 block.

FIG. 23 is a diagram illustrating a sample of a first color component and a sample of a second color component.

FIG. 24 is a diagram illustrating an embodiment for generating a template.

FIG. 25 is a diagram illustrating an embodiment for generating a template matching-based intermediate intra-prediction block on the basis of the MPM list.

FIG. 26 is a view illustrating an embodiment for generating a final intra-prediction block using a template and one intermediate intra-prediction block.

FIG. 27 is a diagram illustrating an embodiment in which a final intra-frame prediction block is generated using the template and a plurality of intermediate intra-prediction blocks.

FIG. 28 is a diagram illustrating a method of scanning a transform coefficient.

MODE FOR INVENTION

A variety of modifications may be made to the present invention and there are various embodiments of the present invention, examples of which will now be provided with reference to drawings and described in detail. However, the present invention is not limited thereto, although the exemplary embodiments can be construed as including all modifications, equivalents, or substitutes in a technical concept and a technical scope of the present invention. The similar reference numerals refer to the same or similar functions in various aspects. In the drawings, the shapes and dimensions of elements may be exaggerated for clarity. In the following detailed description of the present invention, references are made to the accompanying drawings that show, by way of illustration, specific embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to implement the present disclosure. It should be understood that various embodiments of the present disclosure, although different, are not necessarily mutually exclusive. For example, specific features, structures, and characteristics described herein, in connection with one embodiment, may be implemented within other embodiments without departing from the spirit and scope of the present disclosure. In addition, it should be understood that the location or arrangement of individual elements within each disclosed embodiment may be modified without departing from the spirit and scope of the present disclosure. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present disclosure is defined only by the appended claims, appropriately interpreted, along with the full range of equivalents to what the claims claim.

6

Terms used in the specification, 'first', 'second', etc. can be used to describe various components, but the components are not to be construed as being limited to the terms. The terms are only used to differentiate one component from other components. For example, the 'first' component may be named the 'second' component without departing from the scope of the present invention, and the 'second' component may also be similarly named the 'first' component. The term 'and/or' includes a combination of a plurality of items or any one of a plurality of terms.

It will be understood that when an element is simply referred to as being 'connected to' or 'coupled to' another element without being 'directly connected to' or 'directly coupled to' another element in the present description, it may be 'directly connected to' or 'directly coupled to' another element or be connected to or coupled to another element, having the other element intervening therebetween. In contrast, it should be understood that when an element is referred to as being "directly coupled" or "directly connected" to another element, there are no intervening elements present.

Furthermore, constitutional parts shown in the embodiments of the present invention are independently shown so as to represent characteristic functions different from each other. Thus, it does not mean that each constitutional part is constituted in a constitutional unit of separated hardware or software. In other words, each constitutional part includes each of enumerated constitutional parts for convenience. Thus, at least two constitutional parts of each constitutional part may be combined to form one constitutional part or one constitutional part may be divided into a plurality of constitutional parts to perform each function. The embodiment where each constitutional part is combined and the embodiment where one constitutional part is divided are also included in the scope of the present invention, if not departing from the essence of the present invention.

The terms used in the present specification are merely used to describe particular embodiments, and are not intended to limit the present invention. An expression used in the singular encompasses the expression of the plural, unless it has a clearly different meaning in the context. In the present specification, it is to be understood that terms such as "including", "having", etc. are intended to indicate the existence of the features, numbers, steps, actions, elements, parts, or combinations thereof disclosed in the specification, and are not intended to preclude the possibility that one or more other features, numbers, steps, actions, elements, parts, or combinations thereof may exist or may be added. In other words, when a specific element is referred to as being "included", elements other than the corresponding element are not excluded, but additional elements may be included in embodiments of the present invention or the scope of the present invention.

In addition, some of constituents may not be indispensable constituents performing essential functions of the present invention but be selective constituents improving only performance thereof. The present invention may be implemented by including only the indispensable constitutional parts for implementing the essence of the present invention except the constituents used in improving performance. The structure including only the indispensable constituents except the selective constituents used in improving only performance is also included in the scope of the present invention.

Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings. In describing exemplary embodiments of the

present invention, well-known functions or constructions will not be described in detail since they may unnecessarily obscure the understanding of the present invention. The same constituent elements in the drawings are denoted by the same reference numerals, and a repeated description of the same elements will be omitted.

Hereinafter, an image may mean a picture configuring a video, or may mean the video itself. For example, “encoding or decoding or both of an image” may mean “encoding or decoding or both of a moving picture”, and may mean “encoding or decoding or both of one image among images of a moving picture.”

Hereinafter, terms “moving picture” and “video” may be used as the same meaning and be replaced with each other.

Hereinafter, a target image may be an encoding target image which is a target of encoding and/or a decoding target image which is a target of decoding. Also, a target image may be an input image inputted to an encoding apparatus, and an input image inputted to a decoding apparatus. Here, a target image may have the same meaning with the current image.

Hereinafter, terms “image”, “picture”, “frame” and “screen” may be used as the same meaning and be replaced with each other.

Hereinafter, a target block may be an encoding target block which is a target of encoding and/or a decoding target block which is a target of decoding. Also, a target block may be the current block which is a target of current encoding and/or decoding. For example, terms “target block” and “current block” may be used as the same meaning and be replaced with each other.

Hereinafter, terms “block” and “unit” may be used as the same meaning and be replaced with each other. Or a “block” may represent a specific unit.

Hereinafter, terms “region” and “segment” may be replaced with each other.

Hereinafter, a specific signal may be a signal representing a specific block. For example, an original signal may be a signal representing a target block. A prediction signal may be a signal representing a prediction block. A residual signal may be a signal representing a residual block.

In embodiments, each of specific information, data, flag, index, element and attribute, etc. may have a value. A value of information, data, flag, index, element and attribute equal to “0” may represent a logical false or the first predefined value. In other words, a value “0”, a false, a logical false and the first predefined value may be replaced with each other. A value of information, data, flag, index, element and attribute equal to “1” may represent a logical true or the second predefined value. In other words, a value “1”, a true, a logical true and the second predefined value may be replaced with each other.

When a variable i or j is used for representing a column, a row or an index, a value of i may be an integer equal to or greater than 0, or equal to or greater than 1. That is, the column, the row, the index, etc. may be counted from 0 or may be counted from 1.

Description of Terms

Encoder: means an apparatus performing encoding. That is, means an encoding apparatus.

Decoder: means an apparatus performing decoding. That is, means an decoding apparatus.

Block: is an $M \times N$ array of a sample. Herein, M and N may mean positive integers, and the block may mean a sample array of a two-dimensional form. The block may refer to a

unit. A current block may mean an encoding target block that becomes a target when encoding, or a decoding target block that becomes a target when decoding. In addition, the current block may be at least one of an encode block, a prediction block, a residual block, and a transform block.

Sample: is a basic unit constituting a block. It may be expressed as a value from 0 to $2^{Bd}-1$ according to a bit depth (Bd). In the present invention, the sample may be used as a meaning of a pixel. That is, a sample, a pel, a pixel may have the same meaning with each other.

Unit: may refer to an encoding and decoding unit. When encoding and decoding an image, the unit may be a region generated by partitioning a single image. In addition, the unit may mean a subdivided unit when a single image is partitioned into subdivided units during encoding or decoding. That is, an image may be partitioned into a plurality of units. When encoding and decoding an image, a predetermined process for each unit may be performed. A single unit may be partitioned into sub-units that have sizes smaller than the size of the unit. Depending on functions, the unit may mean a block, a macroblock, a coding tree unit, a code tree block, a coding unit, a coding block), a prediction unit, a prediction block, a residual unit), a residual block, a transform unit, a transform block, etc. In addition, in order to distinguish a unit from a block, the unit may include a luma component block, a chroma component block associated with the luma component block, and a syntax element of each color component block. The unit may have various sizes and forms, and particularly, the form of the unit may be a two-dimensional geometrical figure such as a square shape, a rectangular shape, a trapezoid shape, a triangular shape, a pentagonal shape, etc. In addition, unit information may include at least one of a unit type indicating the coding unit, the prediction unit, the transform unit, etc., and a unit size, a unit depth, a sequence of encoding and decoding of a unit, etc.

Coding Tree Unit: is configured with a single coding tree block of a luma component Y , and two coding tree blocks related to chroma components Cb and Cr . In addition, it may mean that including the blocks and a syntax element of each block. Each coding tree unit may be partitioned by using at least one of a quad-tree partitioning method, a binary-tree partitioning method, a ternary-tree partitioning method, etc. to configure a lower unit such as coding unit, prediction unit, transform unit, etc. It may be used as a term for designating a sample block that becomes a process unit when encoding/decoding an image as an input image. Here, a quad-tree may mean a quaternary-tree.

Coding Tree Block: may be used as a term for designating any one of a Y coding tree block, Cb coding tree block, and Cr coding tree block.

Neighbor Block: may mean a block adjacent to a current block. The block adjacent to the current block may mean a block that comes into contact with a boundary of the current block, or a block positioned within a predetermined distance from the current block. The neighbor block may mean a block adjacent to a vertex of the current block. Herein, the block adjacent to the vertex of the current block may mean a block vertically adjacent to a neighbor block that is horizontally adjacent to the current block, or a block horizontally adjacent to a neighbor block that is vertically adjacent to the current block.

Reconstructed Neighbor block: may mean a neighbor block adjacent to a current block and which has been already spatially/temporally encoded or decoded. Herein, the reconstructed neighbor block may mean a reconstructed neighbor unit. A reconstructed spatial neighbor block may be a block

within a current picture and which has been already reconstructed through encoding or decoding or both. A reconstructed temporal neighbor block is a block at a corresponding position as the current block of the current picture within a reference image, or a neighbor block thereof.

Unit Depth: may mean a partitioned degree of a unit. In a tree structure, the highest node (Root Node) may correspond to the first unit which is not partitioned. Also, the highest node may have the least depth value. In this case, the highest node may have a depth of level 0. A node having a depth of level 1 may represent a unit generated by partitioning once the first unit. A node having a depth of level 2 may represent a unit generated by partitioning twice the first unit. A node having a depth of level n may represent a unit generated by partitioning n-times the first unit. A Leaf Node may be the lowest node and a node which cannot be partitioned further. A depth of a Leaf Node may be the maximum level. For example, a predefined value of the maximum level may be 3. A depth of a root node may be the lowest and a depth of a leaf node may be the deepest. In addition, when a unit is expressed as a tree structure, a level in which a unit is present may mean a unit depth.

Bitstream: may mean a bitstream including encoding image information.

Parameter Set: corresponds to header information among a configuration within a bitstream. At least one of a video parameter set, a sequence parameter set, a picture parameter set, and an adaptation parameter set may be included in a parameter set. In addition, a parameter set may include a slice header, and tile header information.

Parsing: may mean determination of a value of a syntax element by performing entropy decoding, or may mean the entropy decoding itself.

Symbol: may mean at least one of a syntax element, a coding parameter, and a transform coefficient value of an encoding/decoding target unit. In addition, the symbol may mean an entropy encoding target or an entropy decoding result.

Prediction Mode: may be information indicating a mode encoded/decoded with intra prediction or a mode encoded/decoded with inter prediction.

Prediction Unit: may mean a basic unit when performing prediction such as inter-prediction, intra-prediction, inter-compensation, intra-compensation, and motion compensation. A single prediction unit may be partitioned into a plurality of partitions having a smaller size, or may be partitioned into a plurality of lower prediction units. A plurality of partitions may be a basic unit in performing prediction or compensation. A partition which is generated by dividing a prediction unit may also be a prediction unit.

Prediction Unit Partition: may mean a form obtained by partitioning a prediction unit.

Reference Picture List: may mean a list including one or more reference pictures used for inter-picture prediction or motion compensation. LC (List Combined), L0 (List 0), L1 (List 1), L2 (List 2), L3 (List 3) and the like are types of reference picture lists. One or more reference picture lists may be used for inter-picture prediction.

Inter-picture prediction Indicator: may mean an inter-picture prediction direction (uni-directional prediction, bi-directional prediction, and the like) of a current block. Alternatively, the inter-picture prediction indicator may mean the number of reference pictures used to generate a prediction block of a current block. Further alternatively, the inter-picture prediction indicator may mean the number of prediction blocks used to perform inter-picture prediction or motion compensation with respect to a current block.

Prediction list utilization flag: may represent whether a prediction block is generated using at least one reference image included in a specific reference picture list. An inter prediction indicator may be derived using a prediction list utilization flag, and reversely, a prediction list utilization flag may be derived using an inter prediction indicator. For example, when a prediction list utilization flag indicates a first value of "0", it represents a prediction block is not generated using a reference picture included in the corresponding reference picture list. When a prediction list utilization flag indicates a second value of "1", it represents a prediction block is generated using a reference picture included in the corresponding reference picture list.

Reference Picture Index: may mean an index indicating a specific reference picture in a reference picture list.

Reference Picture: may mean a picture to which a specific block refers for inter-picture prediction or motion compensation. Alternatively, a reference picture may be a picture including a reference block referred to by a current block for inter prediction or motion compensation. Hereinafter, the term "reference picture" and "reference image" may be used as the same meaning and used interchangeably.

Motion Vector: is a two-dimensional vector used for inter-picture prediction or motion compensation and may mean an offset between a reference picture and an encoding/decoding target picture. For example, (mvX, mvY) may represent a motion vector, mvX may represent a horizontal component, and mvY may represent a vertical component.

Search Range: may be a 2-dimensional region where search for a motion vector during inter prediction is performed. For example, a size of a search range may be $M \times N$. M and N may be a positive integer, respectively.

Motion Vector Candidate: may mean a block that becomes a prediction candidate when predicting a motion vector, or a motion vector of the block. A motion vector candidate may be listed in a motion vector candidate list.

Motion Vector Candidate List: may mean a list configured using one or more motion vector candidates.

Motion Vector Candidate Index: means an indicator indicating a motion vector candidate in a motion vector candidate list. It is also referred to as an index of a motion vector predictor.

Motion Information: may mean information including a motion vector, a reference picture index, an inter-picture prediction indicator, and at least any one among reference picture list information, a reference picture, a motion vector candidate, a motion vector candidate index, a merge candidate, and a merge index.

Merge Candidate List: may mean a list composed of merge candidates.

Merge Candidate: may mean a spatial merge candidate, a temporal merge candidate, a combined merge candidate, a combined bi-prediction merge candidate, a zero merge candidate, or the like. The merge candidate may have an inter-picture prediction indicator, a reference picture index for each list, and motion information such as a motion vector.

Merge Index: may mean an indicator indicating a merge candidate within a merge candidate list. The merge index may indicate a block used to derive a merge candidate, among reconstructed blocks spatially and/or temporally adjacent to a current block. The merge index may indicate at least one item in the motion information possessed by a merge candidate.

Transform Unit: may mean a basic unit when performing encoding/decoding such as transform, inverse-transform, quantization, dequantization, transform coefficient encod-

11

ing/decoding of a residual signal. A single transform unit may be partitioned into a plurality of lower-level transform units having a smaller size. Here, transformation/inverse-transformation may comprise at least one among the first transformation/the first inverse-transformation and the second transformation/the second inverse-transformation.

Scaling: may mean a process of multiplying a quantized level by a factor. A transform coefficient may be generated by scaling a quantized level. The scaling also may be referred to as dequantization.

Quantization Parameter: may mean a value used when generating a quantized level using a transform coefficient during quantization. The quantization parameter also may mean a value used when generating a transform coefficient by scaling a quantized level during dequantization. The quantization parameter may be a value mapped on a quantization step size.

Delta Quantization Parameter: may mean a difference value between a predicted quantization parameter and a quantization parameter of an encoding/decoding target unit.

Scan: may mean a method of sequencing coefficients within a unit, a block or a matrix. For example, changing a two-dimensional matrix of coefficients into a one-dimensional matrix may be referred to as scanning, and changing a one-dimensional matrix of coefficients into a two-dimensional matrix may be referred to as scanning or inverse scanning.

Transform Coefficient: may mean a coefficient value generated after transform is performed in an encoder. It may mean a coefficient value generated after at least one of entropy decoding and dequantization is performed in a decoder. A quantized level obtained by quantizing a transform coefficient or a residual signal, or a quantized transform coefficient level also may fall within the meaning of the transform coefficient.

Quantized Level: may mean a value generated by quantizing a transform coefficient or a residual signal in an encoder. Alternatively, the quantized level may mean a value that is a dequantization target to undergo dequantization in a decoder. Similarly, a quantized transform coefficient level that is a result of transform and quantization also may fall within the meaning of the quantized level.

Non-zero Transform Coefficient: may mean a transform coefficient having a value other than zero, or a transform coefficient level or a quantized level having a value other than zero.

Quantization Matrix: may mean a matrix used in a quantization process or a dequantization process performed to improve subjective or objective image quality. The quantization matrix also may be referred to as a scaling list.

Quantization Matrix Coefficient: may mean each element within a quantization matrix. The quantization matrix coefficient also may be referred to as a matrix coefficient.

Default Matrix: may mean a predetermined quantization matrix preliminarily defined in an encoder or a decoder.

Non-default Matrix: may mean a quantization matrix that is not preliminarily defined in an encoder or a decoder but is signaled by a user.

Statistic Value: a statistic value for at least one among a variable, an encoding parameter, a constant value, etc. which have a computable specific value may be one or more among an average value, a weighted average value, a weighted sum value, the minimum value, the maximum value, the most frequent value, a median value, an interpolated value of the corresponding specific values.

12

FIG. 1 is a block diagram showing a configuration of an encoding apparatus according to an embodiment to which the present invention is applied.

An encoding apparatus **100** may be an encoder, a video encoding apparatus, or an image encoding apparatus. A video may include at least one image. The encoding apparatus **100** may sequentially encode at least one image.

Referring to FIG. 1, the encoding apparatus **100** may include a motion prediction unit **111**, a motion compensation unit **112**, an intra-prediction unit **120**, a switch **115**, a subtractor **125**, a transform unit **130**, a quantization unit **140**, an entropy encoding unit **150**, a dequantization unit **160**, a inverse-transform unit **170**, an adder **175**, a filter unit **180**, and a reference picture buffer **190**.

The encoding apparatus **100** may perform encoding of an input image by using an intra mode or an inter mode or both. In addition, encoding apparatus **100** may generate a bitstream including encoded information through encoding the input image, and output the generated bitstream. The generated bitstream may be stored in a computer readable recording medium, or may be streamed through a wired/wireless transmission medium. When an intra mode is used as a prediction mode, the switch **115** may be switched to an intra. Alternatively, when an inter mode is used as a prediction mode, the switch **115** may be switched to an inter mode. Herein, the intra mode may mean an intra-prediction mode, and the inter mode may mean an inter-prediction mode. The encoding apparatus **100** may generate a prediction block for an input block of the input image. In addition, the encoding apparatus **100** may encode a residual block using a residual of the input block and the prediction block after the prediction block being generated. The input image may be called as a current image that is a current encoding target. The input block may be called as a current block that is current encoding target, or as an encoding target block.

When a prediction mode is an intra mode, the intra-prediction unit **120** may use a sample of a block that has been already encoded/decoded and is adjacent to a current block as a reference sample. The intra-prediction unit **120** may perform spatial prediction for the current block by using a reference sample, or generate prediction samples of an input block by performing spatial prediction. Herein, the intra prediction may mean intra-prediction.

When a prediction mode is an inter mode, the motion prediction unit **111** may retrieve a region that best matches with an input block from a reference image when performing motion prediction, and deduce a motion vector by using the retrieved region. In this case, a search region may be used as the region. The reference image may be stored in the reference picture buffer **190**. Here, when encoding/decoding for the reference image is performed, it may be stored in the reference picture buffer **190**.

The motion compensation unit **112** may generate a prediction block by performing motion compensation for the current block using a motion vector. Herein, inter-prediction may mean inter-prediction or motion compensation.

When the value of the motion vector is not an integer, the motion prediction unit **111** and the motion compensation unit **112** may generate the prediction block by applying an interpolation filter to a partial region of the reference picture. In order to perform inter prediction or motion compensation on a coding unit, it may be determined that which mode among a skip mode, a merge mode, an advanced motion vector prediction (AMVP) mode, and a current picture referring mode is used for motion prediction and motion compensation of a prediction unit included in the corre-

13

sponding coding unit. Then, inter prediction or motion compensation may be differently performed depending on the determined mode.

The subtractor **125** may generate a residual block by using a residual of an input block and a prediction block. The residual block may be called as a residual signal. The residual signal may mean a difference between an original signal and a prediction signal. In addition, the residual signal may be a signal generated by transforming or quantizing, or transforming and quantizing a difference between the original signal and the prediction signal. The residual block may be a residual signal of a block unit.

The transform unit **130** may generate a transform coefficient by performing transform of a residual block, and output the generated transform coefficient. Herein, the transform coefficient may be a coefficient value generated by performing transform of the residual block. When a transform skip mode is applied, the transform unit **130** may skip transform of the residual block.

A quantized level may be generated by applying quantization to the transform coefficient or to the residual signal. Hereinafter, the quantized level may be also called as a transform coefficient in embodiments.

The quantization unit **140** may generate a quantized level by quantizing the transform coefficient or the residual signal according to a parameter, and output the generated quantized level. Herein, the quantization unit **140** may quantize the transform coefficient by using a quantization matrix.

The entropy encoding unit **150** may generate a bitstream by performing entropy encoding according to a probability distribution on values calculated by the quantization unit **140** or on coding parameter values calculated when performing encoding, and output the generated bitstream. The entropy encoding unit **150** may perform entropy encoding of sample information of an image and information for decoding an image. For example, the information for decoding the image may include a syntax element.

When entropy encoding is applied, symbols are represented so that a smaller number of bits are assigned to a symbol having a high chance of being generated and a larger number of bits are assigned to a symbol having a low chance of being generated, and thus, the size of bit stream for symbols to be encoded may be decreased. The entropy encoding unit **150** may use an encoding method for entropy encoding such as exponential Golomb, context-adaptive variable length coding (CAVLC), context-adaptive binary arithmetic coding (CABAC), etc. For example, the entropy encoding unit **150** may perform entropy encoding by using a variable length coding/code (VLC) table. In addition, the entropy encoding unit **150** may deduce a binarization method of a target symbol and a probability model of a target symbol/bin, and perform arithmetic coding by using the deduced binarization method, and a context model.

In order to encode a transform coefficient level (quantized level), the entropy encoding unit **150** may change a two-dimensional block form coefficient into a one-dimensional vector form by using a transform coefficient scanning method.

A coding parameter may include information (flag, index, etc.) such as syntax element that is encoded in an encoder and signaled to a decoder, and information derived when performing encoding or decoding. The coding parameter may mean information required when encoding or decoding an image. For example, at least one value or a combination form of a unit/block size, a unit/block depth, unit/block partition information, unit/block shape, unit/block partition structure, whether to partition of a quad-tree form, whether

14

to partition of a binary-tree form, a partition direction of a binary-tree form (horizontal direction or vertical direction), a partition form of a binary-tree form (symmetric partition or asymmetric partition), whether to partition of a ternary-tree form, a partition direction of a ternary-tree form (horizontal direction or vertical direction), a partition form of a ternary-tree form (symmetric partition or asymmetric partition), whether to partition of a multi-type-tree form, a partition direction of a multi-type-tree form (horizontal direction or vertical direction), a partition form of a multi-type-tree form (symmetric partition or asymmetric partition), a partitioning tree of multi-type-tree form, a prediction mode (intra prediction or inter prediction), a luma intra-prediction mode/direction, a chroma intra-prediction mode/direction, intra partition information, inter partition information, a coding block partition flag, a prediction block partition flag, a transform block partition flag, a reference sample filtering method, a reference sample filter tap, a reference sample filter coefficient, a prediction block filtering method, a prediction block filter tap, a prediction block filter coefficient, a prediction block boundary filtering method, a prediction block boundary filter tap, a prediction block boundary filter coefficient, an intra-prediction mode, an inter-prediction mode, motion information, a motion vector, a motion vector difference, a reference picture index, a inter-prediction angle, an inter-prediction indicator, a prediction list utilization flag, a reference picture list, a reference picture, a motion vector predictor index, a motion vector predictor candidate, a motion vector candidate list, whether to use a merge mode, a merge index, a merge candidate, a merge candidate list, whether to use a skip mode, an interpolation filter type, an interpolation filter tap, an interpolation filter coefficient, a motion vector size, a presentation accuracy of a motion vector, a transform type, a transform size, information of whether or not a primary (first) transform is used, information of whether or not a secondary transform is used, a primary transform index, a secondary transform index, information of whether or not a residual signal is present, a coded block pattern, a coded block flag (CBF), a quantization parameter, a quantization parameter residue, a quantization matrix, whether to apply an intra loop filter, an intra loop filter coefficient, an intra loop filter tap, an intra loop filter shape/form, whether to apply a deblocking filter, a deblocking filter coefficient, a deblocking filter tap, a deblocking filter strength, a deblocking filter shape/form, whether to apply an adaptive sample offset, an adaptive sample offset value, an adaptive sample offset category, an adaptive sample offset type, whether to apply an adaptive loop filter, an adaptive loop filter coefficient, an adaptive loop filter tap, an adaptive loop filter shape/form, a binarization/inverse-binarization method, a context model determining method, a context model updating method, whether to perform a regular mode, whether to perform a bypass mode, a context bin, a bypass bin, a significant coefficient flag, a last significant coefficient flag, a coded flag for a unit of a coefficient group, a position of the last significant coefficient, a flag for whether a value of a coefficient is larger than 1, a flag for whether a value of a coefficient is larger than 2, a flag for whether a value of a coefficient is larger than 3, information on a remaining coefficient value, a sign information, a reconstructed luma sample, a reconstructed chroma sample, a residual luma sample, a residual chroma sample, a luma transform coefficient, a chroma transform coefficient, a quantized luma level, a quantized chroma level, a transform coefficient level scanning method, a size of a motion vector search area at a decoder side, a shape of a motion vector search area at a

15

decoder side, a number of time of a motion vector search at a decoder side, information on a CTU size, information on a minimum block size, information on a maximum block size, information on a maximum block depth, information on a minimum block depth, an image displaying/outputting sequence, slice identification information, a slice type, slice partition information, tile identification information, a tile type, tile partition information, a picture type, a bit depth of an input sample, a bit depth of a reconstruction sample, a bit depth of a residual sample, a bit depth of a transform coefficient, a bit depth of a quantized level, and information on a luma signal or information on a chroma signal may be included in the coding parameter.

Herein, signaling the flag or index may mean that a corresponding flag or index is entropy encoded and included in a bitstream by an encoder, and may mean that the corresponding flag or index is entropy decoded from a bitstream by a decoder.

When the encoding apparatus 100 performs encoding through inter-prediction, an encoded current image may be used as a reference image for another image that is processed afterwards. Accordingly, the encoding apparatus 100 may reconstruct or decode the encoded current image, or store the reconstructed or decoded image as a reference image in reference picture buffer 190.

A quantized level may be dequantized in the dequantization unit 160, or may be inverse-transformed in the inverse-transform unit 170. A dequantized or inverse-transformed coefficient or both may be added with a prediction block by the adder 175. By adding the dequantized or inverse-transformed coefficient or both with the prediction block, a reconstructed block may be generated. Herein, the dequantized or inverse-transformed coefficient or both may mean a coefficient on which at least one of dequantization and inverse-transform is performed, and may mean a reconstructed residual block.

A reconstructed block may pass through the filter unit 180. The filter unit 180 may apply at least one of a deblocking filter, a sample adaptive offset (SAO), and an adaptive loop filter (ALF) to a reconstructed sample, a reconstructed block or a reconstructed image. The filter unit 180 may be called as an in-loop filter.

The deblocking filter may remove block distortion generated in boundaries between blocks. In order to determine whether or not to apply a deblocking filter, whether or not to apply a deblocking filter to a current block may be determined based samples included in several rows or columns which are included in the block. When a deblocking filter is applied to a block, another filter may be applied according to a required deblocking filtering strength.

In order to compensate an encoding error, a proper offset value may be added to a sample value by using a sample adaptive offset. The sample adaptive offset may correct an offset of a deblocked image from an original image by a sample unit. A method of partitioning samples of an image into a predetermined number of regions, determining a region to which an offset is applied, and applying the offset to the determined region, or a method of applying an offset in consideration of edge information on each sample may be used.

The adaptive loop filter may perform filtering based on a comparison result of the filtered reconstructed image and the original image. Samples included in an image may be partitioned into predetermined groups, a filter to be applied to each group may be determined, and differential filtering may be performed for each group. Information of whether or

16

not to apply the ALF may be signaled by coding units (CUs), and a form and coefficient of the ALF to be applied to each block may vary.

The reconstructed block or the reconstructed image having passed through the filter unit 180 may be stored in the reference picture buffer 190. A reconstructed block processed by the filter unit 180 may be a part of a reference image. That is, a reference image is a reconstructed image composed of reconstructed blocks processed by the filter unit 180. The stored reference image may be used later in inter prediction or motion compensation.

FIG. 2 is a block diagram showing a configuration of a decoding apparatus according to an embodiment and to which the present invention is applied.

A decoding apparatus 200 may be a decoder, a video decoding apparatus, or an image decoding apparatus.

Referring to FIG. 2, the decoding apparatus 200 may include an entropy decoding unit 210, a dequantization unit 220, an inverse-transform unit 230, an intra-prediction unit 240, a motion compensation unit 250, an adder 255, a filter unit 260, and a reference picture buffer 270.

The decoding apparatus 200 may receive a bitstream output from the encoding apparatus 100. The decoding apparatus 200 may receive a bitstream stored in a computer readable recording medium, or may receive a bitstream that is streamed through a wired/wireless transmission medium. The decoding apparatus 200 may decode the bitstream by using an intra mode or an inter mode. In addition, the decoding apparatus 200 may generate a reconstructed image generated through decoding or a decoded image, and output the reconstructed image or decoded image.

When a prediction mode used when decoding is an intra mode, a switch may be switched to an intra. Alternatively, when a prediction mode used when decoding is an inter mode, a switch may be switched to an inter mode.

The decoding apparatus 200 may obtain a reconstructed residual block by decoding the input bitstream, and generate a prediction block. When the reconstructed residual block and the prediction block are obtained, the decoding apparatus 200 may generate a reconstructed block that becomes a decoding target by adding the reconstructed residual block with the prediction block. The decoding target block may be called a current block.

The entropy decoding unit 210 may generate symbols by entropy decoding the bitstream according to a probability distribution. The generated symbols may include a symbol of a quantized level form. Herein, an entropy decoding method may be an inverse-process of the entropy encoding method described above.

In order to decode a transform coefficient level (quantized level), the entropy decoding unit 210 may change a one-dimensional vector form coefficient into a two-dimensional block form by using a transform coefficient scanning method.

A quantized level may be dequantized in the dequantization unit 220, or inverse-transformed in the inverse-transform unit 230. The quantized level may be a result of dequantizing or inverse-transforming or both, and may be generated as a reconstructed residual block. Herein, the dequantization unit 220 may apply a quantization matrix to the quantized level.

When an intra mode is used, the intra-prediction unit 240 may generate a prediction block by performing, for the current block, spatial prediction that uses a sample value of a block adjacent to a decoding target block and which has been already decoded.

When an inter mode is used, the motion compensation unit **250** may generate a prediction block by performing, for the current block, motion compensation that uses a motion vector and a reference image stored in the reference picture buffer **270**.

The adder **225** may generate a reconstructed block by adding the reconstructed residual block with the prediction block. The filter unit **260** may apply at least one of a deblocking filter, a sample adaptive offset, and an adaptive loop filter to the reconstructed block or reconstructed image. The filter unit **260** may output the reconstructed image. The reconstructed block or reconstructed image may be stored in the reference picture buffer **270** and used when performing inter-prediction. A reconstructed block processed by the filter unit **260** may be a part of a reference image. That is, a reference image is a reconstructed image composed of reconstructed blocks processed by the filter unit **260**. The stored reference image may be used later in inter prediction or motion compensation.

FIG. 3 is a view schematically showing a partition structure of an image when encoding and decoding the image. FIG. 3 schematically shows an example of partitioning a single unit into a plurality of lower units.

In order to efficiently partition an image, when encoding and decoding, a coding unit (CU) may be used. The coding unit may be used as a basic unit when encoding/decoding the image. In addition, the coding unit may be used as a unit for distinguishing an intra prediction mode and an inter prediction mode when encoding/decoding the image. The coding unit may be a basic unit used for prediction, transform, quantization, inverse-transform, dequantization, or an encoding/decoding process of a transform coefficient.

Referring to FIG. 3, an image **300** is sequentially partitioned in a largest coding unit (LCU), and a LCU unit is determined as a partition structure. Herein, the LCU may be used in the same meaning as a coding tree unit (CTU). A unit partitioning may mean partitioning a block associated with to the unit. In block partition information, information of a unit depth may be included. Depth information may represent a number of times or a degree or both in which a unit is partitioned. A single unit may be partitioned into a plurality of lower level units hierarchically associated with depth information based on a tree structure. In other words, a unit and a lower level unit generated by partitioning the unit may correspond to a node and a child node of the node, respectively. Each of partitioned lower unit may have depth information. Depth information may be information representing a size of a CU, and may be stored in each CU. Unit depth represents times and/or degrees related to partitioning a unit. Therefore, partitioning information of a lower-level unit may comprise information on a size of the lower-level unit.

A partition structure may mean a distribution of a coding unit (CU) within a CTU **310**. Such a distribution may be determined according to whether or not to partition a single CU into a plurality (positive integer equal to or greater than 2 including 2, 4, 8, 16, etc.) of CUs. A horizontal size and a vertical size of the CU generated by partitioning may respectively be half of a horizontal size and a vertical size of the CU before partitioning, or may respectively have sizes smaller than a horizontal size and a vertical size before partitioning according to a number of times of partitioning. The CU may be recursively partitioned into a plurality of CUs. By the recursive partitioning, at least one among a height and a width of a CU after partitioning may decrease comparing with at least one among a height and a width of a CU before partitioning. Partitioning of the CU may be

recursively performed until to a predefined depth or predefined size. For example, a depth of a CTU may be 0, and a depth of a smallest coding unit (SCU) may be a predefined maximum depth. Herein, the CTU may be a coding unit having a maximum coding unit size, and the SCU may be a coding unit having a minimum coding unit size as described above. Partitioning is started from the CTU **310**, a CU depth increases by 1 as a horizontal size or a vertical size or both of the CU decreases by partitioning. For example, for each depth, a CU which is not partitioned may have a size of $2N \times 2N$. Also, in case of a CU which is partitioned, a CU with a size of $2N \times 2N$ may be partitioned into four CUs with a size of $N \times N$. A size of N may decrease to half as a depth increase by 1.

In addition, information whether or not the CU is partitioned may be represented by using partition information of the CU. The partition information may be 1-bit information. All CUs, except for a SCU, may include partition information. For example, when a value of partition information is a first value, the CU may not be partitioned, when a value of partition information is a second value, the CU may be partitioned.

Referring to FIG. 3, a CTU having a depth 0 may be a 64×64 block. 0 may be a minimum depth. A SCU having a depth 3 may be an 8×8 block. 3 may be a maximum depth. A CU of a 32×32 block and a 16×16 block may be respectively represented as a depth 1 and a depth 2.

For example, when a single coding unit is partitioned into four coding units, a horizontal size and a vertical size of the four partitioned coding units may be a half size of a horizontal and vertical size of the CU before being partitioned. In one embodiment, when a coding unit having a 32×32 size is partitioned into four coding units, each of the four partitioned coding units may have a 16×16 size. When a single coding unit is partitioned into four coding units, it may be called that the coding unit may be partitioned (quad-tree partitioned) into a quad-tree form.

For example, when a single coding unit is partitioned into two coding units, a horizontal or vertical size of the two coding units may be a half of a horizontal or vertical size of the coding unit before being partitioned. For example, when a coding unit having a 32×32 size is partitioned in a vertical direction, each of two partitioned coding units may have a size of 16×32 . For example, when a coding unit having a size of 8×32 is horizontally partitioned into two sub-coding units, each of the two sub-coding units may have a size of 8×16 . When a single coding unit is partitioned into two coding units, it may be called that the coding unit is partitioned (binary-tree partitioned) in a binary-tree form.

For example, when one coding unit is partitioned into three sub-coding units, the horizontal or vertical size of the coding unit can be partitioned with a ratio of 1:2:1, thereby producing three sub-coding units whose horizontal or vertical sizes are in a ratio of 1:2:1. For example, when a coding unit having a size of 16×32 is horizontally partitioned into three sub-coding units, the three sub-coding units may have sizes of 16×8 , 16×16 , and 16×8 respectively, in the order from the uppermost to the lowermost sub-coding unit. For example, when a coding unit having a size of 32×32 is vertically split into three sub-coding units, the three sub-coding units may have sizes of 8×32 , 16×32 , and 8×32 , respectively in the order from the left to the right sub-coding unit. When one coding unit is partitioned into three sub-coding units, it can be said that the coding unit is ternary-tree partitioned or partitioned by a ternary tree partition structure.

In FIG. 3, a coding tree unit (CTU) 320 is an example of a CTU to which a quad tree partition structure, a binary tree partition structure, and a ternary tree partition structure are all applied.

As described above, in order to partition the CTU, at least one of a quad tree partition structure, a binary tree partition structure, and a ternary tree partition structure may be applied. Various tree partition structures may be sequentially applied to the CTU, according to a predetermined priority order. For example, the quad tree partition structure may be preferentially applied to the CTU. A coding unit that cannot be partitioned any longer using a quad tree partition structure may correspond to a leaf node of a quad tree. A coding unit corresponding to a leaf node of a quad tree may serve as a root node of a binary and/or ternary tree partition structure. That is, a coding unit corresponding to a leaf node of a quad tree may be further partitioned by a binary tree partition structure or a ternary tree partition structure, or may not be further partitioned. Therefore, by preventing a coding block that results from binary tree partitioning or ternary tree partitioning of a coding unit corresponding to a leaf node of a quad tree from undergoing further quad tree partitioning, block partitioning and/or signaling of partition information can be effectively performed.

The fact that a coding unit corresponding to a node of a quad tree is partitioned may be signaled using quad partition information. The quad partition information having a first value (e.g., "1") may indicate that a current coding unit is partitioned by the quad tree partition structure. The quad partition information having a second value (e.g., "0") may indicate that a current coding unit is not partitioned by the quad tree partition structure. The quad partition information may be a flag having a predetermined length (e.g., one bit).

There may not be a priority between the binary tree partitioning and the ternary tree partitioning. That is, a coding unit corresponding to a leaf node of a quad tree may further undergo arbitrary partitioning among the binary tree partitioning and the ternary tree partitioning. In addition, a coding unit generated through the binary tree partitioning or the ternary tree partitioning may undergo a further binary tree partitioning or a further ternary tree partitioning, or may not be further partitioned.

A tree structure in which there is no priority among the binary tree partitioning and the ternary tree partitioning is referred to as a multi-type tree structure. A coding unit corresponding to a leaf node of a quad tree may serve as a root node of a multi-type tree. Whether to partition a coding unit which corresponds to a node of a multi-type tree may be signaled using at least one of multi-type tree partition indication information, partition direction information, and partition tree information. For partitioning of a coding unit corresponding to a node of a multi-type tree, the multi-type tree partition indication information, the partition direction, and the partition tree information may be sequentially signaled.

The multi-type tree partition indication information having a first value (e.g., "1") may indicate that a current coding unit is to undergo a multi-type tree partitioning. The multi-type tree partition indication information having a second value (e.g., "0") may indicate that a current coding unit is not to undergo a multi-type tree partitioning.

When a coding unit corresponding to a node of a multi-type tree is partitioned by a multi-type tree partition structure, the coding unit may further include partition direction information. The partition direction information may indicate in which direction a current coding unit is to be partitioned for the multi-type tree partitioning. The partition

direction information having a first value (e.g., "1") may indicate that a current coding unit is to be vertically partitioned. The partition direction information having a second value (e.g., "0") may indicate that a current coding unit is to be horizontally partitioned.

When a coding unit corresponding to a node of a multi-type tree is partitioned by a multi-type tree partition structure, the current coding unit may further include partition tree information. The partition tree information may indicate a tree partition structure which is to be used for partitioning of a node of a multi-type tree. The partition tree information having a first value (e.g., "1") may indicate that a current coding unit is to be partitioned by a binary tree partition structure. The partition tree information having a second value (e.g., "0") may indicate that a current coding unit is to be partitioned by a ternary tree partition structure.

The partition indication information, the partition tree information, and the partition direction information may each be a flag having a predetermined length (e.g., one bit).

At least any one of the quad-tree partition indication information, the multi-type tree partition indication information, the partition direction information, and the partition tree information may be entropy encoded/decoded. For the entropy-encoding/decoding of those types of information, information on a neighboring coding unit adjacent to the current coding unit may be used. For example, there is a high probability that the partition type (the partitioned or non-partitioned, the partition tree, and/or the partition direction) of a left neighboring coding unit and/or an upper neighboring coding unit of a current coding unit is similar to that of the current coding unit. Therefore, context information for entropy encoding/decoding of the information on the current coding unit may be derived from the information on the neighboring coding units. The information on the neighboring coding units may include at least any one of quad partition information, multi-type tree partition indication information, partition direction information, and partition tree information.

As another example, among binary tree partitioning and ternary tree partitioning, binary tree partitioning may be preferentially performed. That is, a current coding unit may primarily undergo binary tree partitioning, and then a coding unit corresponding to a leaf node of a binary tree may be set as a root node for ternary tree partitioning. In this case, neither quad tree partitioning nor binary tree partitioning may not be performed on the coding unit corresponding to a node of a ternary tree.

A coding unit that cannot be partitioned by a quad tree partition structure, a binary tree partition structure, and/or a ternary tree partition structure becomes a basic unit for coding, prediction and/or transformation. That is, the coding unit cannot be further partitioned for prediction and/or transformation. Therefore, the partition structure information and the partition information used for partitioning a coding unit into prediction units and/or transformation units may not be present in a bitstream.

However, when the size of a coding unit (i.e., a basic unit for partitioning) is larger than the size of a maximum transformation block, the coding unit may be recursively partitioned until the size of the coding unit is reduced to be equal to or smaller than the size of the maximum transformation block. For example, when the size of a coding unit is 64×64 and when the size of a maximum transformation block is 32×32, the coding unit may be partitioned into four 32×32 blocks for transformation. For example, when the size of a coding unit is 32×64 and the size of a maximum transformation block is 32×32, the coding unit may be

partitioned into two 32×32 blocks for the transformation. In this case, the partitioning of the coding unit for transformation is not signaled separately, and may be determined through comparison between the horizontal or vertical size of the coding unit and the horizontal or vertical size of the maximum transformation block. For example, when the horizontal size (width) of the coding unit is larger than the horizontal size (width) of the maximum transformation block, the coding unit may be vertically bisected. For example, when the vertical size (height) of the coding unit is larger than the vertical size (height) of the maximum transformation block, the coding unit may be horizontally bisected.

Information of the maximum and/or minimum size of the coding unit and information of the maximum and/or minimum size of the transformation block may be signaled or determined at an upper level of the coding unit. The upper level may be, for example, a sequence level, a picture level, a slice level, or the like. For example, the minimum size of the coding unit may be determined to be 4×4. For example, the maximum size of the transformation block may be determined to be 64×64. For example, the minimum size of the transformation block may be determined to be 4×4.

Information of the minimum size (quad tree minimum size) of a coding unit corresponding to a leaf node of a quad tree and/or information of the maximum depth (the maximum tree depth of a multi-type tree) from a root node to a leaf node of the multi-type tree may be signaled or determined at an upper level of the coding unit. For example, the upper level may be a sequence level, a picture level, a slice level, or the like. Information of the minimum size of a quad tree and/or information of the maximum depth of a multi-type tree may be signaled or determined for each of an intra slice and an inter slice.

Difference information between the size of a CTU and the maximum size of a transformation block may be signaled or determined at an upper level of the coding unit. For example, the upper level may be a sequence level, a picture level, a slice level, or the like. Information of the maximum size of the coding units corresponding to the respective nodes of a binary tree (hereinafter, referred to as a maximum size of a binary tree) may be determined based on the size of the coding tree unit and the difference information. The maximum size of the coding units corresponding to the respective nodes of a ternary tree (hereinafter, referred to as a maximum size of a ternary tree) may vary depending on the type of slice. For example, for an intra slice, the maximum size of a ternary tree may be 32×32. For example, for an inter slice, the maximum size of a ternary tree may be 128×128. For example, the minimum size of the coding units corresponding to the respective nodes of a binary tree (hereinafter, referred to as a minimum size of a binary tree) and/or the minimum size of the coding units corresponding to the respective nodes of a ternary tree (hereinafter, referred to as a minimum size of a ternary tree) may be set as the minimum size of a coding block.

As another example, the maximum size of a binary tree and/or the maximum size of a ternary tree may be signaled or determined at the slice level. Alternatively, the minimum size of the binary tree and/or the minimum size of the ternary tree may be signaled or determined at the slice level.

Depending on size and depth information of the above-described various blocks, quad partition information, multi-type tree partition indication information, partition tree information and/or partition direction information may be included or may not be included in a bit stream.

For example, when the size of the coding unit is not larger than the minimum size of a quad tree, the coding unit does not contain quad partition information. Thus, the quad partition information may be deduced from a second value.

For example, when the sizes (horizontal and vertical sizes) of a coding unit corresponding to a node of a multi-type tree are larger than the maximum sizes (horizontal and vertical sizes) of a binary tree and/or the maximum sizes (horizontal and vertical sizes) of a ternary tree, the coding unit may not be binary-tree partitioned or ternary-tree partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but may be deduced from a second value.

Alternatively, when the sizes (horizontal and vertical sizes) of a coding unit corresponding to a node of a multi-type tree are the same as the maximum sizes (horizontal and vertical sizes) of a binary tree and/or are two times as large as the maximum sizes (horizontal and vertical sizes) of a ternary tree, the coding unit may not be further binary-tree partitioned or ternary-tree partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but be derived from a second value. This is because when a coding unit is partitioned by a binary tree partition structure and/or a ternary tree partition structure, a coding unit smaller than the minimum size of a binary tree and/or the minimum size of a ternary tree is generated.

Alternatively, when the depth of a coding unit corresponding to a node of a multi-type tree is equal to the maximum depth of the multi-type tree, the coding unit may not be further binary-tree partitioned and/or ternary-tree partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but may be deduced from a second value.

Alternatively, only when at least one of vertical direction binary tree partitioning, horizontal direction binary tree partitioning, vertical direction ternary tree partitioning, and horizontal direction ternary tree partitioning is possible for a coding unit corresponding to a node of a multi-type tree, the multi-type tree partition indication information may be signaled. Otherwise, the coding unit may not be binary-tree partitioned and/or ternary-tree partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but may be deduced from a second value.

Alternatively, only when both of the vertical direction binary tree partitioning and the horizontal direction binary tree partitioning or both of the vertical direction ternary tree partitioning and the horizontal direction ternary tree partitioning are possible for a coding unit corresponding to a node of a multi-type tree, the partition direction information may be signaled. Otherwise, the partition direction information may not be signaled but may be derived from a value indicating possible partitioning directions.

Alternatively, only when both of the vertical direction binary tree partitioning and the vertical direction ternary tree partitioning or both of the horizontal direction binary tree partitioning and the horizontal direction ternary tree partitioning are possible for a coding tree corresponding to a node of a multi-type tree, the partition tree information may be signaled. Otherwise, the partition tree information may not be signaled but be deduced from a value indicating a possible partitioning tree structure.

FIG. 4 is a view showing an intra-prediction process.

Arrows from center to outside in FIG. 4 may represent prediction directions of intra prediction modes.

Intra encoding and/or decoding may be performed by using a reference sample of a neighbor block of the current block. A neighbor block may be a reconstructed neighbor

block. For example, intra encoding and/or decoding may be performed by using an encoding parameter or a value of a reference sample included in a reconstructed neighbor block.

A prediction block may mean a block generated by performing intra prediction. A prediction block may correspond to at least one among CU, PU and TU. A unit of a prediction block may have a size of one among CU, PU and TU. A prediction block may be a square block having a size of 2×2, 4×4, 16×16, 32×32 or 64×64 etc. or may be a rectangular block having a size of 2×8, 4×8, 2×16, 4×16 and 8×16 etc.

Intra prediction may be performed according to intra prediction mode for the current block. The number of intra prediction modes which the current block may have may be a fixed value and may be a value determined differently according to an attribute of a prediction block. For example, an attribute of a prediction block may comprise a size of a prediction block and a shape of a prediction block, etc.

The number of intra-prediction modes may be fixed to N regardless of a block size. Or, the number of intra prediction modes may be 3, 5, 9, 17, 34, 35, 36, 65, or 67 etc. Alternatively, the number of intra-prediction modes may vary according to a block size or a color component type or both. For example, the number of intra prediction modes may vary according to whether the color component is a luma signal or a chroma signal. For example, as a block size becomes large, a number of intra-prediction modes may increase. Alternatively, a number of intra-prediction modes of a luma component block may be larger than a number of intra-prediction modes of a chroma component block.

An intra-prediction mode may be a non-angular mode or an angular mode. The non-angular mode may be a DC mode or a planar mode, and the angular mode may be a prediction mode having a specific direction or angle. The intra-prediction mode may be expressed by at least one of a mode number, a mode value, a mode numeral, a mode angle, and mode direction. A number of intra-prediction modes may be M, which is equal to or larger than 1, including the non-angular and the angular mode.

In order to intra-predict a current block, a step of determining whether or not samples included in a reconstructed neighbor block may be used as reference samples of the current block may be performed. When a sample that is not usable as a reference sample of the current block is present, a value obtained by duplicating or performing interpolation on at least one sample value among samples included in the reconstructed neighbor block or both may be used to replace with a non-usable sample value of a sample, thus the replaced sample value is used as a reference sample of the current block.

When intra-predicting, a filter may be applied to at least one of a reference sample and a prediction sample based on an intra-prediction mode and a current block size/shape.

In case of a planar mode, when generating a prediction block of a current block, according to a position of a prediction target sample within a prediction block, a sample value of the prediction target sample may be generated by using a weighted sum of an upper and left side reference sample of a current sample, and a right upper side and left lower side reference sample of the current block. In addition, in case of a DC mode, when generating a prediction block of a current block, an average value of upper side and left side reference samples of the current block may be used. In addition, in case of an angular mode, a prediction block may be generated by using an upper side, a left side, a right upper side, and/or a left lower side reference sample of the current

block. In order to generate a prediction sample value, interpolation of a real number unit may be performed.

An intra-prediction mode of a current block may be entropy encoded/decoded by predicting an intra-prediction mode of a block present adjacent to the current block. When intra-prediction modes of the current block and the neighbor block are identical, information that the intra-prediction modes of the current block and the neighbor block are identical may be signaled by using predetermined flag information. In addition, indicator information of an intra-prediction mode that is identical to the intra-prediction mode of the current block among intra-prediction modes of a plurality of neighbor blocks may be signaled. When intra-prediction modes of the current block and the neighbor block are different, intra-prediction mode information of the current block may be entropy encoded/decoded by performing entropy encoding/decoding based on the intra-prediction mode of the neighbor block.

FIG. 5 is a diagram illustrating an embodiment of an inter-picture prediction process.

In FIG. 5, a rectangle may represent a picture. In FIG. 5, an arrow represents a prediction direction. Pictures may be categorized into intra pictures (I pictures), predictive pictures (P pictures), and Bi-predictive pictures (B pictures) according to the encoding type thereof.

The I picture may be encoded through intra-prediction without requiring inter-picture prediction. The P picture may be encoded through inter-picture prediction by using a reference picture that is present in one direction (i.e., forward direction or backward direction) with respect to a current block. The B picture may be encoded through inter-picture prediction by using reference pictures that are preset in two directions (i.e., forward direction and backward direction) with respect to a current block. When the inter-picture prediction is used, the encoder may perform inter-picture prediction or motion compensation and the decoder may perform the corresponding motion compensation.

Hereinbelow, an embodiment of the inter-picture prediction will be described in detail.

The inter-picture prediction or motion compensation may be performed using a reference picture and motion information.

Motion information of a current block may be derived during inter-picture prediction by each of the encoding apparatus 100 and the decoding apparatus 200. The motion information of the current block may be derived by using motion information of a reconstructed neighboring block, motion information of a collocated block (also referred to as a col block or a co-located block), and/or a block adjacent to the co-located block. The co-located block may mean a block that is located spatially at the same position as the current block, within a previously reconstructed collocated picture (also referred to as a col picture or a co-located picture). The co-located picture may be one picture among one or more reference pictures included in a reference picture list.

A method of deriving the motion information of the current block may vary depending on a prediction mode of the current block. For example, as prediction modes for inter-picture prediction, there may be an AMVP mode, a merge mode, a skip mode, a current picture reference mode, etc. The merge mode may be referred to as a motion merge mode.

For example, when the AMVP is used as the prediction mode, at least one of motion vectors of the reconstructed neighboring blocks, motion vectors of the co-located blocks,

motion vectors of blocks adjacent to the co-located blocks, and a (0, 0) motion vector may be determined as motion vector candidates for the current block, and a motion vector candidate list is generated by using the motion vector candidates. The motion vector candidate of the current block can be derived by using the generated motion vector candidate list. The motion information of the current block may be determined based on the derived motion vector candidate. The motion vectors of the collocated blocks or the motion vectors of the blocks adjacent to the collocated blocks may be referred to as temporal motion vector candidates, and the motion vectors of the reconstructed neighboring blocks may be referred to as spatial motion vector candidates.

The encoding apparatus 100 may calculate a motion vector difference (MVD) between the motion vector of the current block and the motion vector candidate and may perform entropy encoding on the motion vector difference (MVD). In addition, the encoding apparatus 100 may perform entropy encoding on a motion vector candidate index and generate a bitstream. The motion vector candidate index may indicate an optimum motion vector candidate among the motion vector candidates included in the motion vector candidate list. The decoding apparatus may perform entropy decoding on the motion vector candidate index included in the bitstream and may select a motion vector candidate of a decoding target block from among the motion vector candidates included in the motion vector candidate list by using the entropy-decoded motion vector candidate index. In addition, the decoding apparatus 200 may add the entropy-decoded MVD and the motion vector candidate extracted through the entropy decoding, thereby deriving the motion vector of the decoding target block.

The bitstream may include a reference picture index indicating a reference picture. The reference picture index may be entropy-encoded by the encoding apparatus 100 and then signaled as a bitstream to the decoding apparatus 200. The decoding apparatus 200 may generate a prediction block of the decoding target block based on the derived motion vector and the reference picture index information.

Another example of the method of deriving the motion information of the current may be the merge mode. The merge mode may mean a method of merging motion of a plurality of blocks. The merge mode may mean a mode of deriving the motion information of the current block from the motion information of the neighboring blocks. When the merge mode is applied, the merge candidate list may be generated using the motion information of the reconstructed neighboring blocks and/or the motion information of the collocated blocks. The motion information may include at least one of a motion vector, a reference picture index, and an inter-picture prediction indicator. The prediction indicator may indicate one-direction prediction (L0 prediction or L1 prediction) or two-direction predictions (L0 prediction and L1 prediction).

The merge candidate list may be a list of motion information stored. The motion information included in the merge candidate list may be at least either one of the zero merge candidate and new motion information that is a combination of the motion information (spatial merge candidate) of one neighboring block adjacent to the current block, the motion information (temporal merge candidate) of the collocated block of the current block, which is included within the reference picture, and the motion information existing in the merge candidate list.

The encoding apparatus 100 may generate a bitstream by performing entropy encoding on at least one of a merge flag and a merge index and may signal the bitstream to the

decoding apparatus 200. The merge flag may be information indicating whether or not to perform the merge mode for each block, and the merge index may be information indicating that which neighboring block, among the neighboring blocks of the current block, is a merge target block. For example, the neighboring blocks of the current block may include a left neighboring block on the left side of the current block, an upper neighboring block disposed above the current block, and a temporal neighboring block temporally adjacent to the current block.

The skip mode may be a mode in which the motion information of the neighboring block is applied to the current block as it is. When the skip mode is applied, the encoding apparatus 100 may perform entropy encoding on information of the fact that the motion information of which block is to be used as the motion information of the current block to generate a bit stream, and may signal the bitstream to the decoding apparatus 200. The encoding apparatus 100 may not signal a syntax element regarding at least any one of the motion vector difference information, the encoding block flag, and the transform coefficient level to the decoding apparatus 200.

The current picture reference mode may mean a prediction mode in which a previously reconstructed region within a current picture to which the current block belongs is used for prediction. Here, a vector may be used to specify the previously-reconstructed region. Information indicating whether the current block is to be encoded in the current picture reference mode may be encoded by using the reference picture index of the current block. The flag or index indicating whether or not the current block is a block encoded in the current picture reference mode may be signaled, and may be deduced based on the reference picture index of the current block. In the case where the current block is encoded in the current picture reference mode, the current picture may be added to the reference picture list for the current block so as to be located at a fixed position or a random position in the reference picture list. The fixed position may be, for example, a position indicated by a reference picture index of 0, or the last position in the list. When the current picture is added to the reference picture list so as to be located at the random position, the reference picture index indicating the random position may be signaled.

FIG. 6 is a diagram illustrating a transform and quantization process.

As illustrated in FIG. 6, a transform and/or quantization process is performed on a residual signal to generate a quantized level signal. The residual signal is a difference between an original block and a prediction block (i.e., an intra prediction block or an inter prediction block). The prediction block is a block generated through intra prediction or inter prediction. The transform may be a primary transform, a secondary transform, or both. The primary transform of the residual signal results in transform coefficients, and the secondary transform of the transform coefficients results in secondary transform coefficients.

At least one scheme selected from among various transform schemes which are preliminarily defined is used to perform the primary transform. For example, examples of the predefined transform schemes include discrete cosine transform (DCT), discrete sine transform (DST), and Karhunen-Loève transform (KLT). The transform coefficients generated through the primary transform may undergo the secondary transform. The transform schemes used for the primary transform and/or the secondary transform may be determined according to coding parameters of the current

block and/or neighboring blocks of the current block. Alternatively, the transform scheme may be determined through signaling of transform information.

Since the residual signal is quantized through the primary transform and the secondary transform, a quantized-level signal (quantization coefficients) is generated. The quantized level signal may be scanned according to at least one of a diagonal up-right scan, a vertical scan, and a horizontal scan, depending on an intra prediction mode of a block or a block size/shape. For example, as the coefficients are scanned in a diagonal up-right scan, the coefficients in a block form change into a one-dimensional vector form. Aside from the diagonal up-right scan, the horizontal scan of horizontally scanning a two-dimensional block form of coefficients or the vertical scan of vertically scanning a two-dimensional block form of coefficients may be used depending on the intra prediction mode and/or the size of a transform block. The scanned quantized-level coefficients may be entropy-encoded to be inserted into a bitstream.

A decoder entropy-decodes the bitstream to obtain the quantized-level coefficients. The quantized-level coefficients may be arranged in a two-dimensional block form through inverse scanning. For the inverse scanning, at least one of a diagonal up-right scan, a vertical scan, and a horizontal scan may be used.

The quantized-level coefficients may then be dequantized, then be secondary-inverse-transformed as necessary, and finally be primary-inverse-transformed as necessary to generate a reconstructed residual signal.

As described above, a video may be divided into a plurality of block units and then encoded/decoded. Units and blocks may be used interchangeably herein.

FIG. 7 is a diagram illustrating various embodiments for dividing one block (e.g., CTU, CTB, or CU) into a plurality of blocks (e.g., CU, PU, or TU). In FIG. 7, w represents a horizontal size of the block, and h represents a vertical size of the block.

For example, one CTU may be recursively divided into a plurality of CUs using a quad tree structure. A prediction method (intra-prediction or inter-prediction) to be applied in units of one CU may be determined.

For example, one CU may be divided into M PUs. Also, for example, one CU may be recursively divided into N TUs using a quad tree structure. Herein, M and N may be positive integers of 2 or more.

As an example of dividing a video into a plurality of blocks, it is possible to perform division using a quad tree structure and then perform division using a binary tree structure. In the following, this division structure is defined as “binary-tree after quad-tree” division.

For example, in the example shown in FIG. 7, one CTU may be divided by performing the binary-tree after quad tree division to be recursively divided into two or four CUs. In binary-tree after quad tree division structure, the quad tree division may be performed earlier than the binary tree division. Here, when one block is divided into two blocks, it may be considered that the block is divided using a binary tree (BT) structure. The size and/or form of the two blocks generated by the binary tree division may be the same or different from each other. When one block is divided into four blocks, it may be considered that the block is divided using a quad tree structure. The size and/or form of some or all of the four blocks generated by quad tree division may be the same or different from each other. The CUs obtained by dividing the CTU using the binary-tree after quad tree division structure may have a square or a non-square (rectangular) form.

At least one of the first flag and the first index may be signaled when the CU is divided using the binary-tree after quad tree division structure.

When the first flag indicates a first value, it may indicate that the corresponding CU is divided by performing the quad tree division. When the first flag indicates a second value, it may indicate that the corresponding CU is no longer divided.

When the first index indicates a first value, it may indicate that the corresponding CU is no longer divided. When the first index indicates a second value, it indicates that the corresponding CU is symmetrically divided in the horizontal direction. When the first index indicates a third value, it indicates that the corresponding CU is symmetrically divided in the vertical direction.

The first flag and the first index indicating whether the division is performed or not and/or what is the division type for the corresponding CU may be signaled as separate syntax elements and signaled as one syntax element. Alternatively, information indicating whether or not to divide the quad tree, information indicating whether or not to divide the binary tree, and information indicating the binary tree division type may be signaled separately, or may be signaled as a syntax element in which some or all of them are integrated.

As in the example shown in FIG. 7, the binary tree symmetric division may have two symmetric division (splitting) types, i.e., symmetric division in the horizontal direction or symmetric division in the vertical direction. In this case, a leaf node in the binary tree may mean a CU. In addition, blocks corresponding to two nodes obtained by the binary tree symmetric division may have the same size.

As in the example shown in FIG. 7, the binary tree asymmetric division may have two asymmetric division (splitting) types, i.e., asymmetric division in the horizontal direction or asymmetric division in the vertical direction. Likewise, a leaf node in the binary tree may mean a CU. In addition, blocks corresponding to two nodes obtained by the binary tree asymmetric division may have different sizes from each other. Unlike the case of binary tree symmetric division, in case of the binary tree asymmetric division, information about division type and/or division ratio may be additionally signaled.

In the binary-tree after quad-tree division structure, a leaf node in the quad tree may be further divided into a binary tree structure. In this case, the leaf node in the quad tree or the leaf node in the binary tree may mean a CU.

In the binary-tree after quad-tree division structure, the CU corresponding to the final leaf node may be a unit of prediction and transform without further division. That is, in the binary-tree after quad-tree division structure, CU, PU and TU may all have the same size. Also, a prediction method (intra-prediction or inter-prediction) may be determined in units of CU. In the binary-tree after quad-tree division structure, intra-prediction, inter-prediction, transform, inverse-transform, quantization, dequantization, entropy encoding/decoding, and in-loop filtering procedure may be performed in units of a square or a non-square (rectangular) block.

A CU may include one luma (Y) component block and two chroma (Cb/Cr) component blocks. In addition, a CU may include only one luma component block or include only two chroma component blocks. In addition, a CU may include only one luma component block, include only Cr chroma component block, or include only Cb chroma component block.

As another embodiment for dividing a video into a plurality of blocks, a quad-tree after binary-tree division

structure may be used. “Quad-tree after binary-tree division” means a division performing a division using a binary tree structure and then performing a division using quad tree structure. “Quad-tree after binary-tree division” and “binary-tree after quad-tree division” may differ only in the order of the tree structures applied to the division. Thus, the above description of “binary-tree after quad-tree division” may be similarly applied to “quad-tree after binary-tree division” except for the order of the tree structures.

As another embodiment of dividing a video into a plurality of blocks, a combined quad-tree and binary-tree division structure may be used. In this case, one CTU may be recursively divided into two or four CUs using the combined quad-tree and binary-tree division structure. In the combined quad-tree and binary-tree division structure, a quad-tree division or a binary-tree division may be performed for one CU.

To a luma signal and a chroma signal within a video or one block (e.g., a CTU), block division structures different from each other may be applied. For example, a luma signal and a chroma signal within a specific slice (I slice) or a block (e.g., a CTU) included in the corresponding slice may be divided by using block division structures different from each other. A luma signal and a chroma signal within other slice (e.g., P or B slice) or a block (e.g., a CTU) included in the corresponding slice may be divided by using an identical block division structure. Herein, when the chroma signal includes a Cb signal and a Cr signal, different intra-prediction modes may be used for the Cb signal and the Cr signal, and intra-prediction modes of the Cb signal and the Cr signal may be entropy encoded/decoded, respectively. An intra-prediction mode of a Cb signal may be entropy encoded/decoded by using an intra-prediction mode of the Cr signal. Conversely, an intra-prediction mode of the Cr signal may be entropy encoded/decoded by using an intra-prediction mode of the Cb signal.

FIG. 8 is a diagram illustrating an embodiment in which one block (for example, a CTU or a CU) is divided by performing a binary-tree after quad-tree division and information on whether or not division is performed and/or a type of division is signaled.

Whether or not quad tree division is performed may be signaled by a QT split flag. Whether or not binary-tree division is performed may be signaled by a BT split flag. A type of binary-tree division may be signaled by a BT split type.

The current block (e.g., CU) has a square, triangular, or rectangular form and may correspond to a leaf node of a quad tree or a leaf node of a binary tree. In addition, intra-prediction, inter-prediction, transform, inverse-transform, quantization, dequantization, entropy encoding/decoding, and in-loop filtering procedure may be performed in units of a size, a form, and/or a depth of the current block.

The current block is divided into at least one symmetric and/or asymmetric sub-blocks, and intra-prediction and/or inter-prediction information different from each other may be derived for each sub-block.

FIG. 9 is a diagram illustrating various embodiments of dividing a current block into two sub-blocks.

In FIG. 9, two sub-blocks obtained by performing division are defined as a first sub-block and a second sub-block respectively, in which the first sub-block is referred to as a sub-block A, and the second sub-block is referred to as a sub-block B.

When the current block is divided into at least one symmetric and/or asymmetric sub-block, the minimum size of the sub-block may be defined as $M \times N$. In this case, M and

N may be the same or different positive integers. For example, a 4×4 block may be defined as a minimum size of a sub-block.

When the current block is divided into at least one symmetric and/or asymmetric sub-block, a flag indicating whether the sub-block is divided or not in units of a block (for example, CU) and/or index information on a division type of the sub-block may be signaled and variably derived on the basis of the encoding parameters of the current block.

When the current block is divided into at least one symmetric and/or asymmetric sub-block, no further block division may be performed for equal to or less than a specific block size or a specific block depth. The information on the specific block size or the specific block depth may be entropy encoded/decoded in one or more units of a video parameter set VPS, a sequence parameter set SPS, a picture parameter set PPS, a tile header, a slice header, CTU, and CU.

When the current block is divided into at least one asymmetric sub-blocks, the divided sub-blocks may have any other form than a square and/or rectangular form.

For example, when the current block is divided into two sub-blocks, the current block may be divided into two sub-blocks of a triangular form obtained by dividing it by a diagonal boundary from the upper left corner to the lower right corner of the current block or divided into two sub-blocks of a triangular form obtained by dividing it by a diagonal boundary from the upper right corner to the lower left corner of the current block. Alternatively, the current block may be divided into four sub-blocks of a triangular form by dividing it by a diagonal boundary from the upper left corner to the lower right corner and dividing it by a diagonal boundary from the upper right corner to the lower left corner. The two sub-blocks of a triangular form may be referred to as asymmetric sub-blocks divided in an asymmetric division type.

FIG. 9(a) shows a current block (CU) in which no division is performed.

For example, when the current block is divided into two sub-blocks as shown in FIG. 9(b), the remaining area excluding the lower right area (the second sub-block (sub-block B)) of the current block is defined as a first sub-block (sub-block A).

The encoder/decoder may store a table or list including a plurality of asymmetric division types. When an asymmetric division type of the current block is determined in the encoder, an index and a flag are determined by referring to the table or list, and the index or the flag may be transmitted to the decoder. Alternatively, the encoder/decoder may determine an asymmetric division type of the current block on the basis of encoding parameters of the current block. In addition, the encoder/decoder may determine an asymmetric division type of the current block from a block neighboring to the current block. In this case, it is possible to determine the asymmetric division type of the current block on the basis of the encoding parameters such as a division type of the neighboring block.

When the current block is divided into at least one asymmetric sub-block, each sub-block may have a horizontal and/or vertical size that is equal to or less than the width w and/or the height h of the current block.

For example, when the current block is divided into two sub-blocks as shown in FIGS. 9(b) to (e), the second sub-block may have a size of $3w/4 \times 3h/4$.

Meanwhile, the ratio of the horizontal and/or vertical length of the second sub-block to the current block may be

set at a predetermined ratio in the encoder and the decoder or may be set on the basis of information signaled from the encoder to the decoder.

The current block (e.g., CU) may mean a leaf node of a quad tree or a leaf node of a binary tree. At least one of encoding/decoding processes such as intra-prediction, inter-prediction, primary/secondary transform, inverse-transform, quantization, dequantization, entropy encoding/decoding, in-loop filtering, and the like that are performed for encoding/decoding the current block may be performed in units of a size, a form, and/or a depth of a sub-block.

When the current block is divided into at least one or more symmetric and/or asymmetric sub-blocks, the prediction information of the current block may be derived by using at least one method of deriving inter-picture prediction information different between sub-blocks, deriving intra-prediction information different between sub-blocks, and deriving combined intra-prediction/inter-prediction information between sub-blocks.

When the current block is divided into at least one symmetric and/or asymmetric sub-block, the flag for indicating whether or not the sub-block is divided in units of a block (for example, CU) and/or index information for a division type of the sub-block may be signaled through a bitstream and variably derived on the basis of the encoding parameter of the current block. In this case, it is possible to define the division type of the sub-block by using at least one type of asymmetric sub-block division types in FIG. 9(b) to (j) and then perform encoding/decoding. For example, when four types of FIG. 9(b) to (e) are used, a flag indicating whether or not sub-block-based encoding and decoding is performed and a sub-block division index may be signaled through a bitstream or be variably derived on the basis of the encoding parameters of the current block. Herein, when explicitly transmitting the index information, at least one of the following entropy encoding/decoding methods may be used, and encoding/decoding may be finally performed by using CABAC(ae(v)) after performing binarization.

Truncated rice binarization method
K-th order exp_golomb binarization method
Restricted K-th order exp_golomb binarization method
Fixed-length binarization method
Unary binarization method
Truncated unary binarization method

Whether the current block is divided into at least one symmetric and/or asymmetric sub-blocks may be determined on the basis of at least one of a slice type, a picture type, and a tile group type of the current block. Here, the slice type may mean at least one of an I slice, a P slice, and a B slice. In addition, the picture type may mean at least one of an I picture, a P picture, and a B picture. Also, the tile group may mean a picture division type including at least one tile, and the tile group may mean at least one of an I tile group, a P tile group, and a B tile group. Also, the slice may be used as the same meaning as a picture and a tile group.

In addition, whether or not the current block is divided into at least one symmetric and/or asymmetric sub-blocks may be determined on the basis of the size of the current block. For example, when the current block has a horizontal size of w and a vertical size of h , it may be determined whether or not to be divided into symmetric and/or asymmetric sub-blocks by comparing values of $w \cdot h$ or $w + h$ with a value for a particular size.

In addition, whether or not the current block is divided into at least one symmetric and/or asymmetric sub-blocks may be determined according to an encoding mode of the current block. For example, when the current block is a skip

mode or a merge mode, the current block may be divided into one or more symmetric and/or asymmetric sub-blocks.

When the current block is divided into at least one symmetric and/or asymmetric sub-block, the motion vector value between each sub-block, a motion vector difference value, a reference picture, a reference picture list, a prediction list utilization flag value, a weighted value used for generating the final prediction block by adding the sub-blocks may be different from each other.

FIG. 10 is a view showing intra-prediction according to the present invention.

Intra-prediction of a current block may include: step S1010 of deriving an intra-prediction mode, step S1020 of configuring a reference sample, and/or step S1030 of performing intra-prediction.

In step S1010, an intra-prediction mode of a current block may be derived. The intra-prediction mode of the current block may be derived by using a method of using an intra-prediction mode of a neighbor block, a method of entropy encoding/decoding an intra-prediction mode of a current block from a bitstream, a method of using a coding parameter of a neighbor block or a method of using intra prediction mode of a color component. According to the method of using the intra-prediction mode of the neighbor block, the intra-prediction mode of the current block may be derived by using at least one intra-prediction mode derived by using an intra-prediction mode of a neighbor block, a combination of at least one intra-prediction mode of a neighbor block, and at least one MPM.

In step S1020, a reference sample may be configured by performing at least one of reference sample selecting, reference sample padding and reference sample filtering.

In step S1030, intra-prediction may be performed by performing at least one of non-angular prediction, angular prediction, positional information based prediction, inter color component prediction, and template matching-based intra-prediction. In step S1030, filtering on a prediction sample may be additionally performed.

In order to derive the intra-prediction mode of the current block, at least one reconstructed neighbor block may be used. A position of the reconstructed neighbor block may be a fixed position that is predefined, or may be a position derived by encoding/decoding. Hereinafter, encoding/decoding may mean entropy encoding and decoding. For example, when a coordinate of a left upper corner side sample of a current block having a $W \times H$ size is $(0, 0)$, a neighbor block may be at least one of blocks adjacent to coordinate of $(-1, H-1)$, $(W-1, -1)$, $(W, -1)$, $(-1, H)$, and $(-1, -1)$, and neighbor blocks of the above blocks. Here, W and H may represent length or the number of samples of width (W) and height (H) of the current block.

An intra-prediction mode of a neighbor block which is not available may be replaced with a predetermined intra-prediction mode. The predetermined intra-prediction mode may be, for example, a DC mode, a planar mode, a vertical mode, a horizontal mode, and/or a diagonal mode. For example, when a neighbor block is positioned outside of a boundary of at least one predetermined unit of a picture, a slice, a tile, and a coding tree unit, the neighbor block is inter-predicted, or when the neighbor block is encoded in a PCM mode, the corresponding block may be determined as non-available. Alternatively, when the neighbor block is unavailable, the intra prediction mode of the corresponding block is not replaced and not used.

The intra-prediction mode of the current block may be derived as a statistical value of an intra-prediction mode of a predetermined positional neighbor block or an intra-

prediction mode of at least two neighbor blocks. In the present description, the statistical value may mean at least one of an average value, a maximum value, a minimum value, a mode, a median value, a weighted average value, and an interpolation value.

Alternatively, the intra-prediction mode of the current block may be derived based on a size of neighbor blocks. For example, an intra-prediction mode of a neighbor block having relatively large size may be derived as the intra-prediction mode of the current block. Alternatively, a statistical value may be calculated by assigning a large weight on an intra-prediction mode of a block having relatively large size. Alternatively, a mode to which a relatively large weight is assigned may be pre-defined or signaled. For example, a relatively large weight may be assigned to at least one among a vertical directional mode, a horizontal directional mode, a diagonal directional mode and non-directional mode. The same weight may be assigned to the above modes.

Alternatively, whether or not the intra-prediction mode of the neighbor block is angular mode may be considered. For example, when the intra-prediction mode of the neighbor block is a non-angular mode, the non-angular mode may be derived as the intra-prediction mode of the current block. Alternatively, an intra-prediction mode of other neighbor block, except for the non-angular mode, may be derived as the intra-prediction mode of the current block.

In order to derive the intra-prediction mode of the current block, it is possible to construct one or more most probable mode (MPM) lists. The MPM list includes one or more MPM candidate modes, and the MPM candidate mode may include an intra-prediction mode of at least one spatial neighboring block in which encoding/decoding is completed and/or a given intra-prediction mode.

FIG. 11 is a diagram illustrating spatial neighboring blocks of a current block used when configuring an MPM list.

For example, assuming that the number of intra-prediction modes configuring the MPM list is six, as shown in FIG. 11, from at least one of the spatial neighboring blocks of the current block, the candidate modes to be included in the MPM list may be sequentially derived up to k at maximum (k is a positive integer of 6 or less). In the following description, for example, k is 5.

The order of deriving the MPM candidate mode from the neighboring blocks may be arbitrarily set by the encoder/decoder. For example, the MPM candidate modes may be derived in the order of a left block L, a top block A, a lower left block BL, an upper right block AR, and an upper left block AL. Further, the MPM candidate modes may be derived in the order of the left block L and the top block A. A Planar mode and/or a DC mode, which is a non-directional mode, may be regarded as an intra-prediction mode having a high probability of occurrence. Therefore, when the Planar mode and/or the DC mode are not included in the five intra-prediction modes derived from the spatial neighboring blocks, the Planar mode and/or DC mode may be included in the MPM list as the MPM candidate mode. That is, the Planar mode and/or DC modes may always be included in the MPM candidate list.

Herein, the order in which the Planar mode and/or the DC mode are located in the MPM list may be arbitrarily set in the encoder/decoder. For example, the MPM list may be configured in the order of the left block L, the top block A, the Planar, the DC, the lower left block BL, the upper right block AR, and the upper left block AL. In addition, the MPM

list may be configured in the order of the left block L, the top block A, the Planar, and the DC.

The redundancy check may be performed to determine whether the intra-prediction modes in the configured MPM list are prediction modes different from each other. When the number of intra-prediction modes included in the MPM list after the redundancy check is smaller than the maximum number (for example, six) of intra-prediction modes that the MPM list may include, an intra-prediction mode in which a predetermined offset is added and/or subtracted to/from the intra-prediction mode having directionality among intra-prediction modes included in the MPM list may be additionally included in the MPM list. Herein, the offset value is not limited to one but may be an integer of two or more.

When the MPM list is not filled through the above process, for example, when the number of the MPM candidate modes is less than six, the MPM list is filled in the order of a vertical mode, a horizontal mode, and a diagonal mode, so that the MPM list may be configured with up to six intra-prediction modes at maximum different from each other. The order in which the default modes (vertical mode, horizontal mode, and diagonal mode) are filled is not limited to the above example and may be any sequence previously defined in the encoder/decoder. When the number of intra-prediction modes is 67 at maximum, mode 0 indicates a Planar mode, mode 1 indicates a DC mode, and modes 2 through 66 may indicate directional modes. In addition, the vertical mode may be mode 50, the horizontal mode may be mode 18, and the diagonal mode may be mode 2, mode 34, and/or mode 66.

For example, the candidate modes included in the MPM list configured as described above may be rearranged according to a predetermined criterion.

FIG. 12 is a view illustrating an embodiment for rearranging candidate modes in the MPM list.

A prediction block is generated for each candidate mode included in the MPM list, and a sum of absolute differences (SAD) is obtained in a boundary region between the generated prediction block and a neighboring reference sample of the current block. Thereafter, the MPM list may be reconfigured by rearranging the MPM candidate modes in the order (increasing order) from the intra-prediction mode having smaller SAD to the intra-prediction mode having higher SAD.

Alternatively, the SAD may be calculated for each of the plurality of MPM candidate modes, and then the MPM list may be configured with candidate modes of the allowed maximum number of MPM candidate modes. Herein, the MPM list may be configured with candidate modes in order of increasing SAD.

When configuring the MPM list, one MPM list may be configured for the current block of a predetermined size. When the current block of the predetermined size is divided into a plurality of sub-blocks, the configured MPM list may be used for each sub-block. In this case, the size of the current block and the sub-block may be, for example, $M \times N$, in which M and N may be a predetermined integer, respectively. For example, the current block and/or the sub-block may have at least one of CTU, CU, signaling unit (SU), QTMax, QTMin, BTMax, BTMin, 4×4 , 8×8 , 16×16 , 32×32 , 64×64 , 128×128 , 256×256 , 4×8 , 8×16 , 16×8 , 32×64 , 32×8 , 4×32 , and the like. In this case, QTMax and QTMin may indicate the maximum size and the minimum size of a block that may be divided into a quad tree, respectively. In addition, BTMax and BTMin may indicate the maximum size and the minimum size of a block that may be divided

35

into a binary tree, respectively. Hereinafter, the size of the sub-block may mean a sub-block division structure.

The intra-prediction mode of the current block may be encoded/decoded using an MPM flag and/or an MPM index (mpm_idx). The MPM flag may indicate whether or not the same candidate mode as the intra-prediction mode of the current block is included in the MPM list. The MPM index may indicate the same candidate as the intra-prediction mode of the current block among the candidate modes included in the MPM list.

For example, when the MPM flag is 1, the intra-prediction mode of the current block (for example, the luminance component) may be derived by using at least one of the MPM index mpm_idx and the intra-prediction mode of neighboring units to the encoded/decoded current block. For example, when the MPM flag is 1, the MPM list may be configured according to the above-described method, and then the intra-prediction mode indicated by the MPM index may be derived as the intra-prediction mode of the current block.

When the configured MPM list is rearranged according to a predetermined criterion, the MPM index is not transmitted, and the candidate mode at any position in the MPM list may be derived as the intra-prediction mode of the current block. The predetermined criterion may be, for example, the Boundary SAD described above. Any position in the MPM list may be, for example, the first (index 0) position of the MPM list.

For example, when the MPM flag is 0, a secondary MPM list (second MPM list) including one or more candidate modes may be configured. In addition, a secondary MPM flag (2nd MPM flag) indicating whether the intra-prediction mode of the current block is the same as the candidate mode included in the secondary MPM list may be encoded/decoded.

When the secondary MPM flag is 1, the secondary MPM index (2nd_mpm_idx) may be additionally encoded/decoded. In this case, the intra-prediction mode of the current block (e.g., luma component) may be derived using at least one of the secondary MPM index and the intra-prediction mode of the encoded/decoded adjacent units.

The candidate mode included in the secondary MPM list may be determined on the basis of the intra-prediction mode of the neighboring blocks of the current block. For example, the intra-prediction mode of the left block or the top block of the current block may be used. Alternatively, when the sub-blocks adjacent to the top of the current block have one or more intra-prediction modes different from each other, the corresponding intra-prediction modes may be included in the secondary MPM list. Similarly, when the sub-blocks adjacent to the left of the current block have one or more intra-prediction modes different from each other, the intra-prediction modes may be included in the secondary MPM list. Alternatively, it is possible to configure the secondary MPM list by using the intra-prediction modes of blocks in which the encoding/decoding is completed earlier than the current block as intra-predicted blocks in the current slice.

When both the MPM flag and the secondary MPM flag are 0, the intra-prediction mode of the current block (for example, luma component) may be encoded/decoded using the luminance component residual intra-prediction mode index (rem_intra_luma_pred_mode).

Also, the intra-prediction mode of chroma component may be derived using at least one of a chroma component residual intra-prediction mode index (intra_chroma_pred_mode) and/or an intra-prediction mode of the corresponding luma block. A method of deriving the intra-prediction

36

mode of the chroma component on the basis of an intra-prediction mode of the luma block will be described later.

Hereinafter, other embodiment will be described in which encoded information around the current block is used to configure one or more candidate modes included in the MPM list (and/or the secondary MPM list).

For example, for each of all the intra-prediction modes, a prediction sample of the current block may be generated. When a prediction block is generated, as described with reference to FIG. 12, a boundary SAD may be calculated between the neighboring reference sample of the current block and the generated prediction sample. Thereafter, the modes are arranged in order of increasing SAD, and the MPM list (and/or the secondary MPM list) may include the candidate modes from the intra-prediction having smaller SAD up to the maximum number of candidate modes that the MPM list (and/or secondary MPM list) may include.

In the embodiment shown in FIG. 12, when 8x8 current blocks are intra-predicted, a prediction block may be generated for each of all the intra-prediction modes allowed in the encoder/decoder. A boundary SAD may be calculated between boundary regions (dotted line region) located at the left column, the upper left sample, and/or the top row and surrounding reference sample (gray region) located in the upper row of each generated prediction block. The intra-prediction modes may be arranged in order of increasing SAD. The intra-prediction modes may be included in the MPM list (and/or the secondary MPM list) as a candidate mode according to the arranged order.

When one reference sample line or more are used, calculation of the boundary SAD may be performed for each of the reference sample lines. Alternatively, when calculating the boundary SAD, one or more columns may be considered for the left column of the prediction block of the current block, and similarly one or more rows may be considered for the top row of the prediction block of the current block.

As other embodiment, after searching other area in the current slice in which encoding/decoding is completed, for a block most similar to the current block, it is possible to apply the above method (a method using Boundary SAD). In this case, after applying the above method or generating the block most similar to the current block and a prediction block according to the intra-prediction mode in the corresponding block, an intra-prediction mode in which the difference value between the two blocks is at minimum is obtained and then included in the MPM list (and/or the secondary MPM list).

Information indicating that the MPM list (and/or the second MPM list) has been configured by using at least one of the above methods may be encoded/decoded or implicitly derived by the decoder. When encoding/decoding the information indicating that the MPM list (and/or the 2nd MPM list) has been configured, at least one of the following entropy encoding/decoding methods may be used, and encoding/decoding may be finally performed by using CABAC (ae(v)) after performing binarization.

Truncated rice binarization method

K-th order exp_golomb binarization method

Restricted K-th order exp_golomb binarization method

Fixed-length binarization method

Unary binarization method

Truncated unary binarization method

According to a further embodiment of the present invention relating to a method of deriving an intra prediction mode, an intra prediction mode of a current block may be derived by using an intra prediction mode of a different color component. For example, when the current block is a

chroma block, an intra prediction mode of a luma block corresponding to the chroma block can be used to derive an intra prediction mode of the chroma block. As the luma block corresponding to the chroma block, there may be one or more luma blocks. The corresponding luma block may be determined depending on at least any one of a position of the luma block, a position of the chroma block, an upper-left sample position of the luma block, an upper-left sample position of the chroma block, the size of the luma block, the size, the shape, and the encoding parameter of the chroma block. Alternatively, the corresponding luma block may be determined depending on at least any one of the size, the shape, and the encoding parameter of a luma block.

The luma block corresponding to the chroma block may be composed of a plurality of partitions. All or part of the plurality of partitions may have different intra prediction modes thereof. An intra prediction mode of the chroma block may be derived on the basis of all or part of the plurality of partitions included in the corresponding luma block. In this case, some partitions may be selectively used, in which the used partitions are selected based on the comparison of the block size, the shape, the depth information, etc. of the chroma block with those of the luma block (all or part of the plurality of partitions). A partition at a position in the luma block corresponding to a predetermined position in the chroma block may be selectively used. The predetermined position may refer to a corner sample (e.g., upper left sample) position in the chroma block or a center sample position in the chroma block. The center sample position may be determined based on an upper-left position of a luma/chroma block, half horizontal size of a luma/chroma block, half vertical size of a luma/chroma block. For example, position of x-axis direction of the center sample may be determined by adding half horizontal size of a luma/chroma block to an upper-left position of the luma/chroma block in horizontal direction. Also, position of y-axis direction of the center sample may be determined by adding half vertical size of a luma/chroma block to an upper-left position of the luma/chroma block in vertical direction. Here, position of a luma block corresponding to a center sample position of the chroma block may mean a center sample position of the luma block.

The method of deriving an intra prediction mode of one color component block using an intra prediction mode of a different color component block (i.e. inter color component intra prediction mode) according to the present invention is not limited to the example in which an intra prediction mode of a luma block corresponding to a chroma block is used. For example, an intra prediction mode of a chroma block may be derived by using or sharing at least any one of an MPM index `mpm_idx` and an MPM list of a luma block corresponding to the chroma block.

FIG. 13 is an exemplary diagram illustrating the relationship between a luma block and a chroma block.

In the example illustrated in FIG. 13, a sample ratio of color components is 4:2:0, and at least one of luma blocks A, B, C, and D corresponds to one chroma block.

With reference to FIG. 13, an intra prediction mode of one chroma block may be derived by using an intra prediction mode of the luma block A corresponding to a sample at an upper left position (0,0) in the chroma block, an intra prediction mode of the luma block D corresponding to a sample at a center position ($nS/2$, $nS/2$) in the chroma block, or an intra prediction mode of the luma block B corresponding to a sample at another center position ($(nS/2)-1$, ($nS/2$)-1) in the chroma block. The predetermined position in the chroma block is not limited to (0, 0), ($(nS/2)-1$, ($nS/2$)-

1) and ($nS/2$, $nS/2$). For example, The predetermined position may be an upper right position, a lower left position, and/or a lower right position. A center position of the chroma block may be ($W/2$, $H/2$) where W is block width and H is block height. Also, a center position of the chroma block may be ($(W/2)-1$, ($H/2$)-1).

The predetermined position may be selected on the basis of the shape of the chroma block. For example, with the chroma block having a square shape, the predetermined position may be a center sample position. With the chroma block having an oblong shape, the predetermined position may be an upper left sample position. Alternatively, the predetermined position may be a position of an upper left sample in the chroma block having a square shape or a position of a center sample in the chroma block having an oblong shape.

According to a further embodiment, an intra prediction mode of a chroma block may be derived by using statistic figures of one or more intra prediction modes of a luma block having an equal size to the chroma block.

In the example illustrated in FIG. 13, a mode corresponding to the average of the intra prediction modes of the luma blocks A and D or a mode corresponding to the average of the intra prediction modes of the luma blocks A, B, C, and D having an equal size to the chroma block is derived as the intra prediction mode of the chroma block.

When there are multiple intra prediction modes of available luma blocks, all or part of them may be selected. The selection is performed based on the predetermined position in the chroma block or based on the size(s), the shape(s), and/or the depth(s) of the chroma block, the luma block, or both. The intra prediction mode of the chroma block can be derived by using the selected intra prediction mode of the luma block.

For example, the size of the luma block A corresponding to the upper left sample position (0,0) in the chroma block and the size of the luminance block D corresponding to the center sample position ($nS/2$, $nS/2$) in the chroma block are compared, and the intra prediction mode of the luma block D having a larger size may be used to derive the intra prediction mode of the chroma block.

Alternatively, when the size of a luma block corresponding to a predetermined position in a chroma block is equal to or larger than the size of the chroma block, an intra prediction mode of the chroma block is derived by using the intra prediction mode of the luma block.

Alternatively, when the size of a chroma block is within a predetermined range, an intra prediction mode of the chroma block is derived by using an intra prediction mode of a luma block corresponding to the upper left sample position (0, 0) in the chroma block.

Alternatively, when the size of a chroma block is within a predetermined range, the size of a luma block corresponding to a predetermined position (0, 0) of the chroma block and the size of a luma block disposed at another predetermined position ($nS/2$, $nS/2$) of the chroma block are compared, and an intra prediction mode of the chroma block is derived by using the intra prediction mode of the luma block having a larger size.

The predetermined range may be derived from at least any one piece of information among information signaled through a bitstream, information of the size (and/or depth) of a block (a chroma block, a luma block, or both), and information predefined in an encoder/decoder.

Alternatively, when a chroma block has an oblong shape, an intra prediction mode of the chroma block may be derived by using an intra prediction mode of a luma block corre-

sponding to a center sample position ($nS/2$, $nS/2$) or an intra prediction mode of a luma block corresponding to another center sample position $((nS/2)-1, (nS/2)-1)$ in the chroma block.

Among the plurality of partitions of the luma block, a partition having the same shape as the chroma block may be used. For example, when the chroma block has a square shape or a non-square shape, a partition having a square shape or a non-square shape, selected among the plurality of partitions of the luma block, may be used.

In the example described with reference to FIG. 13, the method of deriving an intra prediction mode of a chroma block using an intra prediction mode of a luma block also applies to a case in which an intra prediction mode of a luma block is used as an intra prediction mode of a chroma block as it is. The method of deriving an intra prediction mode of a chroma block is not limited to the method of using an intra prediction mode of the corresponding luma block. For example, an intra prediction mode of a chroma block can be derived from information, including an MPM list and an MPM index mpm_idx , which is used to derive an intra prediction mode of a luma block.

Alternatively, the MPM list of the chroma block can be constructed using the intra prediction mode of the luma block corresponding to the sample of the predetermined position in the chroma block. In this case, the mpm_idx information of the chroma block may be encoded and signaled. The MPM list of the chroma block may be constructed in a similar way to the construction of the MPM list of the luma block. MPM candidates of the chroma block may include intra prediction modes of neighbor chroma blocks and/or intra prediction modes of luma blocks corresponding to the chroma block.

When an MPM flag is 0, a second MPM list including at least one intra-prediction mode may be configured, and the intra-prediction mode of the current block may be derived by using a second MPM index ($2nd_mpm_idx$). Herein, a second indicator (for example, a second MPM flag) indicating whether or not the intra-prediction mode of the current block is included in the second MPM list may be encoded/decoded. Similar to a first MPM list, the second MPM list may be configured by using intra-prediction modes of the neighbor block. Herein, the intra-prediction mode included in the first MPM list may not be included in the second MPM list. A number of MPM lists is not limited to 1 or 2, N MPM lists may be used.

When the intra-prediction mode of the current block is not included in one of a plurality of MPM lists, a luma component intra-prediction mode of the current block may be encoded/decoded. In addition, a chroma component intra-prediction mode may be derived and encoded/decoded based on an associated luma component intra-prediction mode.

When the current block is partitioned into a plurality of sub-blocks, in order to derive an intra-prediction mode of each sub-block, at least one of the described methods may be applied.

A size or form or both of a sub-block may be a predetermined size or block or both (for example, 4×4), or may be determined according to a size or form or both of the current block. Alternatively, the size of the sub-block may be determined based on whether or not a neighbor block of the current block is partitioned, or may be determined based on an intra-prediction mode of a neighbor block of the current block. For example, the current block may be partitioned based on a boundary at which an intra-prediction mode of a neighbor block is different. Alternatively, the current block

may be partitioned based on whether the neighbor block is an intra coding block or an inter coding block.

An indicator (for example, $NDIP_flag$) representing that the intra-prediction mode of the current block is derived by using the intra-prediction mode of the neighbor block may be encoded/decoded. The indicator may be encoded/decoded by at least one unit of the current block and the sub-block. Herein, when a size of the current block or the sub-block corresponds to a predetermined size or a predetermined size range, the indicator may be encoded/decoded.

Determining whether or not the size of the current block corresponds to a predetermined size may be performed based on a horizontal or vertical length of the current block. For example, when the horizontal or vertical length is a length capable of being partitioned, it is determined that the size of the current block corresponds to a predetermined size.

Intra-prediction information may be signaled through at least one of a video parameter set (VPS), a sequence parameter set (SPS), a picture parameter set (PPS), an adaptation parameter set (APS), a slice header, and a tile header. In a predetermined block size or less, at least one piece of intra-prediction information may not be signaled. Herein, intra-prediction information of a previously encoded/decoded block (for example, higher block) may be used.

A reference sample for intra-prediction may be configured based on the derived intra-prediction mode. In the description hereinafter, a current block may mean a prediction block or a sub-block having a size/form smaller than a size/form of the prediction block. The reference sample may be configured by using at least one sample reconstructed adjacent to a current block or by using a combination of samples. In addition, filtering may be applied to the configured reference sample.

A number or position or both of reconstructed sample lines used for configuring the reference sample may vary according to a position of a current block within a coding tree block. Each reconstructed sample on a plurality of reconstructed sample lines may be used as a reference sample at it is. Alternatively, a predetermined filter may be applied to the reconstructed sample, and a reference sample may be generated by using the filtered reconstructed sample. Reconstructed samples to which a filter is applied may be included in the same reconstructed sample line or in different reconstructed sample lines.

An indicator indicating whether multiple reference sample lines are utilized for prediction may be signaled. For example, an indicator such as $mrl_enabled_flag$ may be included in at least one of an SPS, a PPS, and a slice header so as to be signaled. The flag may be an indicator indicating whether a single reference sample line is used or multiple reference sample lines are used.

When the indicator indicates that multiple reference sample lines are used, reference sample line indexes are also signaled. For example, mrl_index is signaled. Therefore, it is possible to determine which reference sample lines are used.

When the indicator mrl_index has a value of 0, a first reference sample line which is closest to the current block is utilized. On the other hand, when the indicator mrl_index has a value of 1, a second reference sample line which is second closest to the current block is utilized. When the indicator mrl_index has a value of 2, a third closest reference sample line which is third closest to the current block is

used. The first to fourth reference sample lines respectively correspond to reconstructed sample lines 1 to 4 illustrated in FIG. 14, respectively.

The indicator *mrl_index* is signaled depending on at least one of the intra prediction mode, the MPM information, the size (with and height) of the current block, the presence or absence of an upper boundary of a CTU, and the color component. When the indicator *mrl_index* is not signaled, the first reference sample line adjacent to the current block is used.

For example, when the intra prediction mode is a predetermined mode, the indicator *mrl_index* may be signaled. The intra prediction mode may be the intra prediction mode of the current block or at least one of the intra prediction modes of the respective neighboring blocks. The predetermined mode is at least one of non-directional prediction mode, directional prediction mode, vertical or horizontal mode, even-numbered mode, and odd-numbered mode. For example, when the intra prediction mode of a neighboring block adjacent to the left boundary or the upper boundary of the current block is one of directional modes, the indicator *mrl_index* may be signaled. Alternatively, when the intra prediction mode of the neighboring block is one of even-numbered modes or one of odd-numbered modes, the indicator *mrl_index* may be signaled.

For example, the indicator *mrl_index* may be signaled on the basis of the MPM information of the current block. The MPM information include at least one of an MPM flag, an MPM index, an MPM list, and an MPM candidate. For example, when the MPM flag for the intra prediction mode of the current block indicates matching, the indicator *mrl_index* may be signaled. Alternatively, when any one directional prediction mode is present within an MPM candidate list or only directional prediction modes are present within the MPM candidate list, the indicator *mrl_index* may be signaled. Alternatively, when any one non-directional prediction mode is present in the MPM candidate line, the indicator *mrl_index* may be signaled. Alternatively, the MPM information of the current block is signaled differently depending on the indicator *mrl_index*. For example, when the indicator *mrl_index* has a value other than 0, at least one piece of the MPM information may not be signaled. For example, when the indicator *mrl_index* has a value other than 0, the MPM flag or the remaining mode information may not be signaled. On the other hand, when the indicator *mrl_index* has a value other than 0, the MPM index may be signaled and the intra prediction mode of the current block may be derived using the MPM index. For example, when the indicator *mrl_index* has a value other than 0, the MPM mode may be determined without parsing the MPM flag.

For example, when the size (width or height) of the current block is within a predetermined size range, the indicator *mrl_index* may be signaled. For example, when the size (width or height) is larger than a predetermined size (e.g., 4), the indicator *mrl_index* may be signaled.

For example, the indicator *mrl_index* may be signaled depending on whether the current block is located at the upper boundary of a CTU. For example, when the current block is located at the upper boundary of a CTU, the indicator *mrl_index* may not be signaled.

For example, the indicator *mrl_index* may be signaled when the color component of the current block is a luminance signal, and the indicator *mrl_index* indicator may not be signaled when the color component is a chrominance signal.

Alternatively, the indicator *mrl_index* refers to a reference sample line to be used optionally. For example, the first reference sample line adjacent to the current block may be always used, and the reference sample line indicated by the indicator *mrl_index* indicator may be optionally used.

When multiple reference sample lines are used, whether to apply filtering is determined for each reference sample line. For example, on the basis of the intra prediction mode and the block size/shape, filtering may be applied to the first reference sample line adjacent to the current block but the filtering may not be applied to the second and subsequent reference sample line around the current block. Alternatively, the filtering may be applied only to one reference sample line. For example, the filtering may be applied only to either a left reference sample line or an upper reference sample line. Which reference sample line is subjected to filtering may be determined depending on at least one of the shape, size, and intra prediction mode of the current block. The shape of the current block may be determined depending on a size comparison between the width and the height of the current block or a ratio of the width and the height.

The configured reference sample may be represented as *ref*[*m*, *n*], and a sample obtained by applying a filter to the configured reference sample may be represented as *rec*[*m*, *n*]. Herein, *m* or *n* may be a predetermined integer value representing a position of a sample. When a position of a left upper side sample within the current block is (0, 0), a position of a left upper side reference sample of the current block may be set to (−1, −1).

FIG. 14 is a diagram for describing a plurality of reconstructed sample lines.

A reference sample can be constructed by selecting one or more reconstructed sample lines adjacent to the current block. For example, in FIG. 14, one of the plurality of reconstructed sample lines may be selected so as to construct a reference sample.

For example, a particular reconstructed sample line of the plurality of reconstructed sample lines may be fixedly or adaptively selected, or an arbitrary reconstructed sample line may be adaptively selected, in order to construct a reference sample.

In another embodiment, to construct a reference sample, one or more reconstructed sample lines may be selected from the plurality of reconstructed sample lines illustrated in FIG. 14, and the selected reconstructed sample lines may be combined.

For example, as shown in Equation 1, a reference sample may be constructed using a weighted average of reconstructed samples, in which the weights of the reconstructed samples differ according to the distance between the reconstructed sample and the current block.

$$\begin{aligned} \text{ref}[-1, -1] &= \text{[Equation 1]} \\ &= (\text{rec}[-2, -1] + 2 \times \text{rec}[-1, -1] + \text{rec}[-1, -2] + 2) \gg 2 \\ \text{ref}[x, -1] &= (\text{rec}[x, -2] + 3 \times \text{rec}[x, -1] + 2) \gg 2, \\ &\quad (x = 0 \text{ to } H + W - 1) \\ \text{ref}[-1, y] &= (\text{rec}[-2, y] + 3 \times \text{rec}[-1, y] + 2) \gg 2, \\ &\quad (y = 0 \text{ to } H + W - 1) \end{aligned}$$

Alternatively, a reference sample may be constructed using at least one of a mean value, a maximum value, a minimum value, a median value, and a mode value of a

plurality of reconstructed samples based on at least one of the distance from the current block to the corresponding reconstructed sample and the intra prediction mode of the current block.

Alternatively, a reference sample may be constructed based on a change (change amount) between each of the sample values of the successive reconstructed samples. For example, a reference sample may be constructed based on at least one of a determination of whether the values of two successive reconstructed samples differ by more than a threshold value and a determination of whether the values of successive reconstructed samples change continuously or discontinuously. For example, when the values of a $\text{rec}[-1, -1]$ and a $\text{rec}[-2, -1]$ differ by more than a threshold value, the value of the $\text{ref}[-1, -1]$ is determined as having the value of the $\text{rec}[-1, -1]$, or a value corresponding to a weighted average obtained by applying a predetermined weight to the value of the $\text{rec}[-1, -1]$. For example, each of the values of the successive reconstructed samples changes by n as the distance between the reconstructed sample and the current block decreases, and thus the value of $\text{ref}[-1, -1]$ is represented as " $\text{ref}[-1, -1] = \text{rec}[-1, -1] - n$ ".

In a different embodiment, referring to FIG. 14, two or more reconstructed sample lines may be selected to construct a reference sample. For example, two lines including a reconstructed sample line 1 and a reconstructed sample line 2 may be fixedly selected, or four lines ranging from a reconstructed sample line 1 to a reconstructed sample line 4 may be selected to construct a reference sample.

Alternatively, two or more reconstructed sample lines may be adaptively selected to construct a reference sample. For example, one reconstructed sample line may be fixedly selected, and one or more reconstructed sample lines may be adaptively selected among the other reconstructed sample lines to construct a reference sample.

The fixedly selected reconstructed sample line may be predefined in the encoder/decoder. For the case where the fixedly selected reconstructed sample line is predefined, information on the fixedly selected reconstructed sample line may not be signaled.

The information on the adaptively selected reconstructed sample line(s) may be signaled in the form of an indicator or index. The adaptively selected reconstructed sample line may be determined based on at least one of coding parameters of the current block or a block neighboring the current block. For example, the adaptively selected reconstructed sample line may be determined based on at least one of the size/shape and intra prediction mode of the current block or the block neighboring the current block. In this case, the information necessary for selection may not be signaled.

A reference sample line may include one or more samples. For example, the reference sample line may include samples corresponding to a length equal to the width (that is, the horizontal dimension) or height (that is, the vertical dimension) of the current block. As another example, the reference sample line may include samples corresponding to a length that is two times the width or height of the current block. As a further example, the reference sample line may include samples corresponding to a length equal to N samples (N is 1, 2, 3, . . .) plus two times the sum of the width and height of the current block. That is, the reference sample line may include reference samples corresponding to $2 \times (W + H) + N$ (where W and H are the width and height of the current block, and N is an integer of 1 or more).

The method of constructing a reference sample adjacent to an upper part of the current block and the method of constructing a reference sample adjacent to a left part of the

current block may differ. For example, the number of reference sample lines located above the current block and the number of reference sample lines located to the left of the current block may differ. For example, the number of reference sample lines adjacent to the upper part of the current block may be one and the number of reference sample lines adjacent to the left part of the current block may be two, according to at least one of the width or height of the current block, and the intra prediction mode of the current block. For example, the length of the reference sample line above the current block and the length of the reference sample line located to the left of the current block may differ. For example, the length of the reference sample line may vary according to at least one of the width or height of the current block and the intra prediction mode of the current block.

Each of the reference sample lines may have a different length. For example, referring to FIG. 14, the lengths of the reconstructed sample lines 2 to 4 may be longer than the reconstructed sample line 1 by a length corresponding to one or more samples.

The length of the reference sample line may be different for each of the reconstructed sample lines. For example, a reconstructed sample line n may be longer or shorter than a reconstructed sample line $n-1$ by a length corresponding to m samples. In the example illustrated in FIG. 14, the reconstructed sample line n is longer than the reconstructed sample line $n-1$ by a length corresponding to one sample.

As described above, decision information on whether to construct a reference sample using only the nearest reference sample line or using a plurality of reference sample lines may be encoded/decoded. For example, the decision information may be encoded/decoded at the level of at least one of a sequence, a picture, a slice, a tile, a CTU, a CU, a PU, and a TU. In addition, information on the availability of each of the plurality of reference sample lines may be signaled at a higher level.

At least one of the number, position, and configuration of the reconstructed sample lines used in the reference sample construction may be differently set when the top boundary or the left boundary of the current block corresponds to the boundary of at least one of a picture, a slice, a tile, and a coding tree block (CTB). For example, when two or more reference sample lines are constructed, when the top boundary of the current block corresponds to the boundary of at least one of a picture, a tile, a slice, and a coding tree block (CTB), one reference sample line adjacent to the upper part of the current block may be constructed. For example, one reference sample line may be configured when the top boundary of the current block corresponds to the top boundary of a CTU, and otherwise, two or more reference sample lines may be configured. In this case, since only one reference sample line at the top boundary of the CTU is used, the size of a line buffer for storing data of the reference samples of the reference sample line can be reduced.

When selecting a reference sample, availability determination and reference sample padding may be performed for a block containing the reference sample to be used. For example, when a block containing a reference sample is available, the corresponding reference sample can be used. On the other hand, when a block containing a reference sample is not available, the unavailable reference samples in the block may be padded with one or more available neighboring reference samples.

When a reference sample is located outside the boundary of at least one of a picture, a tile, a slice, or a coding tree block (CTB), the reference sample may be determined to be

unavailable. When the current block is coded with constrained intra prediction (CIP), in the case where the block including the reference sample has been encoded/decoded in an inter prediction mode, the reference sample is determined to be unavailable.

FIG. 15 is a diagram for describing a process of replacing an unavailable sample with an available sample.

When it is determined that the reconstructed neighboring sample is not available, the unavailable sample may be replaced with a reconstructed neighboring sample, which is an available sample. For example, when there are both available samples and unavailable samples as illustrated in FIG. 15, one or more available samples can be used to replace one or more unavailable samples.

The sample values of the unavailable samples may be replaced with the values of the available samples in a predetermined order. The available samples used to replace the unavailable samples may be available samples located adjacent to the unavailable samples. When no available sample is adjacent to the unavailable sample, the earliest or closest available sample may be used to replace the unavailable sample. The replacing order of the unavailable samples may be, for example, from the bottom left to the top right. Alternatively, the replacing order may be from the top right to the bottom left. Specifically, the replacing order may be from the top left corner to the top right and/or to the bottom left. Alternatively, the replacing order may be from the top right and/or from the bottom left to the top left corner.

For example, filling the unavailable samples with the values of available samples may start from the position 0, which is the bottom left sample position. That is, the first four unavailable samples may be filled with a value of "a", and the subsequent 13 unavailable samples may be filled with a value of "b".

For example, the unavailable samples may be filled with a combined value of the available samples. For example, the unavailable samples may be filled with an average value or an interpolated value of the available samples respectively adjacent to both ends of a line of the unavailable samples. That is, the first four unavailable samples are filled with the value "a", and the next 13 unavailable samples may be filled with the average of a value of "b" and a value of "c", or may be filled by interpolating the value "b" and the value "c".

Alternatively, the 13 unavailable samples may be filled with an arbitrary intermediate value between the sample values "b" and "c" of the available samples. In this case, the unavailable samples may be filled with different respective values. For example, as the distance of an unavailable sample to the available sample having the value "a" decreases, the unavailable sample will be filled with a value that is closer to the value "a". For example, the closer an unavailable sample is to an available sample having the value "b", the closer the value that fills the unavailable sample is to the value "b". That is, the value of an unavailable sample may be determined based on the distance between the unavailable sample and the available sample having the value "a" or "b". To replace unavailable samples with available samples, one or more replacement methods including the methods described above may be adaptively used. The method of replacing unavailable samples with available samples may be signaled as information contained in a bitstream, or may be predetermined in the encoder/decoder. Alternatively, a replacement method may be derived according to a predetermined determination method. For example, the replacement method may be determined based on the difference between the values "a" and "b" or based on the number of unavailable samples. More specifi-

cally, the replacement method may be determined by comparing the difference between the values of two available samples with a threshold value and/or by comparing the number of unavailable samples with a threshold value. For example, when the difference between the values of the two available samples is greater than the threshold value, and/or when the number of unavailable samples is greater than the threshold value, the unavailable samples may be replaced to have different values from each other. The selection of the method of replacing unavailable samples with available samples may be performed on a per-predetermined-unit basis. For example, the replacement may be selected on a per-video basis, a per-sequence basis, a per-picture basis, a per-slice basis, a per-tile basis, a per-coding-tree-unit (CTU) basis, a per-coding-unit (CU) basis, a per-prediction-unit (PU) basis, a per-transform-unit (TU) basis, or a per-block basis. At this time, the selection of the method of replacing unavailable samples with available samples may be determined based on the information signaled on a per-predetermined-unit basis or may be derived on a per-predetermined-unit basis. Alternatively, the selection method for the replacement methods may be predetermined in the encoder/decoder.

When a reference sample is located at a predetermined position, padding may be automatically performed without determining whether a block including the reference sample is available or not. For example, referring to FIG. 15, when the position (x, y) of the top left corner sample of the current block is (0, 0), sample availability may not be determined for samples located at (x, y) in which the x coordinate or the y coordinate is equal to or greater than W+H (x=W+H or greater or y=W+H or greater), and the samples may be padded with neighboring reference samples.

For example, a sample $\text{ref}[W+H, -2]$ may be padded with the value of a sample $\text{ref}[W+H-1, -2]$ without performing the availability determination on the sample $\text{ref}[W+H, -2]$. As another example, a sample $\text{ref}[W+H, -3]$ and a sample $\text{ref}[W+H+1, -3]$ may be padded with the value of a sample $\text{ref}[W+H-1, -3]$ without performing the availability determination on the sample $\text{ref}[W+H, -3]$ and the sample $\text{ref}[W+H+1, -3]$. That is, the padding may be performed on the samples located at position (x, y) where x is equal to or greater than W+H or y is equal to or greater than W+H, by using the closest sample on the same sample line without performing the availability determination thereon.

When the position of the top left corner sample of the current block is (0, 0), for samples located at position (x, y) where x is equal to or greater than W and is less than W+H, among the samples located above the current block, the availability determination will be performed, and then the padding will be performed according to the result of the availability determination. For samples located at position (x, y) where y is equal to or greater than H and is less than W+H, among the samples located to the left of the current block, the availability determination will be performed, and the padding will be performed according to the availability determination.

For example, when the position of the top left corner sample of the current block is (0, 0), for samples corresponding to $\text{rec}[x, -1]$ (x ranges from -1 to W+H-1) and/or samples corresponding to $\text{rec}[-1, y]$ (y ranges from 0 to H+W-1), the availability determination and the padding may be performed.

For the padding, a plurality of reference sample lines may be used. For example, when the padding is performed on a first reference sample line adjacent to (that is, the closest to) the current block, a second reference sample line, which is

47

the second closest to the current block, may be used. For example, the padding may be performed according to Equation 2. That is, the sample values of the first reference sample line may be derived by using the weighted average of samples selected from the first reconstructed reference sample line and samples selected from the second reconstructed reference sample line. In this case, the selected reconstructed sample may be one located at a current sample position or at a position adjacent to the current sample position.

$$\text{ref}[x, -1] = (\text{rec}[x, -2] + 3 \times \text{rec}[x, -1] + 2) \gg 2, \quad [\text{Equation 2}]$$

$$(x = 0 \sim H + W - 1)$$

Filtering may be performed on one or more reference samples among the samples constructed as above. The filtering may be adaptively performed based on at least one of the intra prediction mode of the current block, the size of the current block, and the shape of the current block. For example, at least one of a determination of whether to apply filtering, a filter type, a filter strength, and a filter coefficient may be adaptively determined.

For example, whether to apply the filtering may be determined for each of the plurality of reference sample lines. For example, the filtering may be applied to the first reference sample line adjacent to the current block, and may not be applied to the second reference sample line. For example, both a filtered value and an unfiltered value may be used for the same reference sample.

For example, at least one of a 3-tap filter, a 5-tap filter, a 7-tap filter, and an N-tap filter may be selectively applied according to at least one of the intra prediction mode of the current block, the size of the current block, and the shape of the current block. In this case, M is an integer equal to or greater than 3.

For example, filters having different shapes may be selectively used according to at least one of the intra prediction mode, the size, and the shape of the current block. FIG. 16 illustrates various filter shapes.

The shape of the current block may be determined by comparing the width (horizontal dimension) of the current block with the height (vertical dimension) of the current block. For example, at least one of a decision of whether to apply a filter, a filter type, a filter strength, and a filter coefficient may be adaptively determined according to whether the current block is a horizontally oblong block or a vertically oblong block. Alternatively, at least one of a decision of whether to apply filtering, a filter type, a filter strength, and a filter coefficient may be adaptively determined according to whether the current block is a rectangular block or a square block.

Intra prediction for the current block may be performed based on the derived intra prediction mode and the constructed reference sample.

For example, non-directional intra prediction may be performed for the current block. The mode of the non-directional intra prediction may be at least one of a DC mode, a planar mode and an LM mode.

For the DC mode, prediction may be performed using the average value of one or more reference samples among the constructed reference samples. In this case, filtering may be applied to one or more prediction samples (also referred to as predicted samples) located at the boundary of the current block. The DC prediction may be adaptively performed

48

based on at least one of the size of the current block and the shape of the current block. Further, the range of the reference samples used in the DC mode can be determined based on at least one of the size and the shape of the current block.

FIG. 17 is a diagram for describing intra prediction according to the shapes of the current block.

For example, when the current block is a square block, as illustrated in (a) of FIG. 17, DC prediction may be performed by using the average value of the reference sample located above the current block and the reference sample located to the left of the current block.

For example, when the current block is a non-square block, neighboring samples adjacent to the left end and the upper end of the current block may be selectively used. When the current block is a rectangular block, as illustrated in (b) of FIG. 17, the prediction may be performed using the average value of the reference samples adjacent to a longer side among the left side and the upper side of the current block.

For example, when the size of the current block corresponds to a predetermined size or falls within a predetermined range, a predetermined number of reference samples, among the reference samples located above or to the left of the current block, are selected, and the prediction is performed using the average value of the selected reference samples. The predetermined size may be a fixed size of N×M, which is preset in the encoder/decoder. In this case, N and M are integers greater than 0, and N and M may be the same or different from each other. The predetermined range may mean a threshold value for selecting the reference samples for prediction of the current block. The threshold value may be set with at least one of a minimum value and a maximum value. The minimum value and/or the maximum value may be a fixed value or fixed values preset in the encoder/decoder, or a variable value or variable values that is/are encoded and then signaled by the encoder.

For example, one or more average values may be used to perform the prediction. When the current block is a square block or a non-square block, at least one of a first average value or a second average value may be used, in which the first average value is the average of the reference samples located above the current block and the second average value is the average of the reference samples located to the left of the current block. The DC prediction value of the current block may be the first average value or the second average value. Alternatively, the DC prediction value of the current block may be a weighted sum obtained by weighting the first average value and the second average value. For example, the weights for the first and second average values may be the same (that is, 1:1).

According to the above method, a shift operation can be used to calculate all of the DC values. For example, the method can be used even for the case where a sample length, which represents the width, the height, or the sum of the width and height of the current block, is not the power of two. The method may be applied to both luma DC prediction and chroma DC prediction. Alternatively, the method may be applied either to luma DC prediction or to chroma DC prediction.

For example, when the current block is a non-square block, the prediction may be performed based on either the width or the height of the current block. For example, a predicted value may be obtained by dividing the sum of the values of the upper reference sample and the left reference sample by the length of a longer side (namely, the width or the height) of the current block. In this case, the division

operation using the value corresponding to the longer one among the width and the height may be performed by a shift operation.

For example, the DC prediction may be performed using a plurality of reference sample lines. For example, the prediction may be performed using two reference sample lines, as illustrated in (c) of FIG. 17.

For example, the average value of the reference samples included in the two reference sample lines may be determined as the DC prediction value of the current block.

Alternatively, different weights may be applied to the reference samples of the first adjacent line and the reference samples of the second adjacent line of the current block. For example, a weighted average of each sample in the first reference sample line and each sample in the second reference sample line is calculated by applying the weights 3:1 to each sample in the first reference sample line and each sample in the second reference sample line (that is, $(3 \times \text{the first line reference sample} + \text{the second line reference sample}) \gg 2$), and the average of the weighted averages may be determined as the DC prediction value of the current block. Alternatively, the resultant value of $((3 \times \text{the first line reference sample} - \text{the second line reference sample}) \gg 1)$ may be obtained, and the average of these values may be determined as the DC prediction value of the current block. The weights are not limited to the above example, and any weights may be used. In this case, the closer to the current block the reference sample line is, the larger the weight that is applied to the reference sample line. The number of reference sample lines that can be used is not limited to two, and three or more reference sample lines may be used for prediction.

For the planar mode, prediction may be performed with a weighted sum as a function of the distance from at least one reference sample to an intra prediction target sample located in the current block.

Filtering may be performed on reference samples of the current block or prediction samples (that is, predicted samples) of the current block. For example, after filtering is applied to reference samples, planar prediction may be performed, and then filtering may be performed on one or more prediction samples. Among the prediction samples, filtering may be performed on samples in one, two, or N sample lines located at the top boundary or the left boundary of the current block.

To perform the planar prediction, a weighted sum of one or more reference samples may be used. For example, five reference samples may be used, as illustrated in (d) of FIG. 17. For example, to generate a prediction sample for a target position $[x, y]$, the reference samples $r[-1, -1]$, $r[x, -1]$, $r[-1, y]$, $r[W, -1]$, and $r[-1, H]$ may be used. In this case, W and H are the width and the height of the current block, respectively. For example, prediction samples $\text{pred}[x, y]$ can be generated using Equation 3. In Equation 3, a, b, c, d, and e represent weights. N may be $\log_2(a+b+c+d+e)$.

$$\text{pred}[x, y] = (a \times r[-1, -1] + b \times r[x, -1] + c \times r[-1, y] + d \times r[W, -1] + e \times r[-1, H]) \gg N \quad [\text{Equation 3}]$$

As another example, the planar prediction may be performed using a plurality of reference sample lines. For example, the planar prediction may be performed using a weighted sum of two reference sample lines. As another example, the planar prediction may be performed using a

weighted sum of reference samples in the two reference sample lines. In this case, the reference samples selected from the second reference sample line may be samples adjacent to the reference samples selected from the first reference sample line. That is, when the reference sample located at the position $(-1, -1)$ is selected, the reference sample located at the position $(-2, -2)$ may be selected. The planar prediction may be performed by calculating a weighted sum of the selected reference samples, and in this case the same weights as those used for the DC prediction may be used.

A directional prediction mode refers to at least one of a horizontal mode, a vertical mode, and an angular mode having a predetermined angle.

In the horizontal mode or the vertical mode, prediction is performed using one or more reference samples arranged in a linear direction, i.e., in the horizontal direction or the vertical direction. A plurality of reference sample lines may be used. For example, when two reference sample lines are used, prediction may be performed using two reference samples arranged in a horizontal line or a vertical line. Similarly, when N reference sample lines are used, N reference samples in a horizontal line or a vertical line may be used.

For the vertical mode, the statistics of a first reference sample (e.g., $r[x, -1]$) on a first reference sample line and a second reference sample (e.g., $r[x, -2]$) on a second reference sample line may be used to perform the directional prediction.

For example, the predicted value of the vertical mode can be determined by calculating the result value of $(3 \times r[x, -1] + r[x, -2]) \gg 2$. Alternatively, the predicted value of the vertical mode can be determined by calculating the result value of $(3 \times r[x, -1] - r[x, -2]) \gg 1$. In yet another alternative, the predicted value of the vertical mode can be determined by calculating the value of $(r[x, -1] + r[x, -2]) \gg 1$.

For example, the change between each of the sample values on the vertical line may be considered. For example, the predicted value of the vertical mode can be determined by calculating the result value of $(r[x, -1] + (r[x, -1] - r[x, -2]) \gg 1)$. In this case, N may be an integer equal to or greater than 1. As N, a fixed value may be used. Alternatively, N may increase with an increase in the y coordinate of a prediction target sample. For example, $N=y+1$.

Even for the horizontal mode, one or more methods used for the vertical mode can be used.

For an angular mode of a certain angle, prediction may be performed using one or more reference samples arranged in an oblique direction from an intra prediction target sample of the current block, or one or more samples neighboring the reference samples located in the oblique direction. In this case, a total of N reference samples may be used, wherein N may be 2, 3, 4, 5, or 6. It is also possible to perform prediction by applying at least one of an N-tap filter to the N reference samples. Examples of the N-tap filter include a 2-tap filter, a 3-tap filter, a 4-tap filter, a 5-tap filter, and a 6-tap filter. At this time, at least one of the reference samples may be located above the current block and the rest may be located to the left of the current block. The reference samples located above the current block (or the reference samples located to the left of the current block) may be located in the same line or in different lines.

According to another embodiment, intra prediction may be performed based on position information. In this case, the position information may be encoded/decoded, and a reconstructed sample block located at the position described

above may be derived as an intra predicted block of the current block. Alternatively, a block similar to the current block may be searched for by the decoder, and the found block may be derived as the intra predicted block of the current block. The searching for a similar block may be performed in an encoder or a decoder. The range (search range) in which the search is performed may be limited to a predetermined range. For example, the search range may be limited to reconstructed sample blocks within a picture in which the current block is included. Alternatively, the search range may be limited to a CTU in which the current block is included or to a predetermined CU. That is, location information-based intra prediction may be performed by searching for a block similar to the current block among reconstructed samples within a CTU. The searching may be performed using a template. For example, one or more reconstructed samples adjacent to the current block are taken as a template, and a CTU is searched for samples similar to the template.

The location information-based intra prediction may be performed when the CTU consists of only intra coding modes or when the luminance block and the chrominance block have different partition structures. For example, for an inter prediction available slice (e.g., P or B slice), information indicating that the current CTU consists of only intra coding modes may be signaled. In this case, when the information indicates that a current CTU consists of only intra coding modes, the location information-based intra prediction may be performed. Alternatively, when the luminance block and the chrominance block in the current CTU have different partition structures (for example, when dual tree or separate tree is a value of 1), the location information-based intra prediction may be available. On the other hand, when a CTU includes intra coding blocks and inter coding blocks or when the luminance block and the chrominance block have the same partition structure, location information-based intra prediction may not be available.

According to a further embodiment, inter color component intra prediction is performed. For example, it is possible to intra-predict chroma components from the corresponding reconstructed luma component of the current block. Alternatively, it is possible to intra-predict one chroma component Cr from the corresponding reconstructed chroma component Cb of the current block.

An inter color component intra prediction includes a color component block restructuring step, a prediction parameter deriving step, and/or an inter color component prediction execution step. The term 'color component' may refer to at least any one of a luma signal, a chroma signal, Red, Green, Blue, Y, Cb, and Cr. A prediction of a first color component can be performed by using at least any one of a second color component, a third color component, and a fourth color component. The signals of the color components used for the prediction may include at least any one of an original signal, a reconstructed signal, a residual signal, and a prediction signal.

When performing an intra prediction for a second color component target block, a sample of a first color component block corresponding block that corresponds to the second color component target block, a sample of a neighbor block of the first color component corresponding block, or both of the samples may be used. For example, when performing an intra prediction for a chroma component block Cb or Cr, a reconstructed luma component block Y corresponding to the chroma component block Cb or Cr may be used.

When predicting the chroma components on the basis of the luma component, the prediction may be performed according to Equation 4.

$$Pred_c(i, j) = \alpha \cdot rec'_L(i, j) + \beta \quad \text{[Equation 4]}$$

In Equation 4, $Pred_c(i, j)$ represents a predicted chroma sample of the current block, and $rec_L(i, j)$ represents a reconstructed luma sample of the current block. At this time, $rec'_L(i, j)$ may be a down-sampled reconstructed luma sample. Parameters α and β may be derived by minimizing a regression error between the reconstructed neighboring luma sample and the reconstructed neighboring chroma sample around the current block.

There are two modes for predicting the chroma components using the luma component. The two modes may include a single-model mode and a multiple-model mode. The single-model mode may use one linear model when predicting the chroma components from the luma components for the current block. The multiple-model mode may use two linear models.

In the multiple-model mode, the samples adjacent to the current block (that is, adjacent luma samples and adjacent chroma samples) may be classified into two groups. That is, the parameters α and β for each of the two groups may be derived. Further, the luma samples of the current block may be classified according to the rules used for classification of the luma samples adjacent to the current block.

For example, a threshold value for classifying the adjacent samples into two groups may be calculated. The threshold value may be calculated using an average value of the reconstructed adjacent luma samples. However, the calculation of the threshold value is not limited thereto. At least one of various statistical values recognized in the present specification may be used instead of the average value. When the values of the adjacent samples are larger than the threshold value, the adjacent samples may be classified into a first group. Otherwise, the adjacent samples may be classified into a second group.

Although it is described that the multiple-model mode uses two linear modes in the embodiment described above, the present invention is not limited thereto, and may cover other cases in which two or more linear models are used. When N linear models are used, samples may be classified into N groups. To do so, N-1 threshold values may be calculated.

As described above, when predicting a chroma component from a luma component, a linear model can be used. In this case, the linear model may include a simple linear model (hereinafter referred to as "LM1"), a complex linear model (hereinafter referred to as "LM2"), and a complex filter linear model (hereinafter, referred to as "LM3"). Parameters of the models described above may be derived by minimizing regression error between the reconstructed luma samples around the current block and the corresponding reconstructed chroma samples around the current block.

FIG. 18 is a diagram for describing "neighboring samples of a current block" (hereinafter referred to as "adjacent data set") used to derive the parameters of the models.

The adjacent data set for deriving the parameters of the LM1 may be composed of a pair of samples comprising a luma sample and a chroma sample in each of a line area B and a line area C illustrated in FIG. 18. The adjacent data set for deriving the parameters of the LM2 and LM3 may be composed of a pair of samples comprising a luma sample

53

and chroma sample in each of a line area B, a line area C, a line area E, and a line area F illustrated in FIG. 18.

However, the adjacent data set is not limited to the examples described above. For example, to cover various linear relationships between luma and chroma samples in the current block, N adjacent data sets may be used for each mode. For example, N may be an integer of 2 or more, and specifically 3.

The parameters of the linear model may be calculated using both an upper template and a left template. Alternatively, there are two LM modes (an LM_A mode and an LM_L mode), and the upper template and the left template may be used in the LM_A mode and the LM_L mode, respectively. That is, in the LM_A mode, the linear model parameters may be obtained using only the upper template. When the position of the upper left corner sample of the current block is (0, 0), the upper template may be extended to a range from (0, -n) to (W+H-1, -n). In this case, n is an integer equal to or greater than 1. That is, in the LM_L mode, the linear model parameters may be obtained using only the left template. The left template may be extended to a range from (-n, 0) to (-n, H+W-1). In this case, n is an integer equal to or greater than 1.

A power of two numbers of samples can be used to derive the parameters of the linear model. When the current chroma block is a non-square block, the samples used to derive the parameters of the linear model may be determined based on the number of samples on a shorter side, among the horizontal side and the vertical side of the current block. According to one embodiment, when the size of the current block is n×m (where n>m), m samples of the n adjacent samples adjacent to the top boundary of the current block may be selected, for example, by performing sub-sampling uniformly. In this case, the number of samples used to derive the parameters of the linear model may be 2m. As another example, when the size of the current block is n×m (where n>m), m samples of the n adjacent samples adjacent to the top boundary of the current block may not be used. For example, of the n samples, m samples that are farthest from the shorter one of the horizontal side and the vertical side of the current block may not be used. In this case, the number of samples used to derive the parameters of the linear model may be n (n-m samples adjacent to the top boundary of the current block+m samples adjacent to the left boundary of the current block).

Alternatively, when performing an intra prediction for a chroma component block Cr, a chroma component block Cb may be used. Alternatively, when performing an intra prediction for a fourth color component block, at least one of a first color component block, a second color component block, and a third color component, all of which correspond to the fourth color component block, may be used.

Whether or not to perform an inter color component intra prediction may be determined based on at least any one of the size and the shape of a current target block. For example, when the size of the target block is equal to that of a coding tree unit (CTU), larger than a predetermined size, or within a predetermined size range, the inter color component intra prediction for the target block can be performed. Alternatively, when the shape of the target block is a predetermined shape, the inter color component intra prediction for the target block can be performed. The predetermined shape may be a square shape. In this case, when the target block has an oblong shape, the inter color component intra prediction for the target block may not be performed. Meanwhile, when the predetermined shape is an oblong shape, the embodiment described above inversely operates.

54

Alternatively, whether or not to perform an inter color component intra prediction for a prediction target block may be determined based on a coding parameter of at least any one block selected from among a corresponding block corresponding to the prediction target block and neighbor blocks of the corresponding block. For example, when the corresponding block has been predicted through an intra prediction method in a constrained intra prediction (CIP) environment, an inter color component intra prediction for the prediction target block may not be performed. Alternatively, when the intra prediction mode of the corresponding block is a predetermined mode, an inter color component intra prediction for the prediction target block can be performed. Further alternatively, whether or not to perform an inter color component intra prediction may be determined on the basis of at least any one of CBF information of the corresponding block and CBF information of the neighbor blocks thereof. The coding parameter is not limited to a prediction mode of a block but various parameters that can be used for encoding/decoding may be used.

The color component block restructuring step will be described below.

When predicting a second color component block by using a first color component block, the first color component block may be restructured. For example, when an image has an YCbCr color space and when a sampling ratio of color components is one of 4:4:4, 4:2:2, and 4:2:0, the block sizes of color components may differ from each other. Therefore, when predicting a second color component block using a first color component block having a different size from the second color component block, the first color component block may be restructured such that the block sizes of the first color component and the second color component are equalized. The restructured block may include at least any one of a sample in the first color component block that is a corresponding block and a sample in a neighbor block of the first color component block.

FIG. 19 is an exemplary diagram illustrating a process of restructuring a color component block.

In FIG. 19(a), p1[x, y] represents a sample at a position (x, y) in the first color component block. In FIG. 19(b), p1'[x, y] represents a sample at a position (x, y) in the restructured block that is produced by restructuring the first color component block.

When the first color component block has a larger size than the second color component block, the first color component block is down-sampled to have a size equal to that of the second color component block. The down-sampling may be performed by applying an N-tap filter to one or more samples (N is an integer equal to or larger than 1). For the down-sampling, at least any one equation of Equation 5 to Equation 9 may be used. In the case in which any one down-sampling method among various down-sampling methods is selectively used, an encoder may select one down-sampling method as a predetermined down-sampling method. For example, the encoder may select a down-sampling method having optimal effects. The selected down-sampling method is encoded and signaled to a decoder. The signaled information may be index information indicating the down-sampling method.

$$p1'[x, y] = (p1[2x, 2y] + p1[2x, 2y + 1] + 1) \gg 1 \quad [\text{Equation 5}]$$

$$p1'[x, y] = (p1[2x + 1, 2y] + p1[2x + 1, 2y + 1] + 1) \gg 1 \quad [\text{Equation 6}]$$

-continued

$$p1'[x, y] = \quad \text{[Equation 7]}$$

$$(p1[2x-1, 2y] + 2 \times p1[2x, 2y] + p1[2x+1, 2y] + 2) \gg 2$$

$$p1'[x, y] = (p1[2x-1, 2y+1] + \quad \text{[Equation 8] 5}$$

$$2 \times p1[2x, 2y+1] + p1[2x+1, 2y+1] + 2) \gg 2$$

$$p1'[x, y] = (p1[2x-1, 2y] + \quad \text{[Equation 9]}$$

$$2 \times p1[2x, 2y] + p1[2x+1, 2y] + p1[2x-1, 2y+1] +$$

$$2 \times p1[2x, 2y+1] + p1[2x+1, 2y+1] + 4 \gg 3$$

The down-sampling method performed with respect to two or more samples is not limited to any one of the examples of Equation 5 to Equation 9. For example, two or more samples used to calculate a down-sampled value $p1'[x, y]$ may be selected from a sample group consisting of a sample $p1[2x, 2y]$ and neighbor samples thereof. The neighbor samples may be ones selected among $p1[2x-1, 2y-1]$, $p1[2x-1, 2y]$, $p1[2x-1, 2y+1]$, $p1[2x, 2y-1]$, $p1[2x, 2y+1]$, $p1[2x+1, 2y-1]$, $p1[2x+1, 2y]$, and $p1[2x+1, 2y+1]$. The down-sampling can be performed by calculating the average or the weighted average of two or more samples.

Alternatively, the down-sampling may be performed in a manner of selecting a specific sample among one or more samples. In this case, at least any one of the following equations, Equation 10 to Equation 13, may be used for the down-sampling.

$$p1'[x, y] = p1[2x, y] \quad \text{[Equation 10]}$$

$$p1'[x, y] = p1[2x, 2y+1] \quad \text{[Equation 11]}$$

$$p1'[x, y] = p1[2x+1, 2y] \quad \text{[Equation 12] 35}$$

$$p1'[x, y] = p1[2x+1, 2y+1] \quad \text{[Equation 13]}$$

When the first color component block has a smaller size than the second color component block, the first color component block is up-sampled to be restructured such that the sizes of the first color component block and the second color component block are equalized. In this case, the up-scaling is performed according to Equation 14.

$$p1'[2x, 2y] = p1[x, y], \quad \text{[Equation 14]}$$

$$p1'[2x+1, 2y] = (p1[x, y] + p1[x+1, y] + 1) \gg 1,$$

$$p1'[2x, 2y+1] = (p1[x, y] + p1[x, y+1] + 1) \gg 1,$$

$$p1'[2x+1, 2y+1] = (p1[x+1, y] + p1[x, y+1] + 1) \gg 1$$

In the restructuring process, a filter may be applied to one or more samples. For example, the filter may be applied to one or more samples included in at least any one of the first color component block (i.e. corresponding block), neighbor blocks of the corresponding block, the second color component block (i.e. target block), and neighbor blocks of the target block.

In the reference sample restructuring step described above, an indicator corresponding to a predetermined reference sample line among a plurality of reference sample lines may be signaled. In this case, in the restructuring process, the restructuring is performed using the predetermined reference sample line corresponding to the signaled

indicator. For example, when the indicator mrl_index has a value of 0, the reconstruction process is performed using the first and second reference sample lines adjacent to a first color component corresponding block. Alternatively, when the indicator mrl_index has a value of 1, the reconstruction process is performed using the second and third reference sample lines adjacent to the first color component corresponding block. Alternatively, when the indicator mrl_index has a value of 3, the reconstruction process is performed using the third and fourth reference sample lines adjacent to the first color component corresponding block. A reference sample line indicated by the indicator mrl_index may be used for a second color component target block.

In the restructuring process, when a boundary of the second color component block (target block) or a boundary of the first color component block (corresponding block) is a boundary of a predetermined region, the reference samples used for the restructuring may be differently selected. In this case, the number of reference sample lines at the upper side may differ from the number of reference sample lines at the left side. The predetermined region may be at least any one of a picture, a slice, a tile, a CTU, and a CU.

For example, when the upper boundary of the first color component corresponding block is the boundary of the predetermined region, the reference samples at the upper side may not be used for the restructuring but only the reference samples at the left side may be used for the restructuring. When the left boundary of the first color component corresponding block is the boundary of the predetermined region, the reference samples at the left side may not be used for the restructuring but only the reference samples at the upper side may be used for the restructuring. Alternatively, both of N reference sample lines at the upper side and M reference sample lines at the left side may be used for the restructuring, in which N may be smaller than M. For example, when the upper boundary corresponds to the boundary of the predetermined region, N may be 1. Meanwhile, when the left boundary corresponds to the boundary of the predetermined region, M may be 1.

Alternatively, the restructuring may be performed by using N reference sample lines at the upper side and M reference left sample lines at the left side of the first color component corresponding block, regardless of whether the boundary of the predetermined region is the upper boundary or the left boundary of the first color component block.

FIG. 20 is a diagram illustrating an embodiment performing restructuring by using a plurality of upper-side reference sample lines and/or a plurality of left-side reference sample lines.

As illustrated in FIG. 20(a), the restructuring may be performed using four upper-side reference sample lines and four left-side reference sample lines.

For example, when the upper boundary or the left boundary of the first color component corresponding block is the boundary of the predetermined region, the number of the upper-side reference sample lines and the number of the left-side reference sample lines used for the restructuring may differ from each other. For example, as illustrated in FIGS. 20(b) to 20(d), any of the following combinations may be used for the restructuring: two upper-side reference sample lines and four left-side reference sample lines; one upper-side reference sample line and three left-side reference sample lines; and one upper-side reference sample line and two left-side reference sample lines.

The number of reference sample lines used for the restructuring is not limited to the above combinations. That is, N upper-side reference samples lines and M left-side reference

57

sample lines may be used in which N and M are equal to or different from each other. When both of the upper boundary and the left boundary of the corresponding block correspond to the boundary of the predetermined region, N and M may be equal to each other. That is, N and M may be both 1. Alternatively, N may be set smaller than M under the same condition. This is because more resources (memory) are required for the upper-side reference sample lines than for the left-side reference sample lines.

Alternatively, as illustrated in FIG. 20(e), one or more reference samples within a region having a vertical length and a horizontal length not larger than those of the first color component corresponding block may be used for the restructuring.

When performing the restructuring process, the reference samples of the first color component corresponding block may be differently set depending on any one of the block size, the block shape, and the coding parameter of at least any one block selected among the first color component corresponding block, neighbor blocks thereof, the second color component target block, and neighbor blocks thereof.

For example, among samples in the first color component corresponding block and the neighbor blocks thereof, samples in blocks whose encoding mode is an inter frame encoding mode are not used but only samples in blocks whose encoding mode is an intra encoding mode are used for the restructuring.

FIG. 21 is an exemplary diagram illustrating reference samples used for the restructuring in accordance with an intra prediction mode or a coding parameter of a corresponding block.

The restructuring of the reference samples of the first color component block may be differently performed in accordance with the intra prediction modes of the first color component corresponding block. For example, when the intra prediction mode of the corresponding block is a non-angular mode, such as a DC mode and a planar mode, or an angular mode in which both of the upper-side reference samples and the left-side reference samples are used, as illustrated in FIG. 21(a), at least one sample group of the upper-side reference samples and the left-side reference samples is used for the restructuring. Alternatively, when the intra prediction mode of the corresponding block is an angular mode in which both of the upper-side reference samples and the left-side reference samples of the corresponding block are used, as illustrated in FIG. 21(b), the restructuring of the corresponding block is performed using at least one sample group of the upper-side reference samples and the left-side reference samples. Alternatively, when the intra prediction mode of the corresponding block is an angular mode in which both of the left-side reference samples and the upper-side reference samples are used, as illustrated in FIG. 21(c), the corresponding block may be restructured using at least any one sample group of the left-side reference samples and the lower left-side reference samples.

Alternatively, the reference samples used to restructure the first color component corresponding block are differently selected in accordance with the quantization parameter of at least any one of the first color component corresponding block and the neighbor blocks thereof. For example, as illustrated in FIG. 21(d), reference samples in an upper block that is disposed at the upper side of the corresponding block and whose neighbor blocks have a relatively small quantization parameter value QP are used for the restructuring of the corresponding block.

58

Alternatively, when the second color component target block has an oblong shape, reference samples disposed around a first color component corresponding block having a square shape are used for the restructuring.

Alternatively, when the second color component target block is partitioned into two sub-blocks (for example, two 16×8-size sub-blocks) and when the first color component corresponding block is a 32×16-size block, reference samples disposed around a 32×32-size block are used for the restructuring of the corresponding block. In this case, as reference samples of the first color component block corresponding to a second 16×8-size sub-block disposed at a lower side among the partitioned two sub-blocks of the second color component corresponding block, reference samples around a restructured 32×32-size block may be shared.

Hereinbelow, the prediction parameter deriving step will be described.

A prediction parameter can be derived using at least any one of reference samples of the restructured first color component corresponding block and reference samples of the second color component prediction target block. Hereinafter, the terms ‘first color component’ and ‘first color component block’ may respectively refer to a restructured first color component and a restructured first color component block.

FIG. 22 is a diagram illustrating an exemplary restructured first color component corresponding block when a second color component prediction target block is a 4×4 block. In this case, the number of reference sample lines may be N.

The prediction parameter may be derived using reference samples disposed at the upper side and the left side of the restructured first color component corresponding block or of the second color component prediction target block as illustrated in the FIG. 22(a).

For example, the prediction parameter can be derived by adaptively using the reference samples of the restructured first color component, on the basis of the intra prediction mode of the first color component corresponding block. In this case, the reference samples of the second color component can be adaptively used on the basis of the intra prediction mode of the first color component corresponding block.

When the intra prediction mode of the first color component corresponding block is a non-angular mode such as a DC mode or a planar mode, or an angular mode in which both of upper-side reference samples and left-side reference samples are used, reference samples at the upper side and the left side of the first color component corresponding block can be used as illustrated in FIG. 22(a).

When the intra prediction mode of the first color component corresponding block is a non-angular mode in which upper-side reference samples are used, reference samples at the upper side of the first color component corresponding block may be used as illustrated in FIG. 22(b) or 22(c).

When the intra prediction mode of the first color component corresponding block is an angular mode in which left side reference samples are used, reference samples at the left side of the first color component corresponding block may be used as illustrated in FIG. 22(d) or 22(e).

Alternatively, when the intra prediction mode of the first color component corresponding block is an angular mode, reference samples used in each prediction mode can be used as reference samples of the first color component. For example, when the intra prediction mode is a vertical mode, reference samples illustrated in FIG. 22(b) may be used.

When the intra prediction mode is a horizontal mode, reference samples illustrated in FIG. 22(d) may be used. When the intra prediction mode is an up-right diagonal mode, reference samples illustrated in FIG. 22(c) may be used. When the intra prediction mode is a down-left diagonal mode, reference samples illustrated in FIG. 22(e) may be used. When the intra prediction mode is a mode between the vertical mode and the up-right diagonal mode, reference samples illustrated in FIG. 22(f) may be used. When the intra prediction mode is an angular mode of a 45° diagonal direction, upper right reference samples, lower left reference samples, or both are used as illustrated in FIG. 22(g). Reference samples that are differently selected for each intra prediction mode are stored in a format of a look-up table so as to be conveniently used.

The prediction parameter may be derived by adaptively using the reference samples of the first color component or the second color component in accordance with the size and/or the shape of the first color component block and/or the second color component block.

For example, when the second color component target block has a 64×64 size, 32, 16, or 8 reference samples among reference samples at the upper side or the left side of the first color component block or the second color component block may be used. As described above, when the size of the second color component target block is a predetermined size, the reference samples of the first or second color component block may be adaptively used. The predetermined size is not limited to the 64×64 size, but it may be a size signaled through a bitstream or a size derived on the basis of the coding parameter of a current block or a neighbor block thereof.

Alternatively, when the second color component target block has an oblong shape, reference samples adjacent to a longer side, which is a vertical side or a horizontal side, of the second color component target block may be used. For example, when the target block has a block size of 32×8, reference samples at the upper side of the first color component or the second color component block may be used.

Alternatively, when the second color component target block has an oblong shape, reference samples around a square block can be used. For example, when the target block is a 32×8 block, reference samples around a 32×32 block can be used.

The prediction parameter can be derived using reference samples around the restructured first color component block and reference samples around the second color component block. The prediction parameter can be derived on the basis of any one of the factors including a correlation, a change, an average value, and a distribution of color components. In this case, any one of the methods of Least Squares (LS), Least Mean Squares (LMS), etc. may be used.

When deriving the prediction parameters through the LMS method, the prediction parameters may be a and b , α and β , or both. Prediction parameters that can minimize an error between the reference samples of the first color component and the reference samples of the second color component can be derived by Equation 15.

$$E(a, b) = \sum_{n=0}^{N-1} (p2_n - (a \cdot p1'_n + b))^2 \quad [\text{Equation 15}]$$

In Equation 15, $p2_n$ represents a reference sample of the second color component, and $p1'_n$ represents a reference sample of the restructured first color component. N is the

number of used reference samples arranged in a vertical direction or a horizontal direction, and a and b represent prediction parameters.

In this case, a correlation between the reference samples can be calculated by Equation 16.

$$k = \text{Max}(0, \text{BitDepth} + \log_2(N) - 15) \quad [\text{Equation 16}]$$

$$L = \left(\sum_{y=0}^{N-1} p1'[-1, y] + \sum_{x=0}^{N-1} p1'[x, -1] \right) \gg k$$

$$C = \left(\sum_{y=0}^{N-1} p2[-1, y] + \sum_{x=0}^{N-1} p2[x, -1] \right) \gg k$$

$$LL = \left(\sum_{y=0}^{N-1} p1'[-1, y]2 + \sum_{x=0}^{N-1} p1'[x, -1]2 \right) \gg k$$

$$LC = \left(\sum_{y=0}^{N-1} p1'[-1, y] \times p2[-1, y] + \sum_{x=0}^{N-1} p1'[x, -1] \times p2[x, -1] \right) \gg k$$

In Equation 16, BitDepth represents a bit depth. $p1'$ represent a sample of the restructured first color component, and $p2$ represents a sample of the second color component.

FIG. 23 is a diagram illustrating a sample of a first color component and a sample of a second color component.

When there is a region with no reference sample in the process of deriving a prediction parameter, the prediction parameter can be derived using only existing samples.

One or more prediction parameters can be derived. For example, a first prediction parameter may be derived from reference samples having values satisfying a specific requirement among reference samples used to derive prediction parameters. In addition, a second prediction parameter may be derived from referenced samples having values that do not satisfy the specific requirement. The specific requirement may be a condition in which the value of a reference sample is less than a statistic figure (for example, an average value).

According to another embodiment of the present invention, a basic prediction parameter (default parameter) may be used instead of deriving a prediction parameter from values of reference samples. The default parameters may be predefined in the encoder and the decoder. For example, the prediction parameters a and b may be respectively 1 and 0.

Alternatively, when deriving prediction parameters from reference samples, the derived prediction parameters may be encoded and decoded.

When performing an inter color component prediction among color components Y, Cb, and Cr, prediction parameters used to predict color components Cb and Cr can be derived from a color component Y. Prediction parameters used to predict a color component Cr can be derived from a color component Cb. Alternatively, as prediction parameters for predicting a color component Cr, the prediction parameters that have been derived from a color component Y to predict a color component Cb can be used as they are, instead of deriving new prediction parameters for a prediction of the color component Cr.

Hereinbelow, the inter color component prediction execution step will be described.

As described above, after prediction parameters are derived, an inter color component intra prediction can be performed using at least any one of the derived prediction parameters.

61

For example, a prediction of a second color component target block can be performed by applying the derived prediction parameter to a reconstructed signal of the restructured first color component, according to Equation 17.

$$p2[x, y] = a \times p1'[x, y] + b \quad \text{[Equation 17]}$$

In Equation 17, $p2[x, y]$ represents a prediction block of the second color component target block. $p1'[x, y]$ represents the first color component block or the restructured first color component block.

Alternatively, the prediction of the second color component target block can be performed by applying the derived prediction parameter to a residual signal of the restructured first color component, according to Equation 18.

$$p2[x, y] = p2_pred[x, y] + a \times p1'_residual[x, y] \quad \text{[Equation 18]}$$

In Equation 18, $p1'_residual$ represents a residual signal of the first color component and $p2_pred$ represents a prediction signal obtained by performing an intra prediction with respect to the second color component target block.

When the number of the derived prediction parameters is one or more, one or more prediction parameters may be applied to the reconstructed sample of the first color component. For example, when the reconstructed sample of the first color component satisfies a specific requirement, the inter color component intra prediction may be performed by applying the first prediction parameter derived from the reference samples that satisfy the specific requirement. Meanwhile, when the reconstructed sample of the first color component does not satisfy the specific requirement, the inter color component intra prediction may be performed by applying the second prediction parameter derived from the reference samples that do not satisfy the specific requirement. The specific requirement means a condition that the value of a reference sample is less than a statistic figure (for example, an average value) of the reference samples of the first color component.

The inter color component prediction method may be used in an inter prediction mode. For example, when performing the inter prediction on the current block, inter prediction is performed for a first color component, and inter color component prediction or prediction combining inter prediction and inter color component prediction may be performed for a second color component. For example, the first color component may be a luma component, and the second color component may be a chroma component.

The inter-color component prediction may be performed using the prediction sample or the reconstructed sample of the luminance component. For example, after the inter prediction for the luminance component is performed, prediction for a color component may be performed by applying inter-color component prediction parameters to the prediction sample resulting from the inter prediction of the luminance component. Here, the prediction sample refers to a sample that has undergone at least one of motion compensation, motion refinement, overlapped block motion compensation (OBMC), and bi-directional optical flow (BIO).

In addition, the inter color component prediction may be performed adaptively according to the coding parameters of the first color component. For example, it is possible to

62

determine whether to perform inter color component prediction according to CBF information of the first color component. The CBF information may be information indicating whether a residual signal exists or not. That is, when the CBF of the first color component is 1, inter color component prediction may be performed on the second color component. When the CBF of the first color component is 0, inter color component prediction may not be performed on the second color component, and the inter prediction may be performed on the second color component. Alternatively, a flag indicating whether or not to perform the inter color component prediction may be signaled.

When coding parameters of the first color component satisfies a predetermined condition, a flag indicating whether to perform the inter-color component prediction may be signaled. For example, when the CBF of the first color component is 1, the flag may be signaled to determine whether to perform color component prediction.

When performing inter-color component prediction for the second color component, an inter motion prediction or compensation value for the second color component may be used. For example, inter motion prediction or compensation for the second color component may be performed using inter prediction information of the first color component. In addition, prediction may be performed by calculating the weighted sum of the inter-color component prediction value for the second color component and the inter motion compensation value.

According to another embodiment of the present invention, template matching-based prediction may be performed.

When performing the template matching-based prediction, at least one intra-prediction mode (first intra-prediction mode) is explicitly and/or implicitly derived for the current block, and a template may be generated on the basis of the derived intra-prediction mode. Also, at least one intermediate intra-prediction block is generated on the basis of the second intra-prediction mode, and a prediction block for the current block is finally determined by using a weighted sum between the template and the intermediate intra-prediction block. In this case, the first intra-prediction mode and the second intra-prediction mode may be different.

FIG. 24 is a diagram illustrating an embodiment for generating a template.

As shown in FIG. 24, the intra-prediction mode N of the current block may be explicitly and/or implicitly derived. An intra-prediction block may be generated by performing intra-prediction on the basis of the derived intra-prediction mode N, and the intra-prediction block generated may be defined as a template. Herein, N may be zero or a positive integer greater than zero.

In this specification, a template generated in the template matching-based prediction process may be represented by a first intra-prediction block, and an intermediate intra-prediction block may be represented by a second intra-prediction block. In addition, when a final intra-prediction block of the current block is generated by using a weighted sum of the first intra-prediction block and the second intra-prediction block, one or more first intra-prediction blocks and/or second intra-prediction blocks may be used.

The intra-prediction mode of the current block may be explicitly encoded/decoded according to at least one of the various embodiments described herein. Alternatively, the same intra-prediction mode may be implicitly derived in the encoder and the decoder. The first intra-prediction block may be generated using the intra-prediction mode explicitly or implicitly derived as described above.

Alternatively, the first intra-prediction block may be generated using a predetermined intra-prediction mode in the encoder and the decoder.

The predetermined intra-prediction mode may be a Planar mode or a DC mode.

The predetermined intra-prediction mode may be a plurality of intra-prediction modes. In this case, a plurality of prediction blocks corresponding to a plurality of intra-prediction modes may be generated, and a first intra-prediction block (template) may be generated by using a weighted sum of the plurality of intra-prediction blocks generated. For example, after generating two intra-prediction blocks corresponding to the Planar mode and the DC mode, a block obtained by using a weighted sum of the two generated intra-prediction blocks may be defined as a template.

In another embodiment, a template may be generated using an intra-prediction mode of at least one neighboring block of neighboring blocks that are spatially adjacent to a current block. In this case, the neighboring block may be a block in which encoding/decoding is completed by using an intra-prediction mode before the current block.

For example, in the example shown in FIG. 11, when at least one of the neighboring blocks L, A, AR, BL, AL, JL, and JA is encoded/decoded using an intra-prediction mode, it is possible to generate a template using an intra-prediction mode of the corresponding neighboring block.

Alternatively, a predetermined statistical value may be obtained using one or more intra-prediction modes derived from one or more neighboring blocks, and a template may be generated using an intra-prediction mode indicated by the statistical value. Herein, the statistical value may include at least one of an average value, a maximum value, a minimum value, an intermediate value, and a mode value.

Alternatively, after configuring the MPM list for the current block, a template may be generated using one or more intra-prediction modes included in the MPM list. For example, a template may be generated using an intra-prediction mode indicated by the first MPM index or an intra-prediction mode having directionality (angular prediction mode) in the MPM list. When there are a plurality of intra-prediction modes having directionality in the MPM list, a template may be generated using an intra-prediction mode having a smaller MPM index.

An intra-prediction mode for generating a template may be generated by deriving one or more explicit and/or implicit intra-prediction modes.

For example, the intra-prediction mode information transmitted from the encoder may be entropy-encoded/decoded to be explicitly derived, and then the intra-prediction block A corresponding to the intra-prediction mode may be generated. In addition, the intra-prediction block B corresponding to the fixed intra-prediction mode N may be generated, and then a template may be generated by using a weighted sum of the intra-prediction block A and/or the intra-prediction block B. The fixed intra-prediction modes may be a plurality of intra-prediction modes, and thus a plurality of intra-prediction blocks B may be generated. The fixed intra-prediction mode may be a Planar mode or a DC mode.

In another example of generating the intra-prediction block B, it is possible to generate an intra-prediction block B corresponding to an intra-prediction mode of one or more neighboring blocks in which encoding/decoding is completed using an intra-prediction mode, as a block spatially adjacent to the current block. The neighboring block may be a neighboring block L shown in FIG. 11. Alternatively, the

neighboring block is not limited thereto, and may be at least one of the neighboring blocks A, AR, BL, AL, JL and JA shown in FIG. 11.

In another embodiment of generating the intra-prediction block B, after one or more intra-prediction modes are derived from the one or more blocks shown in FIG. 11, it is possible to generate an intra-prediction block B corresponding to the statistical value of the derived one or more intra-prediction modes. The statistical value may be an average value, a median value, a mode value, or the like. However, the statistical value is not limited to this, and may mean at least one of a maximum value, a minimum value, a weighted average value, and an interpolation value.

As another example of generating the intra-prediction block B, it is possible to generate an intra-prediction block B corresponding to at least one intra-prediction mode included in the MPM list for the current block. The intra-prediction mode used to generate the intra-prediction block B may be an intra-prediction mode indicated by the first MPM index of the MPM list. Alternatively, the intra-prediction mode used to generate the intra-prediction block B may be an intra-prediction mode having directionality among modes in the MPM list. In this case, when there are a plurality of modes having directionality in the MPM list, the intra-prediction block B may be generated using an intra-prediction mode having a smaller MPM index.

Thereafter, one or more template matching-based intermediate intra-prediction blocks may be generated using the generated template.

FIG. 25 is a diagram illustrating an embodiment for generating a template matching-based intermediate intra-prediction block on the basis of the MPM list.

The left table shown in FIG. 25 shows the MPM list for the current block. For example, the MPM list includes six candidate modes, each candidate mode being indicated by an index (MPM idx). In the embodiment shown in FIG. 25, the intra-prediction modes included in the MPM list are X, Y, N, Z, K, and L.

It is possible to generate a prediction block using each of intra-prediction modes included in the MPM list. As shown in FIG. 25, a prediction block 0 is a prediction block corresponding to the intra-prediction mode X, and a prediction block 5 is a prediction block corresponding to the intra-prediction mode L. Likewise, although not shown in FIG. 25, prediction blocks 1 to 4 corresponding to the intra-prediction modes Y, N, Z, and K may also be generated. Hereinafter, the prediction block 0 to the prediction block 5 is defined as a prediction block n (n is one of 0 to 5).

Thereafter, template matching between a template and a prediction block n generated by using modes included in the MPM list may be performed. The template matching may be performed by calculating at least one of a sum of absolute differences (SAD) and/or a sum of absolute transformed differences (SATD) between the template and the prediction block n. Herein, the template matching may be omitted for a mode same as the intra-prediction mode (for example, N in FIG. 25) used for generating the template among the intra-prediction modes included in the MPM list.

As described above, after the template matching is performed for each of intra-prediction modes included in the MPM list to calculate SAD_0 (or SARD_0) to SAD_5 (or SARD_5), an intra-prediction block having the smallest SAD and/or SATD may be defined as an intermediate intra-prediction block. For example, when SAD_0 is the smallest, the prediction block 0 corresponding to the intra-prediction mode X may be an intermediate intra-picture block.

65

Alternatively, after the calculated SAD and/or SATD are arranged in increasing order, M intra-prediction blocks in order of increasing SAD and/or SATD may be defined as intermediate intra-prediction blocks. Herein, M is a positive integer equal to or greater than 1 and may be less than or equal to the maximum number of candidate modes configuring the MPM list. For example, when the maximum number of candidate modes configuring the MPM list is 7, the maximum value of M may be defined as 7.

The order of the intra-prediction modes in the MPM list may be rearranged using at least one of the SAD and/or the SATD.

The above description of the template matching may be similarly applied to the case of using a secondary MPM list instead of the MPM list.

That is, a prediction block may be generated using each of intra-prediction modes included in the secondary MPM list. Thereafter, template matching between the template and the prediction block generated using the mode included in the secondary MPM list may be performed. The template matching may be performed by calculating at least one of a sum of absolute differences (SAD) and/or a sum of absolute transformed differences (SATD) between the template and the prediction block. Herein, the template matching may be omitted for a mode same as the intra-prediction mode used when generating the template among the intra-prediction modes included in the secondary MPM list. Then, an intra-prediction block having the smallest SAD and/or SATD may be defined as an intermediate intra-prediction block. Alternatively, the calculated SAD and/or SATD may be arranged in increasing order, and the M intra-prediction blocks in order of increasing SAD and/or SATD may be defined as intermediate intra-prediction blocks. Here, M is a positive integer equal to or greater than 1 and may be less than or equal to the maximum number of candidate modes configuring a secondary MPM list. For example, when the maximum number of candidate modes configuring the secondary MPM list is 16, the maximum value of M may be defined as 16.

The order of the intra-prediction modes in the secondary MPM list may be rearranged using at least one of the SAD and/or the SATD.

In another embodiment, an intra-frame prediction block corresponding to an intra-prediction mode of at least one neighboring block in which encoding/decoding is completed may be defined as an intermediate intra-prediction block. The neighboring blocks may be at least one of the neighboring blocks L, A, AR, BL, AL, JL, and JA shown in FIG. 11.

Alternatively, a predetermined statistical value may be obtained by using one or more intra-prediction modes derived from one or more neighboring blocks, and an intra-prediction block corresponding to the statistical value may be defined as an intermediate intra-prediction block. Herein, the statistical value may include at least one of an average value, a maximum value, a minimum value, an intermediate value, and a mode value. The statistical value may be calculated using intra-prediction modes of all intra-predicted neighboring blocks shown in FIG. 11.

In another embodiment, when the number of intra-prediction modes used when generating the template is N, a prediction block corresponding to the intra-prediction mode in which a predetermined offset is added and/or subtracted to and/or from N may be defined as an intermediate intra-prediction block. Herein, the offset may be a positive integer of 1 or more.

66

After generating the template and at least one intermediate intra-prediction block as described above, the final intra-prediction block for the current block may be generated using the template and at least one intermediate intra-prediction block. Herein, a weighted sum may be used.

FIG. 26 is a view illustrating an embodiment for generating a final intra-prediction block using a template and one intermediate intra-prediction block.

As shown in FIG. 26, a final intra-prediction block may be generated by using a weighted sum of a template and an intermediate intra-prediction block.

FIG. 27 is a diagram illustrating an embodiment in which a final intra-frame prediction block is generated using the template and a plurality of intermediate intra-prediction blocks.

As shown in FIG. 27, a final prediction block may be generated by using a weighted sum of the template and the plurality of intermediate intra-prediction blocks.

In the embodiment described with reference to FIGS. 26 and 27, the weight for performing the weighted sum may be derived as follows. Herein, the sum of all the weights used in the weighted sum may be one or not. In addition, each weight may be greater than or less than one.

For example, all weights used in the weighted sum may be the same.

Alternatively, different weights may be applied for each block according to a predetermined priority between the template and/or at least one intermediate intra-prediction block.

For example, a higher weight may be applied to a template corresponding to a predetermined intra-prediction mode and/or an intermediate intra-prediction block, than an intra-prediction block corresponding to the remaining modes. Conversely, a lower weight may be applied to the template corresponding to the predetermined intra-prediction mode and/or the intermediate intra-prediction block. For example, the predetermined intra-prediction mode may be the first intra-prediction mode of the MPM list. In this case, a relatively high weight may be given to the intra-prediction block corresponding to the first intra-prediction mode in the MPM list.

When encoding a current block by using intra-prediction, the encoder may determine whether or not to perform the method proposed in the present invention and entropy-encode indication information indicating whether or not to perform the method. Whether or not to perform the method proposed in the present invention may be determined through a comparison of a rate-distortion cost value before applying the proposed method and a rate-distortion cost value after application.

The indication information may be conditionally entropy-encoded according to coding parameter of the current block. For example, the indication information may be entropy-encoded only when the intra-prediction mode of the current block is encoded using a mode included in the MPM list and/or the secondary MPM list. The decoder may entropy-decode the indication information from the bit stream and then perform or not perform the method proposed in the present invention according to the received information.

The indication information may be omitted according to the encoding parameter of the current block. For example, it may be determined whether or not to perform the proposed method on the basis of a predetermined size, a form and/or a depth of the current block. For example, the proposed method may be performed or not performed only when the current block is less than or equal to or equal to or greater than the predetermined size, form, or depth. Herein, the

information on the predetermined size, form, and/or depth may be entropy-encoded/decoded in units of at least one of a video parameter set VPS, a sequence parameter set SPS, a picture parameter set PPS, a tile header, a slice header, CTU, and CU.

For example, when the size of the current block is less than or equal to a predetermined first size and/or equal to or greater than a predetermined second size, and the current block is divided into binary tree leaf nodes and/or quad tree leaf nodes, the proposed method may be performed or may be not performed.

Alternatively, the proposed method may be performed when encoded using a mode included in the MPM list and/or the secondary MPM list.

It may be determined whether or not to perform the proposed method on the basis of a comparison of the size and depth of current block with a predetermined threshold value. In this case, the predetermined threshold value may mean a reference size or depth for determining the block structure. The predetermined threshold may be expressed in at least one of a minimum value and a maximum value. The predetermined threshold value may be a fixed value predefined by the encoder/decoder, variably derived on the basis of the encoding parameter of the current block, or may be signaled through a bitstream.

For example, the proposed method may or may not be performed when the size or depth of the current block is less than or equal to a first predetermined threshold value and/or equal to or greater than a second predetermined threshold value.

Alternatively, when a depth of the current block is equal to the predetermined threshold value and the current block corresponds to a quad tree leaf node, the proposed method may not be performed.

In order to determine an encoding mode for the current block, an encoder performs at least one of transform, quantization, entropy encoding, entropy decoding, dequantization, and inverse-transform on a residual signal between the current block and the prediction block, and then determines an encoding mode so that the required bit amount and video quality loss may be minimized. The determined encoding mode may be encoded as encoding mode information, and the encoding mode information may be used with the same meaning as an encoding parameter.

Assuming that the maximum number of candidate blocks in a merge mode and/or a skip mode for the current block is M, a process of determining an encoding mode for the merge mode and/or the skip mode may be performed only on merge modes and/or skip mode candidate blocks of the number that is equal to or less than M according to conditions described below.

In the process of determining the encoding mode for the merge mode and/or the skip mode, when the current block (quad tree leaf node and/or binary tree leaf node) is re-encoded by the encoder, in the case that it is determined as the skip mode in an initial encoding process, it is possible to perform a skip mode encoding determination process for candidate blocks in all merge modes and/or skip modes.

In the process of determining the encoding mode for the merge mode and/or the skip mode, when the current block (quad tree leaf node and/or binary tree leaf node) is re-encoded by the encoder, when it is determined as the merge mode in the initial encoding process, it is possible to perform a merge mode encoding determination process for candidate blocks in all the merge modes and/or skip modes.

In the process of determining the encoding mode for the merge mode and/or the skip mode, when the current block

(quad tree leaf node and/or binary tree leaf node) is re-encoded by the encoder, when it is determined as the skip mode and/or the merge mode in the initial encoding process, it is possible to perform an overlapped block motion compensation (OBMC) process in a motion compensation process for candidate blocks in all merge modes and/or skip modes.

In the process of determining the encoding mode for the current block, the encoding mode determination process for the primary transform and/or the secondary transform may be omitted according to the encoding parameter of the current block and/or the neighboring blocks that are temporally/spatially adjacent to the current block.

For example, when it is assumed that the number of 1D transform kernels that may be used for the primary transform in the primary transform is N at maximum, and thus the primary transform index (ucTrIdx) is entropy-encoded/decoded from 0 to N-1, if CBF for the current block is 0 at index 0, the encoding mode determination process for subsequent indexes (ucTrIdx=1, . . . , N-1) may be omitted. Alternatively, the encoding mode determination process of the current block may be performed only for the primary transform index used for the primary transform of the neighboring blocks of the current block.

In the process of determining the intra-picture and/or inter-picture encoding mode for the current block, when entropy-encoding the primary transform index information (ucTrIdx) for the 1D transform kernel used for the primary transform, a division operation may be replaced with a shift operation.

In the process of determining the intra-picture encoding mode for the current block, when entropy-encoding the primary transform index information (ucTrIdx) for the 1D transform kernel used for the primary transform, a division operation and/or a shift operation may be used. For example, the primary transform index information ucTrIdx may be entropy-encoded using a division operation as shown in Equation 19 below.

$$m_pcBinIf \rightarrow encodeBin \quad [Equation\ 19]$$

$$((ucTrIdx/2)?1:0, m_cEmtTuIdxSCModel.get(0, 0, 1));$$

Alternatively, the primary transform index information ucTrIdx may be entropy-encoded using a shift operation as shown in Equation 20 below.

$$m_pcBinIf \rightarrow encodeBin((ucTrIdx \gg 1)?1:0, m_cEmtTuIdxSCModel.get(0, 0, 1)); \quad [Equation\ 20]$$

In the process of determining the inter-picture encoding mode for the current block, when entropy-encoding the primary transform index information (ucTrIdx) for the 1D transform kernel used for the primary transform, the equations 19 and 20 may be used similarly. However, in this case, get (0,0,1) may be replaced with get (0,0,3) in the equations 19 and 20.

The residual signal generated after performing intra-prediction or inter-prediction may be converted into a frequency domain through a transform process as a part of a quantization process. In this case, a primary transform may be used with various DCT kernels, DST kernels, and the like in addition DCT type 2 (DCT-II). The transform using the

transform kernels may be performed by a separable transform in which one-dimensional transform (1D transform) is performed on the horizontal and vertical directions of the residual signal, or by a two-dimensional non-separable transform (2D Non-separable Transform).

The DCT and DST types used in the transform are shown in Table 1 below.

TABLE 1

Transform set	Transform
0	DCT-VII, DCT-VIII
1	DST-VII, DST-I
2	DST-VII, DCT-V

In addition to DCT-II, at least one of DST-II, DCT-IV, DST-IV, DCT-VIII, and DST-VII may be adaptively used during 1D transform. Alternatively, a transform set may be configured to derive the DCT or DST type used in the transform. For example, according to the intra-prediction mode, different transform sets may be pre-defined for horizontal and vertical directions. The encoder/decoder may perform the transform and inverse-transform using the transform included in the transform set corresponding to the intra-prediction mode of the current block. In this case, the transform set is not transmitted, but is mapped in the encoder/decoder according to the same rule as shown in Table 1, in which information indicating which transform is used among the transforms belonging to the transform set may be signaled. Such a method of using various transforms may be applied to a residual signal generated through intra-prediction or inter-prediction. For example, during inter-picture prediction, the encoder/decoder may perform 1D transform on the vertical and/or horizontal directions using the transform set 0 of Table 1. In this case, information indicating which 1D transform is used for the vertical and/or horizontal directions may be entropy-encoded/decoded. Further, information indicating which one of DST-II, DCT-IV, DST-IV, DCT-VIII and DST-VII is used for the vertical and/or horizontal directional transform may be entropy-encoded/decoded. After the above-described primary transform is completed, the encoder may perform a secondary transform to increase the energy concentration on transform coefficients as shown in FIG. 6. The secondary transform may also perform a separable transform that performs one dimensional transform for each of the horizontal and vertical directions, or may perform a two-dimensional non-separable transform. In addition, the information used for the transform may be transmitted or implicitly derived by the encoder/decoder according to the encoding information of the current block and the neighboring block. For example, a transform set may be defined for the secondary transform, like the first transform, and the transform set may be not entropy-encoded/decoded, but defined according to the same rules by the encoder/decoder. In this case, information indicating which transform is used among transforms belonging to the transform set may be transmitted and may be applied to at least one of residual signals generated through intra-prediction or inter-prediction. The decoder may perform the secondary inverse-transform according to whether or not the secondary inverse-transformation is performed, and perform the primary inverse-transformation on the result of performing the secondary inverse-transform on a result obtained by performing the secondary inverse-transform according to whether or not the primary inverse-transform is performed.

The primary transform and the secondary transform may be applied to at least one signal component of luma/chroma components or may be adaptively applied according to an arbitrary encoding block size/form. Whether or not the primary transform and the secondary transform are used for an arbitrary encoding block and the index indicating the used primary transform and/or secondary transform may be entropy-encoded/decoded or implicitly derived by the encoder/decoder according to at least one of encoding information of the current block and the neighboring block.

FIG. 28 is a diagram illustrating a method of scanning a transform coefficient.

For the residual signal generated through intra-prediction or inter-prediction, the primary and secondary transforms are performed, and then quantization process is performed, in which an entropy encoding process may be performed on the quantized transform coefficients. In this case, as shown in FIG. 28, the quantized transform coefficients may be scanned according to a diagonal direction, a vertical direction, or a horizontal direction on the basis of at least one of an intra-prediction mode of the current block and a minimum size and/or form of the encoding block. In addition, the entropy-decoded quantized transform coefficients may be inverse-scanned and arranged in a block form, and at least one of inverse-quantization and inverse-transform may be performed on the corresponding block. Herein, at least one of diagonal scan, horizontal scan, and vertical scan shown in FIG. 28 may be performed as the inverse-scanning method.

When performing intra or inter prediction, a first color component may undergo intra prediction and a second color component may undergo inter prediction. For example, the first color component is a luminance component, and the second color component is a chrominance component. Conversely, the first color component may be a chrominance component and the second color component may be a luminance component.

Regarding application of filtering to the prediction samples, whether to apply filtering or not may be determined depending on at least one of the intra prediction mode, size (width and height), block shape, multiple sample line-based prediction, and color component of the current block. The filtering refers to a method of filtering one or more prediction samples using one or more reference samples.

For example, when the intra prediction mode of the current block is a predetermined mode, the filtering may be applied to the prediction samples. For example, the predetermined mode is a directional mode, a non-directional, a horizontal mode, or a vertical mode.

For example, when the size of the current block falls within a predetermined size range, the filtering may be applied to the prediction samples. For example, when the current block has a width less than 64 and a height less than 64, the filtering may be applied. Alternatively, when the width or height of the current block is larger or smaller than a predetermined size, the filtering may be applied.

For example, whether to apply filtering to the prediction samples may be determined depending on the reference sample line used for the prediction. For example, when the reference sample line used for the prediction is the first reference sample line adjacent to the current block, the filtering may be applied. On the other hand, when the reference sample line is one of the second and onward reference sample lines positioned around the current block, the filtering may not be applied. The indicator `mrl_index` may be used to determine the reference sample line. For example, when the index for the current block is zero, the

filtering is applied. However, when the index for the current block is a value greater than zero, the filtering is not applied.

For example, when the color component of the block element is a luminance signal, the filtering is applied. However, when the color component of the current block is a chrominance signal, the filtering is not applied.

The prediction for the current block can be performed by combining one or more exemplary prediction methods described above.

For example, the prediction for the current block may be performed by calculating the weighted sum of a prediction value obtained using a predetermined non-directional intra prediction mode and a prediction value obtained using a predetermined directional intra prediction mode. In this case, the weights may vary depending on at least one of the intra prediction mode of the current block, the size/shape of the current block, and the position of the prediction target sample.

For example, the prediction for the current may be performed by calculating the weighted sum of a prediction value obtained using a predetermined intra prediction mode and a prediction value obtained predicted using a predetermined inter prediction mode. In this case, the weights may vary depending on at least one of the encoding mode, the intra prediction mode, the inter prediction mode, and the size/shape of the current block. For example, when the intra prediction mode is a non-directional mode such as DC or Planar, a weight corresponding to $\frac{1}{2}$ may be applied to an intra prediction sample and an inter prediction sample, respectively. Alternatively, when the intra prediction mode is a vertical mode, the weight for the intra prediction sample decreases with distance from the reference sample line above the current block. Similarly, when the intra prediction mode is a horizontal mode, the weight for the intra sample decreases with distance from the reference sample line on the left side of the current block. The sum of the weight applied to the intra prediction sample and the weight applied to the inter prediction sample may be any one of the powers of two. That is, it may be any of 4, 8, 16, 32, and so forth. For example, when the size of the current block is within a predetermined size range, a weight corresponding to $\frac{1}{2}$ may be applied to the intra prediction sample and the inter prediction sample, respectively.

The intra prediction mode may be fixed to DC mode and Planar mode, or may be determined through signaling of information. Alternatively, the intra prediction mode may be any mode selected from among MPM candidate modes, and may be determined through The MPM candidate modes are derived from the intra prediction modes of neighboring blocks. The mode of the neighboring block can be replaced with a predetermined representative mode. For example, the intra prediction mode of a neighboring block is a directional mode of a specific direction categorized into a vertical direction group, the mode of the neighboring block is replaced with the vertical mode. On the other hand, when the intra prediction mode of a neighboring block is a directional mode of a specific direction categorized into a horizontal direction group, the mode of the neighboring block is replaced with the horizontal mode.

The inter prediction may be at least one of DC mode, merge mode, and AMVP mode. When the inter prediction mode of the current block is merge mode, the prediction for the current block may be performed by calculating the weighted sum of the inter prediction value obtained by using motion information corresponding to a merge index and the prediction value obtained by using DC or Planar mode.

For example, the prediction for the current block may be performed by calculating the weighted sum of one or more prediction samples obtained by using multiple sample lines. For example, the prediction may be performed by calculating the weighted sum of a first prediction value obtained by using the first reference sample line near the current block and a second prediction value obtained by using the second and onward reference sample lines near the current block. The reference sample lines used to obtain the second prediction value may be reference sample lines indicated by `mrl_index`. The weights for the first prediction value and the second prediction value may be equal. Alternatively, the weights for the first prediction value and the second prediction value may vary depending on at least one of the intra prediction mode of the current block, the size/shape of the current block, and the position of the sample to be prediction. The first prediction value may be a value predicted using a predetermined mode. For example, the first prediction value may be a value predicted using at least one of DC mode and Planar mode. The second prediction value may be a value predicted using the intra prediction mode of the current block, which is derived in the available intra prediction mode derivation step.

When prediction is performed by calculating the weighted sum of one or more prediction samples, filtering may not be performed on the prediction samples.

The above embodiments may be performed in the same method in an encoder and a decoder.

A sequence of applying to above embodiment may be different between an encoder and a decoder, or the sequence applying to above embodiment may be the same in the encoder and the decoder.

The above embodiment may be performed on each luma signal and chroma signal, or the above embodiment may be identically performed on luma and chroma signals.

A block form to which the above embodiments of the present invention are applied may have a square form or a non-square form.

The above embodiment of the present invention may be applied depending on a size of at least one of a coding block, a prediction block, a transform block, a block, a current block, a coding unit, a prediction unit, a transform unit, a unit, and a current unit. Herein, the size may be defined as a minimum size or maximum size or both so that the above embodiments are applied, or may be defined as a fixed size to which the above embodiment is applied. In addition, in the above embodiments, a first embodiment may be applied to a first size, and a second embodiment may be applied to a second size. In other words, the above embodiments may be applied in combination depending on a size. In addition, the above embodiments may be applied when a size is equal to or greater than a minimum size and equal to or smaller than a maximum size. In other words, the above embodiments may be applied when a block size is included within a certain range.

For example, the above embodiments may be applied when a size of current block is 8×8 or greater. For example, the above embodiments may be applied when a size of current block is 4×4 or greater. For example, the above embodiments may be applied when a size of current block is equal to or less than 16×16 . For example, the above embodiments may be applied when a size of current block is equal to or greater than 16×16 and equal to or smaller than 64×64 .

The above embodiments of the present invention may be applied depending on a temporal layer. In order to identify a temporal layer to which the above embodiments may be

73

applied, a corresponding identifier may be signaled, and the above embodiments may be applied to a specified temporal layer identified by the corresponding identifier. Herein, the identifier may be defined as the lowest layer or the highest layer or both to which the above embodiment may be applied, or may be defined to indicate a specific layer to which the embodiment is applied. In addition, a fixed temporal layer to which the embodiment is applied may be defined.

For example, the above embodiments may be applied when a temporal layer of a current image is the lowest layer. For example, the above embodiments may be applied when a temporal layer identifier of a current image is 1. For example, the above embodiments may be applied when a temporal layer of a current image is the highest layer.

A slice type to which the above embodiments of the present invention are applied may be defined, and the above embodiments may be applied depending on the corresponding slice type.

In the above-described embodiments, the methods are described based on the flowcharts with a series of steps or units, but the present invention is not limited to the order of the steps, and rather, some steps may be performed simultaneously or in different order with other steps. In addition, it should be appreciated by one of ordinary skill in the art that the steps in the flowcharts do not exclude each other and that other steps may be added to the flowcharts or some of the steps may be deleted from the flowcharts without influencing the scope of the present invention.

The embodiments include various aspects of examples. All possible combinations for various aspects may not be described, but those skilled in the art will be able to recognize different combinations. Accordingly, the present invention may include all replacements, modifications, and changes within the scope of the claims.

The embodiments of the present invention may be implemented in a form of program instructions, which are executable by various computer components, and recorded in a computer-readable recording medium. The computer-readable recording medium may include stand-alone or a combination of program instructions, data files, data structures, etc. The program instructions recorded in the computer-readable recording medium may be specially designed and constructed for the present invention, or well-known to a person of ordinary skill in computer software technology field. Examples of the computer-readable recording medium include magnetic recording media such as hard disks, floppy disks, and magnetic tapes; optical data storage media such as CD-ROMs or DVD-ROMs; magneto-optimum media such as floptical disks; and hardware devices, such as read-only memory (ROM), random-access memory (RAM), flash memory, etc., which are particularly structured to store and implement the program instruction. Examples of the program instructions include not only a mechanical language code formatted by a compiler but also a high level language code that may be implemented by a computer using an interpreter. The hardware devices may be configured to be operated by one or more software modules or vice versa to conduct the processes according to the present invention.

Although the present invention has been described in terms of specific items such as detailed elements as well as the limited embodiments and the drawings, they are only provided to help more general understanding of the invention, and the present invention is not limited to the above embodiments. It will be appreciated by those skilled in the

74

art to which the present invention pertains that various modifications and changes may be made from the above description.

Therefore, the spirit of the present invention shall not be limited to the above-described embodiments, and the entire scope of the appended claims and their equivalents will fall within the scope and spirit of the invention.

INDUSTRIAL APPLICABILITY

The present invention may be used in encoding/decoding an image.

The invention claimed is:

1. An image decoding method performed by an image decoding apparatus, the method comprising:
 - determining whether a combination of an intra prediction and an inter prediction is used for prediction on a current block; and
 - in response to that the combination of the intra prediction and the inter prediction is used for prediction on the current block, generating a final prediction block of the current block,
 wherein, generating of the final prediction block of the current block comprises:
 - generating a first prediction block of the current block by performing intra prediction for the current block based on an intra prediction mode and reference samples of a reference sample line;
 - generating a second prediction block of the current block by performing inter prediction for the current block using a merge index; and
 - generating the final prediction block of the current block based on the first prediction block and the second prediction block,
 wherein when the combination of the intra prediction and the inter prediction is applied for prediction on the current block, the intra prediction mode and the reference sample line are determined without obtaining information on intra prediction regarding the intra prediction mode and the reference sample line,
 wherein the intra prediction mode of the current block is fixed as planar mode and the reference sample line is fixed as a first reference sample line adjacent to the current block.
2. The method of claim 1,
 - wherein, the second prediction block of the current block is generated by a merge mode.
3. The method of claim 1,
 - wherein the final prediction block of the current block is generated by a weighted sum of the first prediction block and the second prediction block.
4. The method of claim 3,
 - wherein weights used for the weighted sum comprise a first weight which is applied to the first prediction block and a second weight which is applied to the second prediction block, and
 - wherein a sum of the first weight and the second weight is 4.
5. The method of claim 1,
 - wherein the combination of the intra prediction and the inter prediction is used for prediction on the current block based on a slice type of the current block.
6. An image encoding method performed by an image encoding apparatus, the method comprising:
 - determining whether a combination of an intra prediction and an inter prediction is used for prediction on a current block; and

75

in response to that the combination of the intra prediction and the inter prediction is used for prediction on the current block, generating a final prediction block of the current block,

wherein, generating of the final prediction block of the current block comprises:

generating a first prediction block of the current block by performing intra prediction for the current block based on an intra prediction mode and reference samples of a reference sample line;

generating a second prediction block of the current block by performing inter prediction for the current block using a merge index; and

generating the final prediction block of the current block based on the first prediction block and the second prediction block,

wherein when the combination of the intra prediction and the inter prediction is applied for prediction on the current block, the intra prediction mode and the reference sample line are determined without obtaining information on intra prediction regarding the intra prediction mode and the reference sample line,

wherein the intra prediction mode of the current block is fixed as planar mode and the reference sample line is fixed as a first reference sample line adjacent to the current block.

7. The method of claim 6,

wherein, the second prediction block of the current block is generated by a merge mode.

8. The method of claim 6,

wherein the final prediction block of the current block is generated by a weighted sum of the first prediction block and the second prediction block.

9. The method of claim 8,

wherein weights used for the weighted sum comprise a first weight which is applied to the first prediction block and a second weight which is applied to the second prediction block, and

wherein a sum of the first weight and the second weight is 4.

10. The method of claim 6,

wherein, in response to that the combination of the intra prediction and the inter prediction is used for prediction on the current block, the reference sample for the intra

76

prediction of the current block is determined based on a first reference sample line adjacent to the current block among a plurality of reference sample lines.

11. The method of claim 6,

wherein the combination of the intra prediction and the inter prediction is used for prediction on the current block based on a slice type of the current block.

12. A non-transitory computer-readable recording medium storing a bitstream which is generated by an image encoding method,

wherein the image encoding method comprises

determining whether a combination of an intra prediction and an inter prediction is used for prediction on a current block; and

in response to that the combination of the intra prediction and the inter prediction is used for prediction on the current block, generating a final prediction block of the current block,

wherein, generating of the final prediction block of the current block comprises:

generating a first prediction block of the current block by performing intra prediction for the current block based on an intra prediction mode and reference samples of a reference sample line;

generating a second prediction block of the current block by performing inter prediction for the current block using a merge index; and

generating the final prediction block of the current block based on the first prediction block and the second prediction block,

wherein when the combination of the intra prediction and the inter prediction is applied for prediction on the current block, the intra prediction mode and the reference sample line are determined without obtaining information on intra prediction regarding the intra prediction mode and the reference sample line,

wherein the intra prediction mode of the current block is fixed as planar mode and the reference sample line is fixed as a first reference sample line adjacent to the current block.

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